

Adopted Levels, Gammas

$Q(\beta^-) = -5966.2$ 7; $S(n) = 12644.76$ 5; $S(p) = 6370.81$ 4; $Q(\alpha) = -6997.90$ 4 [2021Wa16](#)
 $S(2n) = 24152.8$ 4, $S(2p) = 17254.1$ 11 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): positive-ray mass spectrograph at Cambridge, UK ([1919As01](#)).

 ^{35}Cl LevelsCross Reference (XREF) Flags

A	$^{35}\text{S} \beta^-$ decay (87.61 d)	K	$^{31}\text{P}(\text{}^6\text{Li}, \text{d})$	U	$^{35}\text{Cl}(\text{p}, \text{p}'\gamma)$
B	$^{35}\text{Ar} \varepsilon$ decay (1.7755 s)	L	$^{32}\text{S}(\alpha, \text{p})$	V	$^{36}\text{Ar}(\text{n}, \text{d})$
C	$^{36}\text{K} \varepsilon \text{p}$ decay (341 ms)	M	$^{32}\text{S}(\alpha, \text{p}\gamma)$	W	$^{36}\text{Ar}(\text{d}, \text{}^3\text{He})$
D	$^1\text{H}(\text{}^{34}\text{S}, \text{}^{35}\text{Cl}\gamma)$	N	$^{33}\text{S}(\alpha, \text{d})$	X	$^{36}\text{Ar}(\text{pol d}, \text{}^3\text{He})$
E	$^2\text{H}(\text{}^{34}\text{S}, \text{n}\gamma)$	O	$^{34}\text{S}(\text{p}, \gamma)$	Y	$^{37}\text{Cl}(\text{p}, \text{t})$
F	$^{12}\text{C}(\text{}^{28}\text{Si}, \alpha \text{p}\gamma)$	P	$^{34}\text{S}(\text{p}, \text{p}), (\text{p}, \text{p}'\gamma): \text{resonances}$	Z	$^{39}\text{K}(\gamma, \alpha)$
G	$^{16}\text{O}(\text{}^{24}\text{Mg}, \alpha \text{p}\gamma)$	Q	$^{34}\text{S}(\text{d}, \text{n})$	Others:	
H	$^{24}\text{Mg}(\text{}^{16}\text{O}, \alpha \text{p}\gamma)$	R	$^{34}\text{S}(\text{}^3\text{He}, \text{d})$	AA	$^{40}\text{Ca}(\mu^-, \nu \alpha \text{n}\gamma)$
I	$^{27}\text{Al}(\text{}^{14}\text{N}, \alpha \text{p}\text{n}\gamma)$	S	$^{35}\text{Cl}(\gamma, \gamma')$	AB	Coulomb excitation
J	$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n}): \text{resonances}$	T	$^{35}\text{Cl}(\text{n}, \text{n}'\gamma)$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}^\ddagger$	XREF	Comments
0.0	$3/2^+$	stable	ABC EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB $\mu = +0.821870$ 2 (2013Ja17 , 2019StZV) $Q = -0.0817$ 8 (2008Py02 , 2021StZZ) J^π : L=2 from 0^+ in (d,n), (n,d), ($^3\text{He}, \text{d}$), (d, ^3He), and (pol d, ^3He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10 , 1949Da14). μ : Nuclear Magnetic Resonance (2013Ja17). Q: Atomic beam magnetic resonance (2008Py02). Others: -0.08249 2 (1972St38), -0.076 5 (1986El09), -0.817 8 (1993Su36), 0.0819 11 (2000Ha64), 0.0850 11 (2004Al08). Evaluated rms nuclear charge radius $R = 3.365$ fm 19 (2013An02).
1219.38 7	$1/2^+$	0.119 ps 14	B E H KLM O QRSTU VWXY	XREF: Others: AA , AB $B(E2)^\ddagger = 0.00076$ 7 $B(E2)^\ddagger$: from Coulomb excitation. $E(\text{level})$: weighted average of 1219.2 2 from $^{35}\text{Ar} \varepsilon$ decay, 1219.39 7 from (p, γ), and 1219.42 10 from (p, p' γ), 1219.0 10 from ($^{16}\text{O}, \alpha \text{p}\gamma$), and 1219.4 6 from ($\alpha, \text{p}\gamma$). J^π : L=0 from 0^+ in (d,n), (n,d), ($^3\text{He}, \text{d}$), (d, ^3He), and (pol d, ^3He). $T_{1/2}$: from lifetime $\tau = 0.172$ ps 20: weighted average of 0.29 ps 4 from ($^{34}\text{S}, \text{n}\gamma$) (1973Wa10 , DSAM), 0.21 ps +10-8 from ($\alpha, \text{p}\gamma$) (1969In04 , DSAM), 80 fs 40 (1971Wi13 , DSAM), 175 fs 20 (1972Hu11 , DSAM), 150 fs 20 (1973Fa07 , DSAM), 145 fs 30 (1976Me12 , DSAM) from (p, γ), 0.13 ps +3-2 from (γ, γ') (1966Ho02 , resonance fluorescence), 200 fs 80 from (n, n' γ) (1989Ge09 , DSAM), 0.215 ps 75 (1972Va06 , DSAM), 0.21 ps +6-4 (1972Br45 , DSAM), 0.27 ps +19-10 (1969Du08 , DSAM) from (p, p' γ), 0.16 ps 5 (1969Ha17 , DSAM), and 0.230 ps 25 (1977Sc36 , DSAM) from Coulomb excitation.
1763.15 4	$5/2^+$	0.36 ps 3	B EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π#</u>	<u>T_{1/2}[‡]</u>	<u>XREF</u>	<u>Comments</u>
				T=1/2 B(E2)↑=0.0106 7 XREF: X(1785)Y(1750) B(E2)↑: from Coulomb excitation. E(level): weighted average of 1763.1 2 from ^{35}Ar ε decay, 1763.0 6 from ($^{28}\text{Si}, \alpha\text{py}$), 1762.95 17 from ($^{24}\text{Mg}, \alpha\text{py}$), 1763.3 2 from ($^{16}\text{O}, \alpha\text{py}$), 1763.19 10 from ($^{14}\text{N}, \alpha\text{pny}$), 1763.14 4 from (p,$\gamma$), 1763.26 10 from (p,p'$\gamma$). J^π: L=2 from 0⁺ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in (α,py), (p,γ), and (p,p'γ). T_{1/2}: lifetime τ=0.52 ps 5: weighted average of 0.54 ps 7 from ($^{34}\text{S}, \text{n}\gamma$) (1973Wa10, DSAM), 0.55 ps 15 from (α,py) (1969In04, DSAM), 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), 0.53 ps 9 (1976Me12, DSAM) from (p,γ), 0.60 ps 21 from (γ,γ') (1962Bo17, resonance fluorescence), 0.205 ps 75 (1972Va06, DSAM), 0.49 ps +9-6 (1972Br45, DSAM), 1.15 ps +45-63 (1969Du08, DSAM) from (p,p'γ), 0.57 ps 7 (1969Ha17, DSAM), and 0.63 ps 5 (1977Sc36, DSAM) from Coulomb excitation.
2645.67 8	7/2 ⁺	0.159 ps 21	B FGHI LM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 18 from ($^{24}\text{Mg}, \alpha\text{py}$), 2645.9 3 from ($^{16}\text{O}, \alpha\text{py}$), 2645.45 12 from ($^{14}\text{N}, \alpha\text{pny}$), 2645.70 8 from (p,$\gamma$), and 2645.74 17 from (p,p'γ). Others: 2645.1 10 from ^{35}Ar ε decay, 2645.8 6 from ($^{28}\text{Si}, \alpha\text{py}$), and 2645.8 8 from ($\alpha$,py). J^π: spin=7/2 from pγ(θ) in (α,py) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to 5/2⁺, 1763 level in (α,py). T_{1/2}: lifetime τ=0.229 ps 30: weighted average of 0.30 ps 9 from (α,py) (1969In04, DSAM), 0.255 ps 65 (1972Hu11, DSAM), 0.200 ps 30 (1976Me12, DSAM) in (p,γ), 0.27 ps 9 in (n,n'γ) (1989Ge09, DSAM), 0.185 ps 50 (1972Va06, DSAM), 0.26 ps 6 (1972Br45, DSAM), 0.35 ps +10-7 (1969Du08, DSAM) in (p,p'γ), and 0.26 ps 9 in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in ($^{28}\text{Si}, \alpha\text{py}$) (2007Ks01, DSAM).
2694.00 20	3/2 ⁺	32 fs 6	B LM O QRSTU Wxy	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 2 from ^{35}Ar ε decay, 2693.6 13 from (α,py), 2693.9 3 from (p,γ), 2694.50 28 from (p,p'γ). J^π: spin=3/2 from γ(θ) in (p,p'γ); L=2 from 0⁺ in ($^3\text{He}, \text{d}$) and ($\text{d}, ^3\text{He}$). T_{1/2}: lifetime τ=46 fs 9: unweighted average of 70 fs 30 in (α,py) from 1972Br33 with DSAM, 21 fs 3 in (α,py) from 1972Hu11 with DSAM, 19 fs 4 in (α,py) from 1973Fa07 with DSAM, 20 fs 4 in (p,γ) from 1976Me12 with DSAM, 65 fs 20 in (p,γ) from 1972Va06 with DSAM, 62 fs 16 in (p,p'γ) from 1969Du08 with DSAM, and 62 fs 8 in (n,n'γ) from 1989Ge09 with DSAM. Others: 115 fs +95-59 in

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π#</u>	<u>T_{1/2}[‡]</u>	<u>XREF</u>	<u>Comments</u>
3002.64 11	5/2 ⁺	22 fs 6	B F HI LM O QRSTU WXY	(p,p'γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs 7 in (p,p'γ) from 1971Ca40 with DSAM is considered an outlier. E(level): weighted average of 3002.3 3 from ^{35}Ar ε decay, 3003.1 8 from ($^{28}\text{Si},\alpha\text{py}$), 3002.0 10 from ($^{16}\text{O},\alpha\text{py}$), 3001.98 22 from ($^{14}\text{N},\alpha\text{pn}\gamma$), 3002.6 15 from ($\alpha,\text{p}\gamma$), 3002.75 8 from (p,γ), 3002.6 5 from (p,p'γ). J^π: L=2 from 0⁺ in ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d,^3He) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0⁺ in (d,n) and the angular distribution fit seems good. T_{1/2}: lifetime τ=32 fs 8: unweighted average of 48 fs 9 in (α,pγ) from 1972Br33 with DSAM, 22 fs 3 in (α,pγ) from 1972Hu11 with DSAM, 14 fs 2 in (α,pγ) from 1973Fa07 with DSAM, 16 fs 4 in (p,γ) from 1976Me12 with DSAM, 72 fs 12 in (n,n'γ) from 1989Ge09 with DSAM, 18 fs +26-15 in (p,p'γ) from 1972Va06 with DSAM, and 31 fs 13 in (p,p'γ) from 1969Du08 with DSAM.
3162.87 9	7/2 ⁻	29.2 ps 12	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 6 from ($^{28}\text{Si},\alpha\text{py}$), 3163.43 19 from ($^{24}\text{Mg},\alpha\text{py}$), 3162.9 3 from ($^{16}\text{O},\alpha\text{py}$), 3162.64 9 from ($^{14}\text{N},\alpha\text{pn}\gamma$), 3163.1 11 from (α,pγ), 3162.99 7 from (p,γ), 3162.75 10 from (p,p'γ). J^π: spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p'γ); L=3 from 0⁺ in (d,n), ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d,^3He). T_{1/2}: lifetime τ=42.1 ps 17: weighted average of 42 ps 2 (1974Va13, RDM), 41.7 ps 17 (1976Ke02, RDM), 45 ps 6 (1991Ja11, RDM) from $^{24}\text{Mg}(^{16}\text{O},\alpha\text{py})$, 41.8 ps 18 from $^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$ (1976Wa11, RDM), 60 ps 7 from (α,pγ) (1969In04, DSAM), 37 ps 4 from (p,γ), (1971Ba23, delayed coincidence), and 46 ps 6 from (p,p'γ) (1974Hu09, RDM). Others: τ=140 ps 40 (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).
3918.47 12	3/2 ⁺	4.9 fs 14	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 5 from ^{35}Ar ε decay, 3915 2 from (α,pγ), 3918.48 10 from (p,γ). J^π: spin=3/2 determined by the feeding 5162γ(θ) from 5/2⁺, 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2⁺ ground state. L=2 from 0⁺ in (pol d,^3He). T_{1/2}: lifetime τ=7 fs 2 from (p,γ), (1972Hu11, DSAM). Others: <22 fs from (α,pγ) (1972Br33, DSAM).
3942.9 2	9/2 ⁺	0.18 ps 3	F HI LM O	E(level): weighted average of 3942.0 7 from ($^{28}\text{Si},\alpha\text{py}$), 3943.4 10 from ($^{16}\text{O},\alpha\text{py}$), 3942.3 10 from ($^{14}\text{N},\alpha\text{pn}\gamma$), 3943.4 12 from (α,pγ), and 3942.9 2 from (p,γ). J^π: spin=9/2 from pγ(θ) in (α,pγ); 2180γ E2 to 5/2⁺. T_{1/2}: lifetime τ=0.26 ps 5: weighted average of 0.33 ps 5 in (α,pγ) (1972Br33, DSAM), 0.39 ps 11 (1972Hu11, DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in ($^{28}\text{Si},\alpha\text{py}$) (2007Ks01, DSAM).

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
3967.6 4	1/2 ⁺	11.8 fs 28	B M O R	E(level): weighted average of 3967.7 4 from ^{35}Ar ε decay, 3968 2 from (α , γ), and 3967.3 6 from (p, γ). J ^π : L=0 from 0 ⁺ in (^3He ,d). T _{1/2} : lifetime τ =17 fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM) in (p, γ). Other: τ <54 fs from (α , γ) (1972Br33, DSAM).
4059.1 4	(3/2) ⁻	13.9 fs 21	LM O QR	E(level): weighted average of 4056.9 27 from (α ,p), 4058 3 from (α , γ), 4059.2 4 from (p, γ). J ^π : L=1 from 0 ⁺ in (^3He ,d); (3/2 ⁻) from FRESKO-DWBA analysis of measured $\sigma(\theta)$ of (α ,p) (2019Se09). T _{1/2} : lifetime τ =20 fs 3: weighted average of 31 fs 13 from (α , γ) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM) from (p, γ).
4112.4 16	7/2 ⁺	49 fs 11	LM O Q	E(level): weighted average of 4111.1 29 from (α ,p), 4108 2 from (α , γ), and 4113.7 10 from (p, γ). J ^π : 4113.4 γ E2(+M3) to 3/2 ⁺ . T _{1/2} : lifetime τ =70 fs 16 from (α , γ) (1972Br33, DSAM).
4173.43 23	(5/2 ⁺)	35 fs 6	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 2 from (α , γ) and 4173.45 19 from (p, γ). J ^π : based on primary transitions from proton resonances in (p, γ): 3827.3 γ from 8001.0, (7/2 ⁺), 4456.4 γ from 8630, 7/2 ⁻ level, 4243.0 γ from 8416.7, 1/2 ⁺ level. Note that primary 3099.1 γ from 7272.7, 1/2 ⁻ resonance disfavors 5/2 ⁺ assignment, but this γ branch is relatively weak. T _{1/2} : lifetime τ =51 fs 8: weighted average of 68 fs 24 in (α , γ) (1972Br33, DSAM), 105 fs +35-30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM) in (p, γ). XREF: q(4170)R(4167) J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); L=1 from 0 ⁺ in (^3He ,d).
4177.9 2	3/2 ⁻	27.0 fs 28	L O qR	T _{1/2} : lifetime τ =39 fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM) from (p, γ). E(level): weighted average of 4347.8 6 from (^{28}Si , α py), 4348.29 24 from (^{24}Mg , α py), 4347.8 3 from (^{16}O , α py), 4347.43 17 from (^{14}N , α pny), 4347.8 12 from (α , γ), and 4347.76 6 from (p, γ). J ^π : spin from $\gamma(\theta)$ in (p, γ); parity from 1184.6 γ , M1+E2 to 7/2 ⁻ , 3163 level in (p, γ) (1974Lo17). T _{1/2} : lifetime τ =1.5 ps 8: unweighted average of 1.31 ps +30-23 from (^{28}Si , α py) (2007Ks01, DSAM), 2.9 ps +14-7 (1972Br33, DSAM), and 0.33 ps 6 (1976Me12, DSAM) from (p, γ). XREF: B(?) E(level): other: 4625.3 28 from (α ,p); 4618 2 from (α , γ) is discrepant. J ^π : from p $\gamma(\theta)$ in (α , γ).
4347.76 7	9/2 ⁻	1.0 ps 6	FGHI LM O	
4624.3 3	3/2,5/2	40 fs 17	B LM O	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π #	T _{1/2} [‡]	XREF			Comments
4769.86 23	(5/2 ⁻)	76 fs 20	LM	O		T _{1/2} : lifetime τ=58 fs 24 from (α,pγ) in 1972Br33 with DSAM. E(level): others: 4770.4 26 from (α,p), 4770 15 from (α,pγ). J ^π : from primary transitions in (p,γ): 4059.2γ from 8829.3, 1/2 ⁻ level, 2778.9γ from 7548.9, 7/2 ⁻ level, 3231.0γ from 8001.0, (7/2 ⁺) level. Other: 7/2,9/2 from pγ(θ) for a group at E=4770 15 in (α,pγ) is inconsistent and disfavored by γ rays to 1/2 ⁺ and 3/2 ⁺ . T _{1/2} : lifetime τ=110 fs 29: weighted average of 270 fs 95 (1972Hu11 , DSAM) and 105 fs 17 (1976Me12 , DSAM) in (p,γ). XREF: x(4839) J ^π : from primary γ transitions in (p,γ): 2339.4γ from 7178.6, 1/2 ⁺ level, 2355.4γ from 7194.6, 1/2 ⁻ level, 2867.2γ from 7706.7, 5/2 ⁺ level. T _{1/2} : lifetime τ=14 fs 5 from (p,γ) (1976Me12 , DSAM). XREF: x(4839) E(level): other: 4859.2 28 from (α,p). J ^π : from primary γ transitions in (p,γ): 2440.1γ from 7194.6, 1/2 ⁻ level, 3183.9γ from 8038.5, 1/2 ⁺ level. T _{1/2} : lifetime τ=7 fs 2 from (p,γ) (1976Me12 , DSAM). E(level): others: 4881.8 29 from (α,p), 4880.8 21 from (α,pγ). J ^π : spin=7/2 from pγ(θ) in (α,pγ); 3417.1 primary γ from 8298.2, 3/2 ⁺ resonance in (p,γ). T _{1/2} : lifetime τ=8 fs 2: weighted average of 7 fs 2 (1972Hu11 , DSAM), 7 fs 3 (1973Fa07 , DSAM), and 10 fs 3 (1976Me12 , DSAM) from (p,γ). Other: τ=0.28 ps 6 from (α,pγ) in 1972Br33 with DSAM. E(level): weighted average of 5006.9 27 from (α,p) and 5010.1 18 from (p,γ). Others: 5015 20 from (α,pγ) and 5010 15 from (³ He,d). J ^π : primary 2184.4γ from 7194.6, 1/2 ⁻ level and 3406.5γ from 8416.7, 1/2 ⁺ level. T _{1/2} : lifetime τ=11 fs 3 from (p,γ) (1976Me12 , DSAM). XREF: m(5175) E(level): weighted average of 5158.9 29 from (α,p), 5163 15 from (³ He,d), 5150 20 from ³⁶ Ar(d, ³ He), and 5155 11 from ³⁶ Ar(pol d, ³ He). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and (d, ³ He). XREF: m(5175) J ^π : spin=7/2 from the feeding 3466γ(θ) from 8630, 7/2 ⁻ level and RUL in (p,γ); parity=- from 3466γ, M1+E2 from 8630, 7/2 ⁻ level. XREF: M(5230) E(level): weighted average of 5209.5 28 from (α,p), 5230 20 from (α,pγ), and 5215.8 15 from (p,γ). J ^π : other: 3/2,5/2 from pγ(θ) in (α,pγ). T _{1/2} : lifetime τ<7 fs from (p,γ) (1976Me12 , DSAM). E(level): weighted average of 5402.0 29 from (α,p)
4839.10 11	(1/2 ⁺ ,3/2)	9.7 fs 35	O		x	
4854.4 19	(1/2,3/2)	4.9 fs 14	L	O	x	
4880.96 15	7/2 ⁽⁺⁾	5.6 fs 14	LM	O		
5009.1 18	(1/2,3/2)	7.6 fs 21	LM	O	R	
5158.6 29	3/2 ⁺ ,5/2 ⁺		Lm	R	WX	
5163.34 22	7/2 ⁻		m	O		
5214.5 20	(5/2 ⁺) [@]	<4.85 fs	LM	O		
5403.3 10	3/2 ⁻	11.8 fs 28	L	O	R	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	J^π	$T_{1/2}^\ddagger$	XREF			Comments
						and 5403.5 10 from (p, γ). Other: 5409 12 from ($^3\text{He},d$).
5407.07 ^a 21	11/2 ⁻	0.28 ps 7	FGHI	LM		J^π: L=1 from 0⁺ in ($^3\text{He},d$); 2400.7γ to 5/2⁺ rules out 1/2⁻ based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL. $T_{1/2}$: lifetime $\tau=17$ fs 4 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 18 from (p,γ). E(level): weighted average of 5407.4 8 from ($^{28}\text{Si},\alpha p\gamma$), 5407.64 27 from ($^{24}\text{Mg},\alpha p\gamma$), 5407.3 3 from ($^{16}\text{O},\alpha p\gamma$), 5406.68 19 from ($^{14}\text{N},\alpha p n\gamma$), and 5407.1 15 from ($\alpha,p\gamma$). J^π: spin=11/2 from $p\gamma(\theta)$ in (p,γ); parity from 1059γ, M1+E2 to 9/2⁻, 4347 level. $T_{1/2}$: lifetime $\tau=0.4$ ps 1 from ($\alpha,p\gamma$) (1974Lo17, DSAM). Other: $\tau<1.1$ ps from ($^{28}\text{Si},\alpha p\gamma$) (2007Ks01, DSAM).
5531 5				LM	0	E(level): weighted average of 5531 5 from (α,p) and 5535 20 from ($\alpha,p\gamma$).
5586.0 20	5/2 ⁺		Lm	0	wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 26 from (α,p) and 5586 2 from (p,γ). J^π: L=2 from 0⁺ in (pol d,^3He) and L+1/2 transfer from analyzing power.
5598.4 22	3/2 ⁺ , 5/2 ⁺	2.1 fs 7	Lm	0	wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 26 from (α,p) and 5599.7 15 from (p,γ). J^π: L(d,^3He)=2 from 0⁺. $T_{1/2}$: lifetime $\tau=3$ fs 1 from (p,γ) (1976Me12, DSAM).
5633.6 30			Lm	r	w y	XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p).
5646 2	(5/2, 7/2, 9/2 ⁺)	2.8 fs 7	Lm	0	r y	XREF: m(5650)r(5660)y(5650) E(level): weighted average of 5645 4 from (α,p) and 5646 2 from (p,γ). J^π: from primary transitions in (p,γ): 1903γ from 7548.9, 7/2⁻ level, 2355γ from 8001.0, (7/2⁺) level, 3261γ from 8906.8, 5/2⁺ level. $T_{1/2}$: lifetime $\tau=4$ fs 1 from (p,γ) in 1976Me12 with DSAM.
5655 2	(3/2) ⁺	13.9 fs 35	Lm	0	r w y	T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 4 from (α,p) and 5655 2 from (p,γ). J^π: L=2 from 0⁺ in ($^3\text{He},d$); primary 2142γ from 7797.2, 1/2⁻ level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2⁻ level in (p,γ) favors 5/2⁺. $T_{1/2}$: lifetime $\tau=20$ fs 5 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 18% of Iγ(2652)=100 6 from (p,γ).
5682.2 20	1/2 ⁻ , 3/2 ⁻		Lm	0	R	XREF: m(5650) E(level): weighted average of 5679.7 34 from

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
				(α ,p), 5683 2 from (p, γ), and 5684 12 from (^3He ,d).
				J ^π : L=1 from 0 ⁺ in (^3He ,d).
				The contribution of I γ to unknown levels adds up to 18% of I γ (1504)=100 from (p, γ).
5723.52 18	5/2 ⁺		L O WX	E(level): others: 5731.2 25 from (α ,p), 5720 20 from ^{36}Ar (d, ^3He), and 5720 11 from ^{36}Ar (pol d, ^3He).
				J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and (d, ^3He) and L+1/2 transfer from analyzing power.
				The contribution of I γ to unknown levels adds up to 100% of I γ (3960)=100 11 from (p, γ).
5758 3	(1/2,3/2 ⁻)		L O R	E(level): weighted average of 5757 4 from (α ,p), 5758 3 from (p, γ), and 5760 12 from (^3He ,d).
				J ^π : L(^3He ,d)=(0,1) from 0 ⁺ .
				The contribution of I γ to unknown levels adds up to 56% of I γ (5758)=100 22 from (p, γ).
5806.6 20	(1/2,3/2,5/2)	3.5 fs 7	L O	E(level): weighted average of 5808.1 31 from (α ,p) and 5806 2 from (p, γ).
				J ^π : primary 2891 γ from 8696.9, 3/2 ⁻ level; 5806 γ to 3/2 ⁺ g.s.
				T _{1/2} : lifetime τ =5 fs 1 from (p, γ) (1976Me12, DSAM).
5830 4	(5/2 ⁻) [@]		LM	XREF: M(5850)
				E(level): weighted average of 5829 4 from (α ,p) and 5850 25 from (α ,p γ).
				J ^π : other: spin=5/2,9/2 from p γ (θ) in (α ,p γ).
5926.65 29	(11/2) ⁻	<0.28 ps	F HI LM	E(level): weighted average of 5927.1 7 from (^{28}Si , α p γ), 5926.9 5 from (^{16}O , α p γ), 5926.47 29 from (^{14}N , α p γ), 5928 3 from (α ,p), and 5927.1 19 from (α ,p γ).
				J ^π : 1579 γ M1+E2, ΔJ =1 to 4348, 9/2 ⁻ in ^{12}C (^{28}Si , α p γ), also assuming spins increase with excitation energies in fusion-evaporation reactions.
				Spin=7/2,11/2 from p γ (θ) in (α ,p γ).
				T _{1/2} : lifetime τ <0.4 ps from (^{28}Si , α p γ) (2007Ks01, DSAM).
6087.31 24	13/2 ⁻	6.2 ps 6	FGHI LM	E(level): weighted average of 6087.6 8 from (^{28}Si , α p γ), 6088.05 29 from (^{24}Mg , α p γ), 6087.2 4 from (^{16}O , α p γ), 6086.89 23 from (^{14}N , α p γ), 6084.2 29 from (α ,p), and 6087.0 18 from (α ,p γ).
				J ^π : spin=13/2 from p γ (θ) in (α ,p γ); parity from 680 γ , M1+E2 to 11/2 ⁻ , 5406 level in (α ,p γ).
				T _{1/2} : lifetime τ =8.9 ps 8: weighted average of 9.3 ps 8 from (α ,p γ) in 1974Lo17 with RDM and 7.7 ps 14 from (^{14}N , α p γ) in 1976Wal1.
6106.2 21	(3/2,5/2 ⁺)	8.3 fs 21	1 O WX	XREF: l(6104)w(6140)x(6136)
				E(level): weighted average of 6104 4 from (α ,p) and 6106.2 21 from (p, γ).
				J ^π : 4886.4 γ to 1/2 ⁺ , 4342.9 γ to 5/2 ⁺ ; primary 6106.2 γ from 7781.7, (5/2 ⁻) level.
				T _{1/2} : lifetime τ =12 fs 3 from (p, γ) in 1976Me12 with DSAM.
6141 4	5/2 ⁺		L WX	XREF: w(6140)x(6136)

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π#</u>	<u>XREF</u>			<u>Comments</u>
6181 3	(1/2 ⁺ to 7/2 ⁺)	L	n0		J ^π : L=2 from 0 ⁺ in (d, ³ He) and (pol d, ³ He), and L+1/2 transfer from analyzing power. XREF: n(6200) E(level): other: 6180 3 from (α,p). J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from 7618.7, 5/2 ⁺ level in (p,γ). Apart from the γ branch I _γ (6180)=100 observed in (p,γ), the contribution of I _γ to unknown levels adds up to 122% of I _γ (6180).
6224.4 30		L	n		XREF: n(6200) J ^π : L(α,d)=6 for a group at 6200 10.
6283 3	(5/2 ⁺)@	L		R	E(level): weighted average of 6283 3 from (α,p) and 6284 4 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d). J ^π : L=(0,1) from 0 ⁺ in (³ He,d).
6329 4	(1/2,3/2 ⁻)			R	XREF: R(6377)
6380.2 9	(9/2 ⁻)@	H	L	R	E(level): weighted average of 6380.8 8 from (¹⁶ O,αpγ) and 6379.3 25 from (α,p). J ^π : other: L=(2,3) from 0 ⁺ in (³ He,d) for a group at E=6377 2 based on very few data points in a very narrow range of angles in the DWBA fit.
6401.2 22	(1/2 ⁻)@		LM		
6428 2	(1/2 ⁺)@	L		R	E(level): weighted average of 6429 3 from (α,p) and 6427 2 from (³ He,d). J ^π : other: L=(3) from 0 ⁺ in (³ He,d) is inconsistent.
6469 3	(1/2 ⁻ ,3/2 ⁻)	L		R	E(level): weighted average of 6475 4 from (α,p) and 6468 2 from (³ He,d). J ^π : L=(1) from 0 ⁺ in (³ He,d).
6492 2	(3/2 ⁺)@	L	0	R	E(level): weighted average of 6492 3 from (α,p), 6492 3 from (p,γ), and 6491 2 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d).
6546 2	(1/2 ⁺)@	L		R	E(level): weighted average of 6549 3 from (α,p) and 6545 2 from (³ He,d). J ^π : other: L=(0,1) from 0 ⁺ in (³ He,d), indicating (1/2,3/2 ⁻).
6643 2	(1/2 ⁻ ,3/2 ⁻)	D		R	E(level): from (³ He,d). J ^π : from L=(1) from 0 ⁺ in (³ He,d).
6659.1 30	(7/2 ⁺)@		L		y
6660.2 8	(11/2 ⁻)	F			XREF: y(6677) J ^π : 2312γ to 9/2 ⁻ , 3497γ to 7/2 ⁻ in (²⁸ Si,αpγ), assuming spin increasing as excitation energy increases.
6675.6 25		D	L	R	y
					XREF: y(6677) E(level): weighted average of 6679.4 31 from (α,p) and 6674 2 from (³ He,d).
6761 2	3/2 ⁺ ,5/2 ⁺	D		R	WX
					XREF: X(6745) E(level): weighted average of 6761 2 from (³ He,d), 6750 20 from ³⁶ Ar(d, ³ He), and 6745 12 from ³⁶ Ar(pol d, ³ He). J ^π : L(d, ³ He)=2 from 0 ⁺ .
6779.0 20		D	L	R	E(level): weighted average of 6781.3 30 from (α,p) and 6778 2 from (³ He,d).
6800 4		D	L		XREF: D(6802) E(level): from (α,p).
6823 2		D		R	E(level): from (³ He,d).
6842 2		D	L	R	E(level): weighted average of 6842 4 from (α,p) and 6842 2 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[‡]</u>	<u>XREF</u>			<u>Comments</u>
						($^3\text{He},d$).
6863 3	(9/2 ⁺) [@]			L		
6866.7 6	(1/2,3/2 ⁻)		D	O R		E(level): other: 6866 2 from ($^3\text{He},d$). J ^π : L($^3\text{He},d$)=(0,1) from 0 ⁺ .
6891.6 30	(9/2 ⁺) [@]			L		
6948.1 34	5/2 ⁺			L	WX	XREF: W(6930) E(level): weighted average of 6948.6 34 from (α,p) and 6930 20 from $^{36}\text{Ar}(d,^3\text{He})$. Other: 6953 43 from $^{36}\text{Ar}(\text{pol } d,^3\text{He})$. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
6986.3 34	(9/2 ⁺) [@]			L		
7066.2 7	(5/2 ⁺) [@]			L O R		E(level): weighted average of 7066.2 7 from (p,γ), and 7066 2 from ($^3\text{He},d$).
7103.4 7	3/2 ⁽⁻⁾			L O R		XREF: L(7105?) E(level): weighted average of 7103.4 7 from (p,γ), and 7103 2 from ($^3\text{He},d$). J ^π : spin=3/2 from $\gamma(\theta)$ in (p,γ); parity from L($^3\text{He},d$)=(1,3) from 0 ⁺ .
7121.7 24				L		
7178.6 8	1/2 ⁽⁺⁾			O		T=3/2 J ^π : spin=1/2 from $\gamma(\theta)$ in (p,γ); parity from IAS in ^{35}S .
7179 4	(7/2) ⁺ [@]			L N		E(level): weighted average of 7180 4 from $^{32}\text{S}(\alpha,p)$ and 7170 10 from $^{33}\text{S}(\alpha,d)$. J ^π : also L=6 from 3/2 ⁺ in $^{33}\text{S}(\alpha,d)$, indicating (7/2;17/2) ⁺ .
7185	5/2 ⁺			O R	X	XREF: R(7178)X(7181) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=7181 60. Other: L($^3\text{He},d$)=(2) from 0 ⁺ for a group at E=7178 2 considered the same level.
7194.6 7	1/2 ⁻	0.027 keV 8		OP R		J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(p,p),(p,p')\gamma$:resonances. Other: 1/2,3/2 from $\gamma(\theta)$ in (p,γ).
7213.7 25				L	y	XREF: y(7250)
7225.5 8	5/2			O	y	XREF: y(7250) J ^π : from $\gamma(\theta)$ in (p,γ).
7228.2 20	(3/2 ⁺) [@]			L R	y	XREF: R(7227)y(7250) E(level): weighted average of 7230.9 30 from $^{32}\text{S}(\alpha,p)$, 7227 2 from $^{34}\text{S}(^3\text{He},d)$. J ^π : other: L=(0,1) from 0 ⁺ in ($^3\text{He},d$) is inconsistent.
7234.0 8	5/2 ⁽⁺⁾			O		J ^π : spin=5/2 from $\gamma(\theta)$ in (p,γ); primary 6014.1γ to 1/2 ⁺ .
7269.2 1				O		
7272.7 9	1/2 ⁻	0.014 keV 4		OP R	y	XREF: P(7274)R(7273)y(7250) J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(p,p),(p,p')\gamma$:resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p,γ).
7348 3	(7/2) [@]			L		
7361.9 8	3/2 ⁽⁺⁾			L O R		E(level): weighted average of 7362 3 from (α,p),

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
7396.4 10	7/2 ⁽⁻⁾		0 R	7362.0 8 from (p,γ), and 7361 2 from (³ He,d). J ^π : spin=3/2 from γ(θ) in (p,γ); 6142.0γ (M1+E2) to 1/2 ⁺ . E(level): weighted average of 7396.0 10 from (p,γ), and 7398 2 from (³ He,d). J ^π : spin=7/2 from γ(θ) in (p,γ); 4232.7γ (M1+E2) to 7/2 ⁻ .
7413.8 30			L	
7450.9 8	3/2		L 0	E(level): weighted average of 7447 3 from (α,p) and 7451.1 6 from (p,γ). J ^π : from γ(θ) in (p,γ).
7496			0	
7501.1 8	(5/2 ⁺ , 7/2, 9/2 ⁺)		1 0	XREF: l(7503) J ^π : primary 3558.0γ to 9/2 ⁺ , 4498.0γ to 5/2 ⁺ . XREF: l(7503)
7502.9 7			1 0	J ^π : from γ(θ) in (p,γ).
7518.8 7	7/2		0	
7548.9 6	7/2 ⁻	8.4 eV 26	OP	T=3/2 XREF: P(7550) J ^π : from resonance analysis of measured σ(θ) in ³⁴ S(p,p),(p,p'γ):resonances. Spin=7/2 also from γ(θ) in (p,γ). T _{1/2} : Γ=5.8 eV 13 from (p,γ), 11.0 eV 15 from ³⁴ S(p,p),(p,p'γ):resonances lifetime τ<1 fs from (p,γ) (1973Fa07, DSAM).
7561.3 6	1/2, 3/2		L 0	E(level): other: 7566.5 35 from (α,p). J ^π : from γ(θ) in (p,γ).
7565 20	5/2 ⁺			X J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and L+1/2 transfer from analyzing power.
7590 4			L	
7600.8 8	5/2 ⁺	<13.9 fs	0	J ^π : spin=5/2 from γ(θ) in (p,γ); 7599.9γ M1+E2 to 3/2 ⁺ . T _{1/2} : lifetime τ<20 fs from (p,γ) (1971Wi13, DSAM), which is equivalent to Γ>0.033 eV.
7618.7 5	5/2 ⁺		0	J ^π : spin=5/2 from γ(θ) in (p,γ); 7617.8γ M1+E2 to 3/2 ⁺ .
7648.1 30			L	
7656.6 8	(1/2, 3/2, 5/2 ⁺)		0	J ^π : primary 3689.1γ to 1/2 ⁺ .
7670 10	(7/2 to 17/2) ⁺		N	J ^π : L(α,d)=6 from 3/2 ⁺ .
7671.9 8	5/2 ⁺		1 Op	XREF: p(7686) J ^π : spin=5/2 from γ(θ) in (p,γ); 4668.8γ, M1+E2 to 5/2 ⁺ ; 5908.2γ, M1+E2 to 5/2 ⁺ ; other: 3324.0γ to 9/2 ⁻ seems to favor 5/2 ⁻ .
7685.5 8	3/2 ⁻		OP	J ^π : spin=3/2 from γ(θ) in (p,γ) and 3/2 ⁻ from R-Matrix analysis in (p,p):resonances. Γ _p =446 eV 15 from (p,p):resonances.
7693.9 8	(1/2 ⁺ , 3/2, 5/2)		0	J ^π : primary 7693.0γ to 3/2 ⁺ , 3634.5γ to (3/2) ⁻ , 4690.8γ to 5/2 ⁺ .
7706.4 8	5/2 ⁺		L OP	E(level): other: 7709.9 25 from (α,p). J ^π : spin=5/2 from γ(θ) in (p,γ) and 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances. Γ _p =4 eV 1 in (p,p).
7744.8 9	7/2 ⁻		L 0	E(level): other: 7744.1 30 from (α,p). J ^π : spin=7/2 from γ(θ) in (p,γ); 4581.5γ M1+E2 to 7/2 ⁻ .

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
7748	(7/2 to 17/2) ⁺		NO	XREF: N(7750) E(level): from (p,γ). J ^π : L(α,d)=6 from 3/2 ⁺ .
7777.1 9	(5/2 ⁺)		L 0	E(level): from (p,γ). Other: 7770.3 33 from (α,p). J ^π : based on primary transitions: 4613.8γ to 7/2 ⁻ , 5131.0γ to 7/2 ⁺ , 6557.1γ to 1/2 ⁺ .
7781.7 11	(5/2 ⁻)		0	J ^π : primary 3433.8γ to 9/2 ⁻ ; relatively weak 6561.7γ to 1/2 ⁺ .
7797.2 9	1/2 ⁻	0.031 keV 10	L OP	E(level): weighted average of 7799.8 30 from (α,p), 7797.1 9 from (p,γ), and 7796 3 from ³⁴ S(p,p),(p,p'γ): resonances. J ^π : spin=1/2 from γ(θ) in (p,γ); (1/2) ⁻ from R-Matrix analysis of in (p,p):resonances.
7837.2 10	3/2 ⁻	3.7 keV 4	OP	T=3/2 E(level): weighted average of 7837.2 10 from (p,γ), and 7837 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and 3/2 ⁻ from R-Matrix analysis in (p,p):resonances.
7839.0 9	(5/2 ⁺)		0	T _{1/2} : other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM). J ^π : primary 6618.9γ to 1/2 ⁺ , 3895.9γ to 9/2 ⁺ .
7868.7 10	(1/2 ⁺ ,3/2,5/2 ⁺)		L 0	E(level): other: 7868.5 35 from (α,p). J ^π : primary 6648.6γ to 1/2 ⁺ , 2652.8γ to (5/2 ⁺).
7873.03 ^b 30	13/2 ⁺		F HI NO	XREF: N(7870) E(level): weighted average of 7873.3 8 from (²⁸ Si,αpγ), 7873.4 4 from (¹⁶ O,αpγ), and 7872.78 30 from (¹⁴ N,αpnγ). J ^π : 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2) ⁻ ; 1786.2γ D(+Q) to 13/2 ⁻ ; band head.
7880.8 9	3/2 ⁺ ,5/2 ⁺	0.008 keV 4	OP	J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885			0	
7889.1 15			0	
7899.2 7	(5/2)		1 0	XREF: l(7901) J ^π : primary 2735.8γ to 7/2 ⁻ , 3018.1γ to 7/2 ⁽⁺⁾ , 3721.1γ to 3/2 ⁻ , 5204.9γ to 3/2 ⁺ .
7903			1 0	XREF: l(7901)
7923.3 9	(3/2 ⁺ ,5/2 ⁺)		0	J ^π : primary 5277.2γ to 7/2 ⁺ , 6703.2γ to 1/2 ⁺ .
7930			0	
7949			0	
7970.3 10	(5/2 ⁻)		0	J ^π : primary 3622.3γ to 9/2 ⁻ , weak 6750.2γ to 1/2 ⁺ .
7978.6 35			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF		Comments
7987.8 10	3/2 ⁺		0		J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); 7986.8 γ M1+E2 to 3/2 ⁺ .
7995.6 10	(5/2)		0		J ^π : primary 2592.0 γ to 3/2 ⁻ , 7994.6 γ to 3/2 ⁺ , 2832.1 γ to 7/2 ⁻ , 5349.5 γ to 7/2 ⁺ .
8001.0 9	(7/2 ⁺)		L	0	E(level): other: 8000.7 41 from (α ,p). J ^π : from primary transitions in (p, γ): 3653.0 γ to 9/2 ⁻ , 4057.9 to 9/2 ⁺ , 3827.3 γ to 5/2 ⁻ , 6237.4 γ to 5/2 ⁺ , 8000.0 γ to 3/2 ⁺ .
8005.0 10	5/2 ⁺	0.011 keV 5	OP	WX	XREF: P(8004)W(8010)X(8007) J ^π : L(d, ³ He)=2 from 0 ⁺ ; spin=5/2 from $\gamma(\theta)$ in (p, γ). J ^π : L(α ,d)=6 from 3/2 ⁺ .
8010 10	(7/2 to 17/2) ⁺		N		
8019.2 35			L		
8035.7 10	1/2 ⁻ , 3/2 ⁻	0.026 keV 7	OP		E(level): other: 8034 3 from ³⁴ S(p,p),(p,p' γ):resonances.
8038.4 11	1/2 ⁺ , 3/2 ⁺	0.30 keV 2	OP		E(level): weighted average of 8038.5 11 from (p, γ), and 8038 3 from ³⁴ S(p,p),(p,p' γ):resonances. J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ), but 1/2 ⁺ from R-Matrix analysis in (p,p); 8037.5 γ M1+E2 to 3/2 ⁺ .
8075.9 10	(5/2)		L	0	E(level): weighted average of 8075.1 46 from (α ,p) and 8075.9 10 from (p, γ). J ^π : primary 3897.8 γ to 3/2 ⁻ , 8074.9 γ to 3/2 ⁺ ; 2912.4 γ to 7/2 ⁻ , 3962.0 γ to 7/2 ⁺ .
8080			0		
8096.1 11	(5/2)		0		J ^π : primary 2932.6 γ to 7/2 ⁻ , 5449.9 γ to 7/2 ⁺ ; 3918.0 γ to 3/2 ⁻ , 4177.4 γ to 3/2 ⁺ .
8100 10	(7/2 to 17/2) ⁺		N		J ^π : L(α ,d)=6 from 3/2 ⁺ .
8106.4 11	3/2 ⁺		L	0	E(level): other: 8103.8 42 from (α ,p). J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); 8105.4 γ M1+E2 to 3/2 ⁺ .
8113.3 11	(1/2 ⁺ , 3/2, 5/2 ⁺)		L	0	XREF: L(8121?) J ^π : primary 2897.4 γ (5/2 ⁺), 4145.7 γ to 1/2 ⁺ .
8147.2 11	3/2 ⁻	0.56 keV 6	OP		XREF: P(8150) J ^π : from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p, γ) based on strengths of primary γ rays. T _{1/2} : from ³⁴ S(p,p),(p,p' γ):resonances in 1977Ou01.
8148 3	1/2 ⁻	2.66 keV 27	P		
8150 3	3/2 ⁻	0.56 keV 6	P		
8156.8 11	(5/2 ⁺)		0		J ^π : primary 3978.7 γ to 3/2 ⁻ , 8155.8 γ to 3/2 ⁺ , 4213.6 γ to 9/2 ⁺ .
8171.7 42			L		
8179.4 12	(3/2 ⁻ , 5/2, 7/2 ⁺)		0		J ^π : primary 5016.0 γ to 7/2 ⁻ , 5485.0 γ to 3/2 ⁺ .
8208.2 10	5/2 ⁺	0.033 keV 10	OP	WX	T=3/2 XREF: W(8210)X(8204) E(level): other: 8209 3 from ³⁴ S(p,p),(p,p' γ):resonances. J ^π : L(pol d, ³ He)=2 and L+1/2 from analyzing power.
8211 3	1/2 ⁺	0.094 keV 15	L	OP	XREF: O(8209)

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF			Comments
8216.3 10	5/2 ⁺	0.014 keV 3	1	OP	wx	E(level): from (p,p):resonances. Other: 8210.2 46 from (α,p). XREF: w(8210)x(8204) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances. XREF: l(8246.0)P(8243) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances. XREF: l(8246.0) J ^π : primary 4137γ to 7/2 ⁺ , 7031γ to 1/2 ⁺ . XREF: L(8270.6) E(level): other: 8271 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2 ⁻ rules out 3/2 ⁺ , since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8242.4 12	3/2 ⁻	0.140 keV 15	1	OP		XREF: P(8279) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2 ⁻ disfavors both 3/2 ⁺ and 5/2 ⁺ , but 3/2 ⁺ is much less likely.
8251 5	(3/2 ⁺ , 5/2 ⁺)		1	0		J ^π : primary 3118.9γ to 7/2 ⁻ , 4104.2γ to 3/2 ⁻ , 8281.4γ to 3/2 ⁺ . XREF: L(8292)P(8290) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances. XREF: P(8300) XREF: l(8292) J ^π : spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2 ⁺ . XREF: L(8321.0) J ^π : primary 4350.5γ to 1/2 ⁺ , 5154.7γ to 7/2 ⁻ , 5671.9γ to 7/2 ⁺ .
8269.0 5	5/2 ⁺	0.005 keV 3	L	OP		E(level): weighted average of 8319.4 9 from (²⁸ Si,αpγ) and 8319.7 5 from (¹⁶ O,αpγ). J ^π : 2911.9γ E2, ΔJ=2 to 11/2 ⁻ , 2232.7γ to 13/2 ⁻ ; band assignment. T _{1/2} : lifetime τ=18 fs 1 from (²⁸ Si,αpγ) (2013Bi10, DSAM). T _{1/2} : from (HI,xny). XREF: L(8321.0) J ^π : primary 4209.1γ to 7/2 ⁺ , 7102.9γ to 1/2 ⁺ .
8277.2 5	(5/2 ⁺)	0.006 keV 3		OP		XREF: P(8383) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8282.4 5	(3/2 ⁻ , 5/2)			0		XREF: L(8398.8)P(8404) E(level): others: 8398.8 38 from (α,p), and 8404 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; its primary 5708.5γ to 3/2 ⁺ , 2694 level rules out 7/2 ⁻ based on RUL and Γ=0.002 keV.
8287.82 30	1/2 ⁻	0.04 keV 1	L	OP		
8297.2 9	3/2 ⁻	0.073 keV 15		OP		
8298.2 5	3/2 ⁺		1	0		
8318.1 3	(5/2 ⁺)		L	0		
8319.6 ^a 5	15/2 ⁻	12.5 fs 7	F H			
8323.1 13	(3/2 ⁺ , 5/2 ⁺)		L	0		
8346 5				0		
8381.8 8	5/2 ⁺	0.023 keV 7		OP		
8389.4 16				0		
8402.9 5	5/2 ⁻	0.002 keV 1	L	OP		

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Adopted Levels, Gammas (continued)

³⁵ Cl Levels (continued)						
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF			Comments
8405.8 14	(5/2 ⁻)	0.001 keV 1	OP			E(level): weighted average of 8405.7 14 from (p,γ), and 8406 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : (5/2 ⁻ , 7/2 ⁻) from R-Matrix analysis in (p,p):resonances; primary 8404.6γ to 3/2 ⁺ .
8409.4 18	1/2 ⁻	0.125 keV 15	OP			E(level): weighted average of 8409.6 18 from ³⁴ S(p,γ) and 8409 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8416.7 5	1/2 ⁺	0.026 keV 7	OP			E(level): weighted average of 8416.7 5 from (p,γ), and 8418 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 1/2 ⁺ from R-Matrix analysis in (p,p):resonances; but primary 4473.5γ to 9/2 ⁺ and 7196.5γ to 1/2 ⁺ in (p,γ) gives (5/2 ⁺), which may indicate two different resonances with close energies.
8430.3 5	(3/2) ⁺ (1/2 ⁺ , 3/2, 5/2 ⁺)	0.090 keV 15	L	O	E(level): other: 8429.2 40 from (α,p).	
8434.8 5			OP		XREF: P(8435)	
8464.3 5			O		J ^π : primary 4496.7γ to 1/2 ⁺ , 3248.3γ to (5/2 ⁺).	
8484.4 5	(3/2) ⁺	0.012 keV 5	1	OP	E(level): weighted average of 8484.4 5 from (p,γ), and 8486 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; 3/2 ⁺ proposed by 2001Vo24 in (p,γ) based on proposed isobaric analog to 2940, 3/2 ⁺ level in ³⁵ S.	
8486.2 5	3/2 ⁻	0.150 keV 15	1	OP	E(level): weighted average of 8486.1 5 from (p,γ), and 8488 3 from ³⁴ S(p,p),(p,p'γ):resonances.	
8487.5 5	15/2 ⁻	<69 fs	F	H	E(level): weighted average of 8487.8 10 from (²⁸ Si,αpγ) and 8487.4 5 from (¹⁶ O,αpγ). J ^π : 3080.0γ E2, ΔJ=2 to 11/2 ⁻ , 2399.8γ D(+Q) to 13/2 ⁻ . T _{1/2} : lifetime τ<0.1 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM).	
8506.5 5	1/2 ⁻	0.150 keV 15	L	O	E(level): other: 8505.1 47 from (α,p).	
8514.3 5			OP		XREF: P(8515)	
8533.9 5			L	O	E(level): weighted average of 8532.4 43 from (α,p) and 8533.9 5 from (p,γ).	
8558.2? 13	(13/2 ⁻)	<0.14 ps	F	XREF: F(?) J ^π : 2631γ D(+Q), ΔJ=1 to 11/2 ⁻ in (²⁸ Si,αpγ). T _{1/2} : lifetime τ<0.2 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM).		
8571.9 5	(5/2) ⁺	0.08 keV 1	L	OP	X	T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 5 from (p,γ), and 8573 3 from ³⁴ S(p,p),(p,p'γ):resonances. Other: 8567.9 37 from (α,p).
8580.5 5	1/2 ⁺	0.75 keV 8	OP			XREF: P(8582)
8585.8 5	5/2 ⁺	0.003 keV 2	O			XREF: P(8592)
8590.1 5			OP			

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF		
			Comments		
					E(level): weighted average of 8590.0 5 from (p,γ), and 8592 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
					J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2 ⁻ ruled out 3/2 ⁺ based on RUL and Γ=0.003 keV.
8612.0 5	3/2 ⁺ , 5/2 ⁺		1 0 W		XREF: W(8610)
8613.7 5	5/2 ⁺	0.175 keV 20	1 OP		J ^π : L(d, ^3He)=2 from 0 ⁺ .
8618.2 5	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	1 OP		XREF: P(8616)
					XREF: P(8620)
					E(level): weighted average of 8618.1 5 from (p,γ), and 8620 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8630.1 3	7/2 ⁻	0.001 keV 1	0		J ^π : 4282.1γ M1+E2 to 9/2 ⁻ , 6866.4γ D+Q to 5/2 ⁺ , spin=3/2, 7/2 from γ(θ) in (p,γ).
8633 3	(3/2, 5/2 ⁺)	0.001 keV 1	P		
8640	(5/2 ⁺)		0		J ^π : spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2 ⁺ .
8641.4 5	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	OP		XREF: P(8643)
8654 4			L		
8686.2 5	5/2 ⁻	0.001 keV 1	OP		E(level): weighted average of 8686.2 5 from (p,γ), and 8687 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
					J ^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2 ⁺ disfavors both 5/2 ⁻ and 7/2 ⁻ , but 7/2 ⁻ is much less likely.
8688.3 5	1/2 ⁺	0.20 keV 2	OP		XREF: P(8689)
8691 3	1/2 ⁻	6.4 keV 7	P		
8696.9 5	3/2 ⁻	0.8 keV 1	L OP		XREF: L(8698)P(8698)
8700 10	(7/2 to 17/2) ⁺		N		J ^π : L(α,d)=6 from 3/2 ⁺ .
8706.3 5			0		
8717.8 5			L 0		E(level): weighted average of 8720.9 37 from (α,p) and 8717.7 5 from (p,γ).
8750.9 5	3/2 ⁻	0.30 keV 3	OP		E(level): weighted average of 8750.9 5 from (p,γ), and 8750 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8767.0 5			0		
8772.9 5	1/2 ⁻	0.57 keV 6	OP		XREF: P(8775)
8779.8 5	3/2 ⁻	0.214 keV 25	OP		XREF: P(8781)
					E(level): weighted average of 8779.8 5 from (p,γ), and 8781 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8787.2 5	(5/2 ⁺)	0.001 keV 1	L OP		E(level): others: 8786.3 38 from (α,p), and 8788 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
					J ^π : (3/2 ⁺ , 5/2 ⁺) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2 ⁻ .
8788.3 ^b 6	(15/2 ⁺)	<0.28 ps	F H		XREF: F(?)
					J ^π : 915.4γ D, ΔJ=1 to 13/2 ⁺ ; band assignment.
					T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2007Ks01, DSAM).
8798.4 5	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	OP		E(level): weighted average of 8798.4 5 from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF		Comments
8820.9 5	(1/2 ⁺ to 7/2 ⁺)		0		(p,γ), and 8799 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8824.2 5	1/2 ⁺	1.70 keV 17	OP		J ^π : primary 5817.6γ and 7057.0γ to 5/2 ⁺ ; 6126.3γ and 8819.7γ to 3/2 ⁺ , which yields (1/2 ⁺ , 3/2 [±] , 5/2 [±] , 7/2 ⁺).
8829.3 5	1/2 ⁻	12.2 keV 12	OP		XREF: P(8825) E(level): weighted average of 8824.2 5 from (p,γ), and 8825 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8830 3	1/2 ⁺	0.080 keV 15	P		E(level): weighted average of 8829.3 5 from (p,γ), and 8829 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8833.1 5	(5/2, 7/2)		0		J ^π : 1/2 ⁻ from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2 ⁺ gives J≥3/2, which may indicate a different level.
8838.1 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	1 OP		J ^π : primary 3617.1γ to (5/2 ⁺), 4063.0γ to (5/2 ⁻); 4719.1γ to 7/2 ⁺ , 5669.6γ to 7/2 ⁻ .
8844.4 ^b 4	17/2 ⁺	5.9 ps 11	F HI N		XREF: P(8839) XREF: N(8840) E(level): weighted average of 8845.1 9 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 8844.6 5 from ($^{16}\text{O},\alpha\text{p}\gamma$), and 8844.2 4 from ($^{14}\text{N},\alpha\text{p}\text{p}\gamma$).
8856.2 5	5/2 ⁺	0.010 keV 5	OP		J ^π : 971.38γ E2, ΔJ=2 to 13/2 ⁺ , 526γ to 15/2 ⁻ ; band assignment. T _{1/2} : weighted average of 6.1 ps 11 from ($^{16}\text{O},\alpha\text{p}\gamma$) (1991Ja11, DSAM) and 5.5 ps 14 from ($^{14}\text{N},\alpha\text{p}\text{p}\gamma$) (1976Wa11, DSAM).
8868.6 5	3/2 ⁺ , 5/2 ⁺	0.027 keV 10	OP		E(level): weighted average of 8856.1 5 from (p,γ), and 8858 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8884.1 5			0		J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.010 keV.
8886.1 5	(5/2)		L 0		E(level): weighted average of 8868.6 5 from (p,γ), and 8870 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8893.3 5	(5/2 ⁺ , 7/2 ⁺)		0		E(level): other: 8887.5 41 from (α,p).
8904.8 5	(1/2, 3/2, 5/2 ⁺)		0		J ^π : primary 3722.6γ to 7/2 ⁻ , 6239.8γ to 7/2 ⁺ ; 4707.9γ to 3/2 ⁻ , 4967.2γ to 3/2 ⁺ .
8906.8 5	5/2 ⁺	0.002 keV 1	OP	x	J ^π : primary 4950.0γ to 9/2 ⁺ , 6198.8γ to 3/2 ⁺ .
8919.8 5	(5/2 ⁻ , 7/2, 9/2 ⁺)		0	x	J ^π : primary 4726.6γ to 3/2 ⁻ , 6210.3γ to 3/2 ⁺ , 7684.5γ to 1/2 ⁺ .
					XREF: P(8909)x(8921) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2 ⁺ rules out 3/2 ⁺ based on RUL and Γ=0.002 keV.
					XREF: x(8921) J ^π : primary 4571.7γ to 9/2 ⁻ , 7156.0γ to 5/2 ⁺ .

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
8933.3 5			0	
8953.1 5	3/2 ⁺	0.075 keV 15	OP	XREF: P(8955) E(level): weighted average of 8953.0 5 from (p,γ), and 8955 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and (3/2) ⁺ from R-Matrix analysis in (p,p):resonances.
8957.9 5	(1/2 ⁻ ,3/2 ⁻)	0.04 keV 1	L OP	XREF: P(8959) E(level): others: 8957.7 44 from (α,p) and 8959 3 from (p,p):resonances.
8981.9 5	5/2 ⁻ ,7/2 ⁻	0.003 keV 2	OP	XREF: P(8983) E(level): weighted average of 8981.9 5 from (p,γ), and 8983 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances.
8984.2 5	5/2 ⁺	0.025 keV 10	OP	XREF: P(8985) J ^π : 3/2 ⁺ ,5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.025 keV.
8988.4 5			0	
8992.4 5			0	
8996.7 5	5/2 ⁻ ,7/2 ⁻	0.002 keV 1	L OP	XREF: L(8996.9)P(8998)
9001.1 5			0	
9019.3 5	3/2 ⁻	3.27 keV 30	OP	XREF: P(9021) T _{1/2} : weighted average of 3.1 keV 3 from (p,γ) and 3.50 keV 35 from (p,p):resonances.
9024.5 5			1 0	
9029.9 5	(5/2) ⁺	0.04 keV 1	1 OP	XREF: P(9031)
9033.1 5			0	
9038.4 5	1/2 ⁻	0.292 keV 30	OP	XREF: P(9040) E(level): weighted average of 9038.3 5 from (p,γ), and 9040 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances.
9048.4 5	5/2 ⁻ ,7/2 ⁻	0.001 keV 1	OP	XREF: P(9048)
9050 3	(5/2) ⁺	0.095 keV 15	P	
9081.5 4	5/2 ⁺	59 eV 10	L OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 4 from (p,γ), and 9084 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances. J ^π : spin=5/2 from γ(θ) in (p,γ) and (5/2) ⁺ from R-Matrix analysis in (p,p):resonances. T _{1/2} : weighted average of 65 eV 20 from (p,γ) and 57 eV 10 from (p,p):resonances.
9088.5 5			L OP	XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 5 from (p,γ), and 9089 4 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances.
9099.3 5			0	
9100.4 5	3/2 ⁻	0.20 keV 2	OP	XREF: P(9101)
9107.4 5			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 5 from (p,γ), and 9108 4 from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances.
9110.1 5			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 5 from (p,γ), and 9114 4 from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
9124.0 5			OP	$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. XREF: P(9120)
9127 4	5/2 ^{&}		J P	E(level): from (p,p):resonances. Other: 9127 9 from $^{31}\text{P}(\alpha,\text{p})$:resonances.
9135.1 5		2.0 keV 4	Op	XREF: p(9136)
9138.3 5			Op	T _{1/2} : from (p,γ).
9147 4	(7/2 to 17/2) ⁺		N P	XREF: p(9136) XREF: N(9150)P(9147)
9155.7 5			0	J ^π : L(α,d)=6 from 3/2 ⁺ .
9157.1 5	5/2 ⁻		OP	XREF: P(9159)
				J ^π : spin=5/2 from γ(θ) in (p,γ); 5993.6γ M1+E2 to 7/2 ⁻ .
9161 4	1/2 ^{&}		J 1	XREF: l(9160.0)
9163.3 5	(3/2 ⁻)		1 OP	XREF: l(9160.0)P(9168)
9184.2 5			Op	E(level): other: 9168 4 from (p,p):resonances.
9188.8 5			Op	XREF: p(9185)
9194.1 5			L OP	XREF: p(9185) E(level): others: 9194.0 39 from (α,p), and 9196 4 from (p,p):resonances.
9207 4	9/2 ⁺		P	
9222 4	5/2 ⁺		P	X XREF: X(9220)
				J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=9220 120.
9227 4	5/2 ⁻		P	
9241 4	7/2 ⁺		P	
9255 4	1/2 ^{&}		J P	E(level): from (p,p):resonances. Other: 9256 4 from $^{31}\text{P}(\alpha,\text{p})$:resonances.
9264.9 39			L	
9271 4			P	
9276 4			P	
9284 4	5/2 ⁻		P	
9294 4			P	
9299 4			P	
9316.2 39			L	
9325 4	5/2 ⁺		P	
9335 4	7/2 ⁺		L P	E(level): weighted average of 9336 4 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances and 9334 5 from $^{32}\text{S}(\alpha,\text{p})$.
9345 4			P	
9349 4			P	
9355 4			P	
9376 4			L	
9400 9	1/2 ^{&}		J	
9450 10	(7/2 to 17/2) ⁺		N	J ^π : L(α,d)=6 from 3/2 ⁺ .
9456 4	3/2 ^{&}		J	
9476 4			L	
9481.0 18	3/2 ^{&}		J	
9508 4	(1/2 ⁻ ,3/2,5/2 ⁺)		L	X XREF: X(9514)
				J ^π : L(pol d, ^3He)=(1,2) from 0 ⁺ .
9551 6	5/2 ^{&}		J	
9673 6	1/2 ^{&}		J	
9712.7 27	1/2 ^{&}		J L	E(level): weighted average of 9713.0 27 from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
				$^{31}\text{P}(\alpha, p), (\alpha, n)$: resonances and 9711.8 44 from (α, p).
9740 4			L	
9751.1 27	7/2 ^{&}		J	
9799 5			L	
9814.0 27	5/2 ⁽⁺⁾		J	XREF: X(9820) J ^π : spin=5/2 from angular distributions in $^{31}\text{P}(\alpha, p)$: resonances; L=(1,2) from 0 ⁺ in (pol d, ^3He) gives (1/2 ⁻ , 3/2, 5/2 ⁺).
9836 5			L	
9869.8 27	1/2 ^{&}		J L	XREF: L(9870)
9900.8 27	3/2 ^{&}		J	
9923 4	3/2 ⁻ &		J	
9958 6	3/2 ^{&}		J	
9969 5	(1/2, 3/2) &		J L	XREF: J(9968) E(level): from (α, p) and (α, n): resonances.
10031 5	(1/2 ⁺ , 3/2 ⁻) &		J	
10058 5	3/2 ^{&}		J	
10075 5	(1/2) &		J L	E(level): from (α, p) and (α, n): resonances.
10090 5	5/2 ⁺ &		J	
10132 5	3/2 ⁻ &		J	
10163 5	&		L	
10172 5	3/2 ⁻ , 5/2 ⁻ &		J	
10179 5			L	
10181.1 ^a 5	19/2 ⁻	42.3 fs 28	F H	E(level): weighted average of 10181.1 10 from ($^{28}\text{Si}, \alpha p \gamma$) and 10181.1 5 from ($^{16}\text{O}, \alpha p \gamma$). J ^π : 1693.5γ E2, ΔJ=2 to 15/2 ⁻ ; 1336.3γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=61 fs 4 from ($^{28}\text{Si}, \alpha p \gamma$) (2013Bi10, DSAM).
10215 5	3/2 ^{&}		J L	XREF: L(10218.0)
10221.9 11	(17/2 ⁻)	<0.28 ps	F H	XREF: F(?) E(level): from ($^{16}\text{O}, \alpha p \gamma$). J ^π : 1377.8γ to 17/2 ⁺ consistent with ΔJ=0. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha p \gamma$) (2007Ks01, DSAM).
10236 5	3/2 ⁻ &		J	
10278 5	1/2 ^{&}		J	
10295 5			J	
10322 5	5/2 ⁺ &		J	
10340 5	(3/2, 5/2) &		J	
10359 5	3/2 ^{&}		J	
10385 5			J	
10395 5			L	
10399 5	3/2 ⁺ , 7/2 ⁺ &		J	
10431 5	5/2 ⁺ &		J	
10465 5	5/2 ⁺ &		J L	E(level): weighted average of 10467 5 from $^{31}\text{P}(\alpha, p), (\alpha, n)$: resonances and 10463.2 49 from

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
				(α ,p).
10486 5	5/2 ⁺ &		J	
10517 6			L	
10534 5	1/2 ⁺ ,5/2 ⁺ &		J	
10548 6			L	
10584 5	3/2 ⁺ &		J L	E(level): weighted average of 10587 5 from $^{31}\text{P}(\alpha,p),(\alpha,n)$:resonances and 10579.5 54 from (α ,p).
10642 5	5/2 ⁺ &		J L	E(level): weighted average of 10641 5 from $^{31}\text{P}(\alpha,p),(\alpha,n)$:resonances and 10643.0 50 from (α ,p).
10678 5	3/2 ⁺ ,5/2 ⁺ &		J	
10696 5	7/2 ⁻ &		J	
10733 5	3/2 ⁺ &		J L	E(level): weighted average of 10734 5 from $^{31}\text{P}(\alpha,p),(\alpha,n)$:resonances and 10732.1 50 from (α ,p).
10760 5	3/2 ⁺ ,7/2 ⁻ &		J L	E(level): weighted average of 10760 5 from $^{31}\text{P}(\alpha,p),(\alpha,n)$:resonances and 10759.1 55 from (α ,p).
10800 5	5/2 ⁺ &		J	
10816 5	1/2 ⁺ ,3/2 ⁻ &		J	
10844 5	3/2 ⁻ &		J	
10858.5 ^b 8	19/2 ⁺	<0.28 ps	F H	XREF: F(?) J ^π : 2069.8γ E2, ΔJ=2 to 15/2 ⁺ , 2014.7γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si},\alpha p\gamma$) (2007Ks01, DSAM).
10870 5	(5/2,7/2)&		J	
10886 5			J	
10909 5	3/2 ⁺ ,5/2 ⁻ &		J	
10928 5	7/2&		J	
10942 5			J	
10964 5	5/2 ⁺ &		J	
10973 5			J	
10998 5	3/2 ⁺ ,7/2 ⁻ &		J	
11034 5	3/2 ⁺ &		J	
11063 5	1/2 ⁺ ,9/2 ⁺ &		J	
11082 5	5/2 ⁺ ,9/2 ⁺ &		J	
11105 5			J	
11123 5	3/2&		J	
11142 5	5/2 ⁺ &		J	
11154 5			J	
11169 5			J	
11179 5	3/2 ⁻ ,5/2 ⁺ &		J	
11185 5			J	
11198 5	3/2&		J	
11230 5			J	
11246 5	3/2&		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π	T _{1/2} [‡]	XREF	Comments
11265 5	3/2 ⁻ &		J	
11287 5			J	
11307 5			J	
11314 5	1/2 ⁻ , 5/2 ⁺ &		J	
11327 5	1/2 ⁻ , 5/2 ⁺ &		J	
11357 5			J	
11375 5			J	
11390 5			J	
11410 5			J	
11421 5	3/2 ⁻ &		J	
11434 5			J	
11459.1 ^b 7	21/2 ⁺	43.0 fs 21	F H J	E(level): weighted average of 11459.2 12 from ($^{28}\text{Si}, \alpha\gamma$) and 11459.0 7 from ($^{16}\text{O}, \alpha\gamma$). Other: 11460 5 from $^{31}\text{P}(\alpha, p), (\alpha, n)$: resonances. J ^π : 2614.5γ E2, ΔJ=2 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=62 fs 3 from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
11473 5	1/2 ⁻ , 5/2 ⁻ &		J	
11492 5			J	
11505 5	3/2 ⁻ , 5/2 ⁻ &		J	
11525 5			J	
11540 5			J	
11551 5	3/2 ⁻ , 5/2 ⁺ &		J	
11565 5			J	
11589 5			J	
11609 5	5/2 ⁺ , 7/2 ⁻ &		J	
11627 5	1/2 ⁺ , 5/2 ⁺ &		J	
11638 5			J	
11651 5			J	
11684 5	1/2 ⁺ , 5/2 ⁺ &		J	
11773 5	3/2 ⁻ , 7/2 ⁻ &		J	
11783 5	3/2 ⁻ &		J	
1.25×10 ⁴ 5			J	X T=3/2 XREF: J(12900) E(level): from (pol d, ^3He). E(level): weighted average of 12572.2 12 from ($^{28}\text{Si}, \alpha\gamma$) and 12572.2 6 from ($^{16}\text{O}, \alpha\gamma$). J ^π : 2390.8γ E2, ΔJ=2 to 19/2 ⁻ , 1113.3γ D, ΔJ=1 to 21/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=48 fs 6 from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
12572.2 ^a 6	23/2 ⁻	33 fs 4	F H	T=3/2 J ^π : 3734γ to 23/2 ⁻ ; band assignment. T _{1/2} : lifetime τ<13 fs from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
13900			J	
16306.4 ^a 16	(27/2 ⁻)	<9.0 fs	F	

[†] Most of Adopted Levels are reported in (p,γ) dataset with precise E(level) values, deduced based on measured γ-ray energies which however are not reported in references in (p,γ) dataset, while E(level) values from E_γ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p,γ) datasets. Therefore, a least-squares fit to

Adopted Levels, Gammas (continued)

 ^{35}Cl Levels (continued)

available $E\gamma$ values with uncertainties would not produce $E(\text{level})$ values more precise and complete than those already reported in (p, γ) and the evaluators have adopted $E(\text{level})$ values from (p, γ) or averages of values based on $E\gamma$ data from all available datasets where applicable, as noted.

‡ Sources of $T_{1/2}$ values are listed under comments; Γ from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances, unless otherwise noted.

For proton-resonance levels populated in (p,p):resonances, J^π assignments are from R-Matrix analysis, unless otherwise noted.

@ From FRESKO-DWBA analysis of measured $\sigma(\theta)$ of (α ,p) ([2019Se09](#)).

& From angular distributions in $^{31}\text{P}(\alpha, \text{p})$:resonances.

^a Band(A): Band based on $f_{7/2}$ orbital.

^b Band(B): Band based on $13/2^+$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
1219.38	1/2 ⁺	1219.4 1	100	0.0	3/2 ⁺	M1+E2	0.093 10	3.66×10 ⁻⁵ 5	B(M1)(W.u.)=0.101 +14-11; B(E2)(W.u.)=2.2 +6-5 E _γ : weighted average of 1219.3 2 from ³⁵ Ar ε decay and 1219.4 1 from (p,p'γ), and 1219.4 6 from (α,pγ). Mult.,δ: ΔJ=1 from γ(θ) in (p,p'γ), E1+M2 ruled out by RUL. δ from adopted B(E2)†=0.00076 7 measured in Coulomb excitation and adopted lifetime τ=0.172 ps 20.
1763.15	5/2 ⁺	543.77 1763.16 10	<0.2 100	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[E2] M1+E2	-2.85 10	0.000323 5 0.0002124 30	I _γ : from (p,γ). Other: <0.5 from (α,pγ). B(M1)(W.u.)=0.00122 +14-12; B(E2)(W.u.)=12.1 +11-9 E _γ : weighted average of 1763.0 2 from ³⁵ Ar ε decay, 1763.1 2 from (²⁴ Mg,αpγ), 1763.3 2 from (¹⁶ O,αpγ), 1763.15 10 from (¹⁴ N,αpnγ), and 1763.2 1 from (p,p'γ). Other: 1763.2 6 from (α,pγ). Mult.: from γγ(ADO) in (¹⁶ O,αpγ) with ΔJ=1 and γ(lin pol) in (²⁸ Si,αpγ). δ: weighted average of -3.0 1 from (α,pγ), -2.66 12 from (p,p'γ), -2.95 45 from Coulomb excitation, -2.6 4 from (¹⁶ O,αpγ), all from γ(θ) or pγ(θ). Other: -0.42 17 from γγ(θ) in (p,γ); 1.8 +10-4; deduced by the evaluators from adopted B(E2)†=0.0106 7 and T _{1/2} =0.36 ps 3.
2645.67	7/2 ⁺	882.80 11	11.4 10	1763.15	5/2 ⁺	M1+E2	+0.21 5	5.29×10 ⁻⁵ 9	B(M1)(W.u.)=0.0196 +35-27; B(E2)(W.u.)=4.2 +23-18 E _γ : weighted average of 882.9 1 from (²⁴ Mg,αpγ), 882.7 5 from (¹⁶ O,αpγ), 882.32 35 from (¹⁴ N,αpnγ), 882.4 10 from (α,pγ), and 882.3 3 from (p,p'γ). I _γ : unweighted average of 8.7 6 from (²⁸ Si,αpγ), 15.2 13 from (²⁴ Mg,αpγ), 11.08 28 from (¹⁶ O,αpγ), 11.1 22 from (α,pγ), 9.5 11 from (p,γ), and 13 5 from (p,p'γ). Mult.: D+Q from pγ(θ) in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 5 from (α,pγ) and +0.17 5 from (p,p'γ). I _γ : from (α,pγ). Other: <2.2 from (p,γ). B(E2)(W.u.)=3.6 +6-4 E _γ : weighted average of 2645.5 4 from (²⁴ Mg,αpγ), 2645.8 5 from (¹⁶ O,αpγ), 2645.79 30 from
		1426.26 2645.70 20	<1.2 100.0 11	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[M3] E2		7.52×10 ⁻⁵ 11 0.000633 9	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
									($^{14}\text{N},\alpha\text{p}\gamma$), and 2645.7 2 from (p,p' γ). Other: 2646.2 15 from ($\alpha,\text{p}\gamma$).
									I_γ : others: 100.0 31 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 100.0 22 from ($^{16}\text{O},\alpha\text{p}\gamma$), 100.0 22 from ($\alpha,\text{p}\gamma$), and 100.0 20 from (p,p' γ). Other: 100 6 from ($^{28}\text{Si},\alpha\text{p}\gamma$).
									Mult.: from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si},\alpha\text{p}\gamma$); $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ).
2694.00	3/2 ⁺	930.9 2	16.3 13	1763.15	5/2 ⁺	M1+E2	+0.09 3	4.68×10 ⁻⁵ 7	B(M1)(W.u.)=0.110 +26−19; B(E2)(W.u.)=3.9 +33−22
									E_γ : weighted average of 930.7 2 from ^{35}Ar ε decay, 930.5 15 from ($\alpha,\text{p}\gamma$), and 931.3 3 from (p,p' γ).
									I_γ : weighted average of 14.3 18 from ^{35}Ar ε decay, 18.2 26 from ($\alpha,\text{p}\gamma$), 16.3 13 from (p, γ), and 18.6 25 from (p,p' γ).
									Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ).
		1474.8 3	9.5 10	1219.38	1/2 ⁺	M1+E2	−0.63 21	8.4×10 ⁻⁵ 4	B(M1)(W.u.)=0.0116 +37−29; B(E2)(W.u.)=8.0 +45−39
									E_γ : weighted average of 1474.8 3 from ^{35}Ar ε decay and 1475 2 from (p,p' γ).
									I_γ : weighted average of 9.3 10 from ^{35}Ar ε decay, 11.7 26 from ($\alpha,\text{p}\gamma$), 9.5 13 from (p, γ), and 8.3 25 from (p,p' γ).
									Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ).
		2693.7 2	100.0 17	0.0	3/2 ⁺	M1+E2	+0.26 2	0.000553 8	B(M1)(W.u.)=0.026 +6−4; B(E2)(W.u.)=0.92 +26−19
									E_γ : weighted average of 2693.7 2 from ^{35}Ar ε decay and 2694.1 6 from (p,p' γ). Other: 2693.2 20 from ($\alpha,\text{p}\gamma$).
									I_γ : weighted average of 100.0 17 from ^{35}Ar ε decay, 100 4 from ($\alpha,\text{p}\gamma$), 100.0 25 from (p, γ), and 100.0 30 from (p,p' γ).
									Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ).
									δ : weighted average of +0.17 8 from ($\alpha,\text{p}\gamma$) and +0.26 2 from (p,p' γ).
3002.64	5/2 ⁺	308.64	<5	2694.00	3/2 ⁺	[M1,E2]		0.0014 9	I_γ : from ($\alpha,\text{p}\gamma$).
		356.97	<1	2645.67	7/2 ⁺	[M1,E2]		9	I_γ : from ($\alpha,\text{p}\gamma$).
		1239.47	2.2 11	1763.15	5/2 ⁺	[M1,E2]		4.4×10 ⁻⁵ 6	I_γ : from (p,p' γ).
									B(M1)(W.u.)=0.011 +8−5 if M1, B(E2)(W.u.)=26 +19−12 if E2.
		1783.21	3.3 11	1219.38	1/2 ⁺	[E2]		0.0002268 32	B(E2)(W.u.)=6.4 +35−22
									I_γ : from (p,p' γ).
		3002.3 4	100.0 10	0.0	3/2 ⁺	M1+E2	+0.08 2	0.000671 9	B(M1)(W.u.)=0.034 +14−7; B(E2)(W.u.)=0.09 +7−4
									E_γ : weighted average of 3002.1 4 from ^{35}Ar ε decay,

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
3162.87	7/2 ⁻	160.67 20	1.5 4	3002.64	5/2 ⁺	[E1]		0.00330 5	3002.5 15 from ($\alpha,\text{p}\gamma$), and 3002.5 5 from ($\text{p},\text{p}'\gamma$). I_γ : weighted average of 100.0 19 from ^{35}Ar ε decay and 100.0 10 from ($\text{p},\text{p}'\gamma$). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ($\text{p},\text{p}'\gamma$). δ : weighted average of +0.09 3 from ($\alpha,\text{p}\gamma$) and +0.07 2 from ($\text{p},\text{p}'\gamma$). Other: -0.22 1 from (p,γ). $B(\text{E1})(\text{W.u.})=6.6\times 10^{-5}$ 17 E_γ : weighted average of 161 1 from ($^{16}\text{O},\alpha\text{p}\gamma$) and 160.66 20 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$). I_γ : unweighted average of 1.10 30 from ($^{28}\text{Si},\alpha\text{p}\gamma$) and 1.89 22 from (p,γ). $B(\text{E1})(\text{W.u.})=1.61\times 10^{-5}$ 20 E_γ : weighted average of 517.7 1 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 517.3 5 from ($^{16}\text{O},\alpha\text{p}\gamma$), 517.26 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 517.4 15 from ($\alpha,\text{p}\gamma$), and 517.0 8 from ($\text{p},\text{p}'\gamma$). I_γ : unweighted average of 9.2 5 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 14.3 8 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 11.1 13 from ($^{16}\text{O},\alpha\text{p}\gamma$), 11.1 33 from ($\alpha,\text{p}\gamma$), 8.9 11 from (p,γ), and 19 8 from ($\text{p},\text{p}'\gamma$). Mult.: D from $\gamma\gamma(\theta)(\text{DCO})$ in ($^{28}\text{Si},\alpha\text{p}\gamma$); $\Delta\pi=\text{yes}$ from level scheme.
		517.47 11	12.3 16	2645.67	7/2 ⁺	(E1)		0.0001097 15	$B(\text{E1})(\text{W.u.})=1.88\times 10^{-8}$ +31-34; $B(\text{M2})(\text{W.u.})=0.0085$ +43-38 I_γ : weighted average of 0.50 20 from ($^{16}\text{O},\alpha\text{p}\gamma$) and 0.33 5 from (p,γ). Mult., δ : D+Q from $\gamma(\theta)$ in (p,γ) (1971Pr11); $\Delta\pi=\text{yes}$ from level scheme.
		1399.9 7	0.34 5	1763.15	5/2 ⁺	(E1+M2)	+0.44 12	0.000181 12	$B(\text{M2})(\text{W.u.})=0.255$ 9; $B(\text{E3})(\text{W.u.})=4.2$ +11-9 E_γ : weighted average of 3162.7 4 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 3162.6 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3162.43 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), and 3162.6 1 from ($\text{p},\text{p}'\gamma$), and 3162.6 15 from ($\alpha,\text{p}\gamma$). I_γ : weighted average of 100 5 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 100.0 30 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 100.0 20 from ($^{16}\text{O},\alpha\text{p}\gamma$), 100.0 33 from ($\alpha,\text{p}\gamma$), 100.0 11 from (p,γ), and 100.0 20 from ($\text{p},\text{p}'\gamma$). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,γ). δ : weighted average of +0.16 1 from ($^{28}\text{Si},\alpha\text{p}\gamma$), +0.18 7 from ($^{16}\text{O},\alpha\text{p}\gamma$), and +0.25 4 from (p,γ). Others: -0.14 2 from ($\alpha,\text{p}\gamma$) and -0.22 5 from ($\text{p},\text{p}'\gamma$).
		1943.43	<0.22	1219.38	1/2 ⁺	[E3]		0.0001742 24	
		3162.53 10	100.0 11	0.0	3/2 ⁺	M2+E3	+0.17 2	0.000523 7	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
3918.47	3/2 ⁺	1272.78	<3.7	2645.67	7/2 ⁺	[E2]	5.56×10 ⁻⁵ 8	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : weighted average of 29 12 from ³⁵ Ar ε decay and 23.9 28 from (p, γ). B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2. I_γ : from (α ,p γ). B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2. I_γ : weighted average of 100 6 from ³⁵ Ar ε decay, 100 5 from (α ,p γ), and 100.0 19 from (p, γ). δ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, γ). B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2. E_γ : from (α ,p γ). I_γ : weighted average of 6.2 31 from (²⁸ Si, α p γ), 8.1 21 from (α ,p γ), and 8.7 22 from (p, γ). B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2. B(E2)(W.u.)=8.7 +17-13 E_γ : weighted average of 2180 1 from (¹⁶ O, α p γ) and 2180.2 15 from (α ,p γ). I_γ : weighted average of 100.0 21 from (α ,p γ) and 100.0 22 from (p, γ). Other: 100 11 from (²⁸ Si, α p γ). B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2. B(M1)(W.u.)=0.071 +23-14 E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2155.1 15	24.2 28	1763.15	5/2 ⁺	[M1,E2]	0.00036 4	
		2698.98	19 5	1219.38	1/2 ⁺	[M1,E2]	0.00060 6	
		3918.24	100.0 19	0.0	3/2 ⁺	M1+E2	0.00108 7	
3942.9	9/2 ⁺	1297.4 15	8.0 21	2645.67	7/2 ⁺	[M1,E2]	5.2×10 ⁻⁵ 8	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : weighted average of 29 12 from ³⁵ Ar ε decay and 23.9 28 from (p, γ). B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2. I_γ : from (α ,p γ). B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2. I_γ : weighted average of 100 6 from ³⁵ Ar ε decay, 100 5 from (α ,p γ), and 100.0 19 from (p, γ). δ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, γ). B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2. E_γ : from (α ,p γ). I_γ : weighted average of 6.2 31 from (²⁸ Si, α p γ), 8.1 21 from (α ,p γ), and 8.7 22 from (p, γ). B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2. B(E2)(W.u.)=8.7 +17-13 E_γ : weighted average of 2180 1 from (¹⁶ O, α p γ) and 2180.2 15 from (α ,p γ). I_γ : weighted average of 100.0 21 from (α ,p γ) and 100.0 22 from (p, γ). Other: 100 11 from (²⁸ Si, α p γ). B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2. B(M1)(W.u.)=0.071 +23-14 E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2180.1 10	100.0 21	1763.15	5/2 ⁺	E2	0.000414 6	
3967.6	1/2 ⁺	1273.6	2.7 6	2694.00	3/2 ⁺	[M1,E2]	4.9×10 ⁻⁵ 7	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2204.4	≤1.3	1763.15	5/2 ⁺	[E2]	0.000426 6	
		2748.5 6	100 6	1219.38	1/2 ⁺	[M1]	0.000569 8	
		3967.4	23 4	0.0	3/2 ⁺	[M1,E2]	0.00109 7	
4059.1	(3/2) ⁻	1365.1	1.3 4	2694.00	3/2 ⁺	[E1]	0.0001824 26	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2295.9	5.5 16	1763.15	5/2 ⁺	[E1]	0.000848 12	
		2839.6	100.0 21	1219.38	1/2 ⁺	[E1]	1.17×10 ⁻³ 2	
		4058.9	<2.1	0.0	3/2 ⁺	[E1]	1.70×10 ⁻³ 2	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
4112.4	7/2 ⁺	2349.2	89 6	1763.15	5/2 ⁺	M1+E2	-0.16 2	0.000405 6	B(M1)(W.u.)=0.0159 +48-31; B(E2)(W.u.)=0.28 +12-8 I _γ : other: 16 5 from (α,pγ).
		4112.1	100 6	0.0	3/2 ⁺	E2(+M3)	0.00 10	1.21×10 ⁻³ 2	B(E2)(W.u.)=0.76 +22-15 I _γ : other: 100 5 from (α,pγ).
4173.43	(5/2 ⁺)	1479.40	45 14	2694.00	3/2 ⁺	[M1,E2]		9.1×10 ⁻⁵ 13	B(M1)(W.u.)=0.048 +20-12 if M1, B(E2)(W.u.)=82 +34-21 if E2. B(E2)(W.u.)=82 +34-21 upper bound exceeds RUL=100 if E2.
		1527.72	<17	2645.67	7/2 ⁺	[M1,E2]		0.000106 15	
		2410.19	28 9	1763.15	5/2 ⁺	[M1,E2]		0.00048 5	B(M1)(W.u.)=0.0068 +33-19 if M1, B(E2)(W.u.)=4.5 +21-13 if E2.
		2953.92	≤5.2	1219.38	1/2 ⁺	[E2]		0.000771 11	
		4173.16	100 17	0.0	3/2 ⁺	[M1,E2]		0.00116 7	B(M1)(W.u.)=0.0047 +15-7 if M1, B(E2)(W.u.)=1.02 +33-14 if E2.
4177.9	3/2 ⁻	1175.2	<0.82	3002.64	5/2 ⁺	[E1]		6.06×10 ⁻⁵ 8	
		1483.9	13.1 33	2694.00	3/2 ⁺	[E1]		0.000265 4	B(E1)(W.u.)=5.7×10 ⁻⁴ +16-14
		1532.2	<0.82	2645.67	7/2 ⁺	[M2]		6.18×10 ⁻⁵ 9	
		2414.7	<1.6	1763.15	5/2 ⁺	[E1]		0.000921 13	
		2958.4	51 8	1219.38	1/2 ⁺	(E1+M2)	+0.11 4	1.22×10 ⁻³ 2	B(E1)(W.u.)=2.8×10 ⁻⁴ +5-4; B(M2)(W.u.)=1.7 +16-11 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme. B(M2)(W.u.)=1.7 +16-11 upper bound exceeds RUL=3.
		4177.6	100 8	0.0	3/2 ⁺	(E1+M2)	+0.06 3	1.74×10 ⁻³ 3	B(E1)(W.u.)=1.93×10 ⁻⁴ +28-20; B(M2)(W.u.)=0.18 +24-14 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme.
4347.76	9/2 ⁻	1184.90 20	100.0 29	3162.87	7/2 ⁻	M1+E2	-0.36 3	3.57×10 ⁻⁵ 5	B(M1)(W.u.)=0.007 +7-3; B(E2)(W.u.)=2.5 +27-10 E _γ : weighted average of 1185.0 2 from (²⁴ Mg,αpγ), 1184.9 3 from (¹⁶ O,αpγ), and 1184.80 20 from (¹⁴ N,αpnγ). Other: 1184.6 10 from (α,pγ). I _γ : weighted average of 100 4 from (²⁸ Si,αpγ), 100.0 31 from (²⁴ Mg,αpγ), 100 4 from (¹⁶ O,αpγ), 100.0 31 from (α,pγ), and 100.0 29 from (p,γ). Mult.: from γγ(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=1. δ: weighted average of -0.36 3 from (α,pγ), -0.3 1 from (²⁸ Si,αpγ), and -0.65 30 from (¹⁶ O,αpγ). B(E1)(W.u.)=3.7×10 ⁻⁵ +39-14; B(M2)(W.u.)=0.019
		1701.88 25	46.8 24	2645.67	7/2 ⁺	E1+M2	-0.018 12	0.000434 6	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
								+47–17 E_γ : weighted average of 1702.0 3 from ($^{24}\text{Mg}, \alpha\text{py}$), 1701.7 5 from ($^{16}\text{O}, \alpha\text{py}$), and 1701.85 25 from ($^{14}\text{N}, \alpha\text{pny}$). Other: 1702.1 15 from (α, py). I_γ : unweighted average of 45.6 24 from ($^{28}\text{Si}, \alpha\text{py}$), 39.5 24 from ($^{24}\text{Mg}, \alpha\text{py}$), 52.0 14 from ($^{16}\text{O}, \alpha\text{py}$), 52 7 from (α, py), and 44.9 29 from (p, γ). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha\text{py}$) with $\Delta J=1$ and $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (α, py). δ : from (p, γ). Others: +0.5 +5–3 from ($^{28}\text{Si}, \alpha\text{py}$), +0.5 4 from ($^{16}\text{O}, \alpha\text{py}$).
4347.76	9/2 [−]	2584.51 4347.8 8	≤ 4.4 12.4 11	1763.15 0.0	5/2 ⁺ 3/2 ⁺	[M2] [E3]	0.000329 5 0.001000 14	B(E3)(W.u.)=44 +46–17 E_γ, I_γ : from ($^{16}\text{O}, \alpha\text{py}$). E_γ : other: 4618 2 from (α, py). Mult., δ : from $\text{py}(\theta)$ with −0.08 14 for J=3/2; +0.36 15 for J=5/2 in (α, py). If J(4624=5/2), Mult would be M1+E2 based on RUL.
4624.3	3/2, 5/2	4624.0	100	0.0	3/2 ⁺	D+Q		B(M1)(W.u.)=0.040 +23–5 if M1, B(E2)(W.u.)=58 +34–7 if E2. B(E1)(W.u.)=4.6×10 ^{−4} +30–9
4769.86	(5/2 [−])	1606.95 1767.17 2075.79 3550.29 4769.51	100 16 54 16 <15 <15 <15	3162.87 3002.64 2694.00 1219.38 0.0	7/2 [−] 5/2 ⁺ 3/2 ⁺ 1/2 ⁺ 3/2 ⁺	[M1, E2] [E1] [E1] [M2] [E1]	0.000133 19 0.000483 7 0.000702 10 0.000645 9 1.95×10 ^{−3} 3	
4839.10	(1/2 ⁺ , 3/2)	1836.41 2145.03 2193.36 ^b 3075.81 3619.52 4838.74	<12 <6.8 <5.1 100 9 <12 70 9	3002.64 2694.00 2645.67 1763.15 1219.38 0.0	5/2 ⁺ 3/2 ⁺ 7/2 ⁺ 5/2 ⁺ 1/2 ⁺ 3/2 ⁺			
4854.4	(1/2, 3/2)	3634.8 4854.0	100 7 33 7	1219.38 0.0	1/2 ⁺ 3/2 ⁺			
4880.96	7/2 ⁽⁺⁾	1878.27 2186.89 2235.21	15 5 <16 47 8	3002.64 2694.00 2645.67	5/2 ⁺ 3/2 ⁺ 7/2 ⁺	[M1, E2] [E2] [M1, E2]	0.000240 30 0.000417 6 0.00040 4	B(M1)(W.u.)=0.051 +30–16 if M1, B(E2)(W.u.)=55 +32–17 if E2. I_γ : other: 52 9 from (α, py) for a doublet. B(M1)(W.u.)=0.095 +43–18 if M1, B(E2)(W.u.)=72 +33–14 if E2. B(E2)(W.u.)=72 +33–14 upper bound exceeds RUL=100 if E2. E_γ : from (α, py). I_γ : weighted average of 100 9 from (α, py) and 100 8 from (p, γ). B(M1)(W.u.)=0.075 +33–12 if M1, B(E2)(W.u.)=29 +13–5 if E2.
		3117.4 20	100 8	1763.15	5/2 ⁺	[M1, E2]	0.00078 6	
		4880.60	<8.1	0.0	3/2 ⁺	[E2]	1.46×10 ^{−3} 2	

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>									
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	<u>$\delta^\#$</u>	<u>α^a</u>	<u>Comments</u>
5009.1	(1/2,3/2)	835.7	<25	4173.43	(5/2 ⁺)				
		950.0	<10	4059.1	(3/2) ⁻				
		1846.2 ^b	<50	3162.87	7/2 ⁻				
		2006.4	<20	3002.64	5/2 ⁺				
		2315.0	<25	2694.00	3/2 ⁺				
		2363.3 ^b	<5	2645.67	7/2 ⁺				
		3245.8	<10	1763.15	5/2 ⁺				
		3789.5	<7	1219.38	1/2 ⁺				
		5008.7	100	0.0	3/2 ⁺				
5163.34	7/2 ⁻	2000.41	93 12	3162.87	7/2 ⁻	D+Q	+0.44 20		
		2160.63	23 9	3002.64	5/2 ⁺				
		3400.01	100 12	1763.15	5/2 ⁺	D(+Q)	0.00 3		
		3943.72 ^b	<23	1219.38	1/2 ⁺				
		5162.93	<23	0.0	3/2 ⁺				
5214.5	(5/2 ⁺)	205.4	<23	5009.1	(1/2,3/2)				
		1041.1	<60	4173.43	(5/2 ⁺)				
		1155.4	<10	4059.1	(3/2) ⁻				
		2051.6	<15	3162.87	7/2 ⁻				
		2211.8	<25	3002.64	5/2 ⁺				
		2520.4	<15	2694.00	3/2 ⁺				
		2568.7	<15	2645.67	7/2 ⁺				
		3451.2	<6	1763.15	5/2 ⁺				
		3994.9	<4	1219.38	1/2 ⁺				
		5214.1	100	0.0	3/2 ⁺	[M1,E2]		0.00147 8	
5403.3	3/2 ⁻	2400.6	46 9	3002.64	5/2 ⁺	[E1]		0.000913 13	B(E1)(W.u.)=0.0011 +6-2
		3640.0	<18	1763.15	5/2 ⁺	[E1]		1.54×10 ⁻³ 2	
		4183.7	100 18	1219.38	1/2 ⁺	[E1]		1.75×10 ⁻³ 2	B(E1)(W.u.)=4.5×10 ⁻⁴ +20-6
		5402.9	<13	0.0	3/2 ⁺	[E1]		2.14×10 ⁻³ 3	
5407.07	11/2 ⁻	1059.32 20	14.5 22	4347.76	9/2 ⁻	M1+E2	+0.14 3	3.67×10 ⁻⁵ 5	B(M1)(W.u.)=0.0082 +30-20; B(E2)(W.u.)=0.54 +35-23
									E_γ : weighted average of 1059.3 2 from (²⁴ Mg, $\alpha\gamma$), 1059.5 2 from (¹⁶ O, $\alpha\gamma$), and 1059.17 20 from (¹⁴ N, $\alpha p n\gamma$). Other: 1059.3 10 from ($\alpha, p\gamma$).
									I_γ : unweighted average of 12.0 12 from (²⁸ Si, $\alpha\gamma$), 16.9 17 from (²⁴ Mg, $\alpha\gamma$), 19.2 5 from (¹⁶ O, $\alpha\gamma$), and 9.9 22 from ($\alpha, p\gamma$).
									Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in (²⁸ Si, $\alpha\gamma$), $\Delta J=1$.
									δ : weighted average of +0.12 3 from (¹⁶ O, $\alpha\gamma$),

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	
5407.07	11/2 ⁻	2243.71 23	100.0 22	3162.87	7/2 ⁻	E2		0.000444 6	+0.2 1 from ($^{28}\text{Si}, \alpha p \gamma$), and +0.25 8 from ($\alpha, p \gamma$). B(E2)(W.u.)=4.6 +15-9 E _γ : weighted average of 2244.2 3 from ($^{24}\text{Mg}, \alpha p \gamma$), 2243.3 2 from ($^{16}\text{O}, \alpha p \gamma$), 2244.00 25 from ($^{14}\text{N}, \alpha p n \gamma$), and 2244 3 from ($\alpha, p \gamma$). I _γ : weighted average of 100.0 33 from ($^{28}\text{Si}, \alpha p \gamma$), 100.0 32 from ($^{24}\text{Mg}, \alpha p \gamma$), 100.0 23 from ($^{16}\text{O}, \alpha p \gamma$), and 100.0 22 from ($\alpha, p \gamma$). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha p \gamma$), $\Delta J=2$.
5586.0	5/2 ⁺	2940.2 3822.6 4366.3 5585.5	67 17 100 17 <17 <17	2645.67 7/2 ⁺ 1763.15 5/2 ⁺ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5598.4	3/2 ⁺ , 5/2 ⁺	3835.0	37 14	1763.15 5/2 ⁺		[M1, E2]		0.00105 7	B(M1)(W.u.)=0.050 +29-19 if M1, B(E2)(W.u.)=13 +8-5 if E2.
		5597.9	100 14	0.0 3/2 ⁺		[M1, E2]		0.00157 8	B(M1)(W.u.)=0.044 +23-12 if M1, B(E2)(W.u.)=5.3 +28-14 if E2.
5646	(5/2, 7/2, 9/2 ⁺)	2483 2643 3883	100	3162.87 7/2 ⁻ 3002.64 5/2 ⁺ 1763.15 5/2 ⁺					
5655	(3/2) ⁺	5646 2652	<8 100 6	0.0 3/2 ⁺ 3002.64 5/2 ⁺		[M1, E2]		0.00058 5	B(M1)(W.u.)=0.076 +30-14 if M1, B(E2)(W.u.)=41 +16-7 if E2.
		3892	7.5 25	1763.15 5/2 ⁺		[M1, E2]		0.00107 7	B(M1)(W.u.)=0.0018 +10-6 if M1, B(E2)(W.u.)=0.45 +25-15 if E2.
5682.2	1/2 ⁻ , 3/2 ⁻	4435 1504.3	<7.5 100	1219.38 1/2 ⁺ 4177.9 3/2 ⁻		[M1, E2]		0.00125 8	
5723.52	5/2 ⁺	3029.38 3960.13	63 8 100 11	2694.00 3/2 ⁺ 1763.15 5/2 ⁺					
5758	(1/2, 3/2 ⁻)	1580 4538 5758	67 22 <13 100 22	4177.9 3/2 ⁻ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5806.6	(1/2, 3/2, 5/2)	5806.1	100	0.0 3/2 ⁺					
5830	(5/2 ⁻)	2667	100	3162.87 7/2 ⁻		D+Q			I _γ : from ($\alpha, p \gamma$). δ : -2.2 5 or -0.6 1 for J=5/2 from $\gamma(\theta)$ in ($\alpha, p \gamma$).
5926.65	(11/2) ⁻	1579.09 30	100 11	4347.76 9/2 ⁻		M1+E2	-0.2 1	0.0001074 22	E _γ : weighted average of 1578.9 5 from ($^{16}\text{O}, \alpha p \gamma$), 1579.15 30 from ($^{14}\text{N}, \alpha p n \gamma$), and 1579.3 15 from ($\alpha, p \gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult.[#]</u>	<u>δ[#]</u>	<u>α^a</u>	<u>Comments</u>
									Mult., δ : D+Q and δ from $\gamma\gamma(\theta)$ (DCO) in (²⁸ Si, $\alpha\gamma$) with $\Delta J=1$; E1+M2 ruled out by RUL. Other: -0.8 4 from $\gamma(\theta)$ in ($\alpha,\text{p}\gamma$).
5926.65	(11/2) ⁻	1983.7	11.1 22	3942.9	9/2 ⁺	[E1]		0.000639 9	I _{γ} : from (²⁸ Si, $\alpha\gamma$).
		2763.66		3162.87	7/2 ⁻				
6087.31	13/2 ⁻	680.35 10	100.0 21	5407.07	11/2 ⁻	M1+E2	+0.020 15	8.70×10 ⁻⁵ 12	B(M1)(W.u.)=0.0107 10 E _{γ} : weighted average of 680.4 1 from (²⁴ Mg, $\alpha\text{p}\gamma$), 680.5 5 from (¹⁶ O, $\alpha\text{p}\gamma$), and 680.22 15 from (¹⁴ N, $\alpha\text{p}\text{n}\gamma$). Other: 679.9 10 from ($\alpha,\text{p}\gamma$). I _{γ} : weighted average of 100.0 31 from (²⁸ Si, $\alpha\text{p}\gamma$), 100.0 21 from (¹⁶ O, $\alpha\text{p}\gamma$), and 100 6 from ($\alpha,\text{p}\gamma$). Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ($\alpha,\text{p}\gamma$). Other: $\delta=+0.1$ 1 from $\gamma\gamma(\text{DCO})$ in (²⁸ Si, $\alpha\text{p}\gamma$), $\Delta J=1$.
		1739.4 4	6.9 9	4347.76	9/2 ⁻	E2		0.0002076 29	B(E2)(W.u.)=0.054 +9-8 E _{γ} : weighted average of 1739.6 5 from (¹⁶ O, $\alpha\text{p}\gamma$) and 1739.3 4 from (¹⁴ N, $\alpha\text{p}\text{n}\gamma$). I _{γ} : weighted average of 6.3 6 from (²⁸ Si, $\alpha\text{p}\gamma$), 7.9 9 from (¹⁶ O, $\alpha\text{p}\gamma$), and 18 6 from ($\alpha,\text{p}\gamma$). Mult.: from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\text{lin pol})$ in (²⁸ Si, $\alpha\text{p}\gamma$), $\Delta J=2$.
6106.2	(3/2,5/2 ⁺)	4342.8	10.3 35	1763.15	5/2 ⁺				
		4886.5	62 17	1219.38	1/2 ⁺				
		6105.6	100 17	0.0	3/2 ⁺				
6181	(1/2 ⁺ to 7/2 ⁺)	6180	100	0.0	3/2 ⁺				
6380.2	(9/2 ⁻)	296 1		6087.31	13/2 ⁻				I _{γ} : from (¹⁶ O, $\alpha\text{p}\gamma$).
		971 1		5407.07	11/2 ⁻				I _{γ} : from (¹⁶ O, $\alpha\text{p}\gamma$).
6492	(3/2 ⁺)	6491	100	0.0	3/2 ⁺				
6660.2	(11/2 ⁻)	2312.4		4347.76	9/2 ⁻				
		3497.1		3162.87	7/2 ⁻				
7066.2	(5/2 ⁺)	1902.8	1.8	5163.34	7/2 ⁻				
		2892.6	1.0	4173.43	(5/2 ⁺)				
		2953.7	1.2	4112.4	7/2 ⁺				
		3007.0	2.9	4059.1	(3/2) ⁻				
		3903.1	12	3162.87	7/2 ⁻				
		4063.3	6.1	3002.64	5/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[#]</u>	<u>δ[#]</u>
7066.2	(5/2 ⁺)	4371.9	10	2694.00	3/2 ⁺		
		4420.2	3.3	2645.67	7/2 ⁺		
		5302.6	29	1763.15	5/2 ⁺		
		5846.3	37	1219.38	1/2 ⁺		
		7065.4	100	0.0	3/2 ⁺		
7103.4	3/2 ⁽⁻⁾	2925.4	2.3	4177.9	3/2 ⁻		
		3184.8	2.3	3918.47	3/2 ⁺		
		4100.5	4.7	3002.64	5/2 ⁺		
		4409.1	17	2694.00	3/2 ⁺	D(+Q)	0.00 3
		5339.8	6.3	1763.15	5/2 ⁺		
		5883.5	100	1219.38	1/2 ⁺	D(+Q)	0.00 3
		7102.6	23	0.0	3/2 ⁺		
7178.6	1/2 ⁽⁺⁾	1072.4	7.9	6106.2	(3/2,5/2 ⁺)		
		1775.3	4.5	5403.3	3/2 ⁻		
		2339.4	4	4839.10	(1/2 ⁺ ,3/2)		
		3005.0	2.1	4173.43	(5/2 ⁺)		
		3119.4	58	4059.1	(3/2) ⁻		
		3210.8	24	3967.6	1/2 ⁺		
		3260.0	7.9	3918.47	3/2 ⁺		
		4484.3	11	2694.00	3/2 ⁺		
		5958.7	45	1219.38	1/2 ⁺		
		7177.8	100	0.0	3/2 ⁺		
		1970	<1.1	5214.5	(5/2 ⁺)		
		2176	<1.1	5009.1	(1/2,3/2)		
7185	5/2 ⁺	3011	<11	4173.43	(5/2 ⁺)		
		3126	57 13	4059.1	(3/2) ⁻		
		4022	<6.5	3162.87	7/2 ⁻		
		4182	<6.5	3002.64	5/2 ⁺		
		4491	4.4 22	2694.00	3/2 ⁺		
		4539	13 5	2645.67	7/2 ⁺		
		5421	9 7	1763.15	5/2 ⁺		
		5965	35 15	1219.38	1/2 ⁺		
		7184	100 22	0.0	3/2 ⁺		
		2185.4	1.2	5009.1	(1/2,3/2)		
		2340.1	3.0	4854.4	(1/2,3/2)		
		2355.4	2.7	4839.10	(1/2 ⁺ ,3/2)		
7194.6	1/2 ⁻	3016.6	24	4177.9	3/2 ⁻		
		3135.4	4.5	4059.1	(3/2) ⁻		
		3226.8	2.7	3967.6	1/2 ⁺		
		3276.0	6	3918.47	3/2 ⁺		
		4500.3	1.6	2694.00	3/2 ⁺		
		5431.0	0.75	1763.15	5/2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
7194.6	1/2 ⁻	5974.7	100	1219.38	1/2 ⁺			
		7193.8	3.0	0.0	3/2 ⁺			
7225.5	5/2	1119.3	3.1	6106.2	(3/2,5/2 ⁺)			
		1418.9		5806.6	(1/2,3/2,5/2)			
		1543.3	3.4	5682.2	1/2 ⁻ ,3/2 ⁻			
		1571	1.4	5655	(3/2) ⁺			
		1822.2	1.5	5403.3	3/2 ⁻			
		2010.9	12	5214.5	(5/2 ⁺)			
		3047.5	14	4177.9	3/2 ⁻			
		3166.3	3.4	4059.1	(3/2) ⁻			
		3306.9	2.5	3918.47	3/2 ⁺			
		4222.6		3002.64	5/2 ⁺			
		4531.2	17	2694.00	3/2 ⁺	D+Q	+0.36 4	
		5461.9	12	1763.15	5/2 ⁺	D+Q	-0.36 7	
		7224.7	100	0.0	3/2 ⁺	D+Q	+0.36 4	
7234.0	5/2 ⁽⁺⁾	1579	0.22	5655	(3/2) ⁺			
		3056.0		4177.9	3/2 ⁻			
		4539.7	1.9	2694.00	3/2 ⁺			
		4588.0	3.2	2645.67	7/2 ⁺			
		5470.4	0.22	1763.15	5/2 ⁺			
		6014.1	1.9	1219.38	1/2 ⁺			
		7233.2	100	0.0	3/2 ⁺	D+Q	-0.10	
7272.7	1/2 ⁻	1515	1.7	5758	(1/2,3/2 ⁻)			
		1869.4	0.29	5403.3	3/2 ⁻			
		2263.5	2.0	5009.1	(1/2,3/2)			
		2433.5	1.5	4839.10	(1/2 ⁺ ,3/2)			
		3099.1	0.58	4173.43	(5/2 ⁺)			
		3213.4	1.3	4059.1	(3/2) ⁻			
		3304.9	1.2	3967.6	1/2 ⁺			
		4578.4	1.6	2694.00	3/2 ⁺			
		6052.8	35	1219.38	1/2 ⁺			
		7271.9	100	0.0	3/2 ⁺			
7361.9	3/2 ⁽⁺⁾	1707	0.41	5655	(3/2) ⁺			
		2352.7	0.82	5009.1	(1/2,3/2)			
		2522.7	1.4	4839.10	(1/2 ⁺ ,3/2)			
		3183.8	0.69	4177.9	3/2 ⁻			
		3302.6	2.7	4059.1	(3/2) ⁻			
		3394.1	4.1	3967.6	1/2 ⁺			
		3443.3	1.1	3918.47	3/2 ⁺			
		4359.0	0.41	3002.64	5/2 ⁺			
		4667.6	0.69	2694.00	3/2 ⁺			
		5598.3	11	1763.15	5/2 ⁺	D+Q	δ : >+2.5 or -0.01 10 from $\gamma(\theta)$ in (p, γ).	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7361.9	3/2 ⁽⁺⁾	6141.9	100	1219.38	1/2 ⁺	(M1+E2)			Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition. δ : -0.05 3 or $+2.5$ 3 in (p, γ). δ : $-0.16 < \delta(Q/D) < -0.12$ or $+7.5 < \delta(Q/D) < +10$ in (p, γ).
		7361.1	14	0.0	3/2 ⁺	D+Q			
7396.4	7/2 ⁽⁻⁾	3048.5	23	4347.76	9/2 ⁻				Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		3283.8	17	4112.4	7/2 ⁺				
		3453.3	11	3942.9	9/2 ⁺				
		4233.3	100	3162.87	7/2 ⁻	(M1+E2)	+0.28 25	1.12×10^{-3} 3	
		4393.5	30	3002.64	5/2 ⁺	D(+Q)	+0.011 18		
		4750.4	28	2645.67	7/2 ⁺				
		5632.8	4.3	1763.15	5/2 ⁺				
7450.9	3/2	3532.2	6.9 34	3918.47	3/2 ⁺				
		4448.0	13.7 14	3002.64	5/2 ⁺				
		4756.6	11.0 11	2694.00	3/2 ⁺				
		5687.3	5.5 28	1763.15	5/2 ⁺				
		6230.9	100 10	1219.38	1/2 ⁺				
7501.1	(5/2 ⁺ , 7/2, 9/2 ⁺)	3558.0	46 6	3942.9	9/2 ⁺				
		4498.2	23.6 24	3002.64	5/2 ⁺				
		4855.1	100 11	2645.67	7/2 ⁺				
		5737.5	12.7 13	1763.15	5/2 ⁺				
7502.9		1396.7 @	14 @	6106.2	(3/2, 5/2 ⁺)				
		1904.4 @	4 @	5598.4	3/2 ⁺ , 5/2 ⁺				
		2493.7 @	4 @	5009.1	(1/2, 3/2)				
		3324.8 @	6 @	4177.9	3/2 ⁻				
		3443.6 @	100 @	4059.1	(3/2) ⁻				
		3559.8 @	4 @	3942.9	9/2 ⁺				
		4339.7 @	8 @	3162.87	7/2 ⁻				
		4500.0 @	8 @	3002.64	5/2 ⁺				
		4856.9 @	14 @	2645.67	7/2 ⁺				
		5739.2 @	1.4 @	1763.15	5/2 ⁺				
		6282.9 @	36 @	1219.38	1/2 ⁺				
		7502.0 @	0.6 @	0.0	3/2 ⁺				
7518.8	7/2	1920.3	2.9 15	5598.4	3/2 ⁺ , 5/2 ⁺				
		1932.7	4.4 22	5586.0	5/2 ⁺				
		3170.9	2.9 15	4347.76	9/2 ⁻				
		3575.7	2.9 15	3942.9	9/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
7518.8	7/2	4355.6	100 10	3162.87	7/2 ⁻	D(+Q)	+0.1 2		
		4872.8	4.4 22	2645.67	7/2 ⁺				
		5755.1	29.4 30	1763.15	5/2 ⁺	D+Q	+0.098 22		
		1903	0.59 7	5646	(5/2,7/2,9/2 ⁺)				
		1962.8	0.11	5586.0	5/2 ⁺	[E1]		0.000624 9	
		2778.9	1.49 11	4769.86	(5/2 ⁻)	[M1,E2]		0.00064 6	
		4385.7	100.0 11	3162.87	7/2 ⁻	M1+E2	-0.07 2	1.16×10 ⁻³ 2	
		4545.9	2.13 11	3002.64	5/2 ⁺	(E1+M2)	+0.6 4	0.00163 22	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4902.9	0.72 8	2645.67	7/2 ⁺	[E1]		1.99×10 ⁻³ 3	
		5785.2	0.35 4	1763.15	5/2 ⁺	[E1]		2.24×10 ⁻³ 3	
7561.3	1/2,3/2	7548.0	0.287 32	0.0	3/2 ⁺	(M2)			Mult.: Q(+O) with $\delta(\text{O/Q})=0.1$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; E3 component is less likely based on RUL.
		3383.2	2.6	4177.9	3/2 ⁻				
		3502.0	17	4059.1	(3/2 ⁻)				
		3593.5	1.4	3967.6	1/2 ⁺				
		3642.6	4.6	3918.47	3/2 ⁺				
		4866.9	63	2694.00	3/2 ⁺				
		6341.3	100	1219.38	1/2 ⁺				
		7560.4	97	0.0	3/2 ⁺				
		2437.4	5.8	5163.34	7/2 ⁻	[E1]		0.000934 13	
		2719.7	6.1	4880.96	7/2 ⁽⁺⁾				
7600.8	5/2 ⁺	2761.6	6.1	4839.10	(1/2 ⁺ ,3/2)				
		2830.8	3.9	4769.86	(5/2 ⁻)				
		2976.4	5.5	4624.3	3/2,5/2				
		3422.7	9.1	4177.9	3/2 ⁻	[E1]		1.44×10 ⁻³ 2	
		3427.2		4173.43	(5/2 ⁺)				
		3541.5	6.1	4059.1	(3/2 ⁻)	[E1]		1.50×10 ⁻³ 2	
		4437.6	15	3162.87	7/2 ⁻	[E1]		1.85×10 ⁻³ 3	
		4597.8	24	3002.64	5/2 ⁺	M1+E2	+0.50 22	1.25×10 ⁻³ 3	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	3/2 ⁺	(M1+E2)		0.00139 8	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =no from level scheme.
		4954.8	5.8	2645.67	7/2 ⁺	[M1,E2]		0.00140 8	
7618.7	5/2 ⁺	5837.1	58	1763.15	5/2 ⁺	M1+E2	+0.31 7	1.57×10 ⁻³ 2	
		6380.8	3.3	1219.38	1/2 ⁺	[E2]			
		7599.9	100	0.0	3/2 ⁺	M1+E2	-0.19 4		
		1438	0.77	6181	(1/2 ⁺ to 7/2 ⁺)				
		2404.1	2.6	5214.5	(5/2 ⁺)				
		2455.3	0.9	5163.34	7/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)										
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. $\#$	$\delta^\#$	α^a	Comments	
7618.7	$5/2^+$	2779.5	0.9	4839.10	$(1/2^+, 3/2)$					
		3440.6	3.9	4177.9	$3/2^-$					
		3559.4	2.6	4059.1	$(3/2)^-$					
		4455.5	3.9	3162.87	$7/2^-$					
		5855.0	13	1763.15	$5/2^+$	M1+E2	+0.48 11	1.59×10^{-3} 3		
		7617.8	100	0.0	$3/2^+$	M1+E2	+2.2 3			
7656.6	$(1/2, 3/2, 5/2^+)$	1550.4	5.3 26	6106.2	$(3/2, 5/2^+)$					
		1974.3	1.8 9	5682.2	$1/2^-, 3/2^-$					
		2002	1.8 9	5655	$(3/2)^+$					
		2817.4	5.3 26	4839.10	$(1/2^+, 3/2)$					
		3478.5	1.8 9	4177.9	$3/2^-$					
		3597.3	56 6	4059.1	$(3/2)^-$					
		3688.8	3.5 18	3967.6	$1/2^+$					
		7655.7	100 11	0.0	$3/2^+$					
		2073.4	2.1	5598.4	$3/2^+, 5/2^+$					
7671.9	$5/2^+$	2508.5	7	5163.34	$7/2^-$					
		2790.8	2.1	4880.96	$7/2^{(+)}$					
		2901.9	1.8	4769.86	$(5/2^-)$					
		3324.0	11	4347.76	$9/2^-$					
		3498.3	8.8	4173.43	$(5/2^+)$					
		4508.7	3.2	3162.87	$7/2^-$					
		4668.9	16	3002.64	$5/2^+$	M1+E2	-1.4 5	0.00134 4		
		5025.8	23	2645.67	$7/2^+$	D(+Q)	+0.3 22			
		5908.2	100	1763.15	$5/2^+$	M1+E2	+0.86 14	1.64×10^{-3} 3		
		7671.0	1.4	0.0	$3/2^+$					
		1927		5758	$(1/2, 3/2^-)$					
7685.5	$3/2^-$	2087.0	2.9	5598.4	$3/2^+, 5/2^+$					
		2282.1	2.9	5403.3	$3/2^-$					
		2470.9	1	5214.5	$(5/2^+)$					
		2676.3	0.86	5009.1	$(1/2, 3/2)$					
		3507.4	4.3	4177.9	$3/2^-$					
		3511.9	5.7	4173.43	$(5/2^+)$					
		3626.2	8.6	4059.1	$(3/2)^-$					
		4682.5	14	3002.64	$5/2^+$	D(+Q)	-0.01 5			
		5921.8	1	1763.15	$5/2^+$					
		6465.5	1.4	1219.38	$1/2^+$					
		7684.6	100	0.0	$3/2^+$	D+Q	+6.0 7			
		Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);								

Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);
 $\Delta\pi$ =yes from level scheme; but E1+M2 with such a large $\delta(M2/E1)$ is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p, γ).

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>									
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_γ [†]</u>	<u>I_γ [‡]</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult. #</u>	<u>δ [#]</u>	<u>α ^a</u>	<u>Comments</u>
7693.9	(1/2 ⁺ , 3/2, 5/2)	2095.4	1.0 5	5598.4	3/2 ⁺ , 5/2 ⁺				
		3634.6	1.0 5	4059.1	(3/2) ⁻				
		4690.9	2.1 11	3002.64	5/2 ⁺				
		7693.0	100 11	0.0	3/2 ⁺				
7706.4	5/2 ⁺	2051	4.8	5655	(3/2) ⁺	M1+E2		0.00032 4	δ : <-2 or ≥ 13 from $\gamma(\theta)$ in (p, γ).
		2825.3	0.48	4880.96	7/2 ⁽⁺⁾				
		2867.2	1.6	4839.10	(1/2 ⁺ , 3/2)				
		3082.0	0.96	4624.3	3/2, 5/2				
		3528.3	2.4	4177.9	3/2 ⁻				
		3593.8	2.4	4112.4	7/2 ⁺				
		3787.7	0.72	3918.47	3/2 ⁺				
		4703.4	1.2	3002.64	5/2 ⁺				
		5060.3	4.8	2645.67	7/2 ⁺	M1+E2		0.00143 8	δ : -1.01 10 or ≥ 13 from (p, γ).
		5942.7	0.48	1763.15	5/2 ⁺				
		6486.4	0.6	1219.38	1/2 ⁺				
		7705.5	100	0.0	3/2 ⁺	D(+Q)	+0.1 1		
		2863.7	47	4880.96	7/2 ⁽⁺⁾				
		2974.8	4.7	4769.86	(5/2 ⁻)				
7744.8	7/2 ⁻	3632.2	14	4112.4	7/2 ⁺				
		3801.7	23	3942.9	9/2 ⁺	(E1)		1.60 $\times 10^{-3}$ 2	Mult.: D(+Q) with $\delta = -0.16$ 17 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2 ⁻	M1+E2	+0.18 8	1.23 $\times 10^{-3}$ 2	
		5098.7	100	2645.67	7/2 ⁺	(E1)		2.05 $\times 10^{-3}$ 3	Mult.: D(+Q) with $\delta = +0.1$ 30 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.15	5/2 ⁺				
		6524.8 ^b	1.4	1219.38	1/2 ⁺	[E3]			This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2 ⁺)	2122	1.3	5655	(3/2) ⁺				
		2613.7	2.8	5163.34	7/2 ⁻				
		2937.9	5	4839.10	(1/2 ⁺ , 3/2)				
		3007.1	7.5	4769.86	(5/2 ⁻)				
		3599.0	18	4177.9	3/2 ⁻				
		3717.8	2.8	4059.1	(3/2) ⁻				
		4613.9	5	3162.87	7/2 ⁻				
		4774.1	10	3002.64	5/2 ⁺				
		5082.7	23	2694.00	3/2 ⁺				
		5131.0	3.3	2645.67	7/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7777.1	(5/2 ⁺)	6013.4	100	1763.15	5/2 ⁺				
		6557.1	28	1219.38	1/2 ⁺				
		7776.2	45	0.0	3/2 ⁺				
7781.7	(5/2 ⁻)	1675.5	3.9	6106.2	(3/2,5/2 ⁺)				
		2183.2		5598.4	3/2 ⁺ ,5/2 ⁺				
		2378.3		5403.3	3/2 ⁻				
		2567.1	4.2	5214.5	(5/2 ⁺)				
		3011.7	19	4769.86	(5/2 ⁻)				
		3433.8	5.6	4347.76	9/2 ⁻				
		3608.1	100	4173.43	(5/2 ⁺)				
		3669.1	17	4112.4	7/2 ⁺				
		3722.4	3.3	4059.1	(3/2) ⁻				
		3863.0	31	3918.47	3/2 ⁺				
		4618.5	14	3162.87	7/2 ⁻				
		5135.6	22	2645.67	7/2 ⁺				
		6018.0	50	1763.15	5/2 ⁺				
		6561.7	3.3	1219.38	1/2 ⁺				
		7780.8	4.7	0.0	3/2 ⁺				
7797.2	1/2 ⁻	2039		5758	(1/2,3/2 ⁻)				
		2142	1.6	5655	(3/2) ⁺				
		2393.8	2.4	5403.3	3/2 ⁻				
		2788.0	2.4	5009.1	(1/2,3/2)				
		2942.7	1.6	4854.4	(1/2,3/2)				
		2958.0	1.3	4839.10	(1/2 ⁺ ,3/2)				
		3737.9	0.36	4059.1	(3/2) ⁻				
		6577.2	11	1219.38	1/2 ⁺				
		7796.3	100	0.0	3/2 ⁺				
7837.2	3/2 ⁻	1656 ^{&}	1.4 ^{&}	6181	(1/2 ⁺ to 7/2 ⁺)				
		2238.7 ^{&}	1.4 ^{&}	5598.4	3/2 ⁺ ,5/2 ⁺				
		2622.6 ^{&}	3.8 ^{&}	5214.5	(5/2 ⁺)				
		3659.1 ^{&}	76 ^{&}	4177.9	3/2 ⁻	(M1+E2)	-0.05 3	0.000917 13	Mult.: D+Q from (p, γ); $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=1.03 $\times 10^3$ 19 exceeds RUL=10. B(E2)(W.u.)=7 $\times 10^2$ +11-6 exceeds RUL=100.
		3663.6 ^{&}	8.1 ^{&}	4173.43	(5/2 ⁺)				
		3777.9 ^{&}	5.4 ^{&}	4059.1	(3/2) ⁻				
		3894.1 ^{&b}	1.4 ^{&}	3942.9	9/2 ⁺				This γ is a primary γ from the unresolved 7837+7839 doublet goes to 9/2 ⁺ , which is unlikely from a 3/2 ⁻ , 7837 level. It is likely from the (5/2 ⁺), 7839.0 level.
		3918.5 ^{&}	0.54 ^{&}	3918.47	3/2 ⁺				

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7837.2	$3/2^-$	4834.2 &	11 &	3002.64	$5/2^+$				
		6073.5 &	5.1 &	1763.15	$5/2^+$				
		6617.2 &	100 &	1219.38	$1/2^+$	(E1)			Mult.: D(+Q) with $\delta=+0.06$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 +10-8 exceeds RUL=0.1.
		7836.3 &	57 &	0.0	$3/2^+$	(E1)			Mult.: D(+Q) with $\delta=+0.02$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 +47-44 exceeds RUL=0.1.
7839.0	$(5/2^+)$	1658 &	1.4 &	6181	$(1/2^+ \text{ to } 7/2^+)$				
		2240.5 &	1.4 &	5598.4	$3/2^+, 5/2^+$				
		2624.4 &	3.8 &	5214.5	$(5/2^+)$				
		3660.9 &	76 &	4177.9	$3/2^-$				
		3665.4 &	8.1 &	4173.43	$(5/2^+)$				
		3779.7 &	5.4 &	4059.1	$(3/2)^-$				
		3895.9 &	1.4 &	3942.9	$9/2^+$				
		3920.3 &	0.54 &	3918.47	$3/2^+$				
		4836.0 &	11 &	3002.64	$5/2^+$				
		6075.3 &	5.1 &	1763.15	$5/2^+$				
		6619.0 &	100 &	1219.38	$1/2^+$				
		7838.1 &	57 &	0.0	$3/2^+$				
		7868.7							
7868.7	$(1/2^+, 3/2, 5/2^+)$	2654.1	1.6	5214.5	$(5/2^+)$				
		2859.5	4.9	5009.1	$(1/2, 3/2)$				
		3695.1	8.2	4173.43	$(5/2^+)$				
		6105.0	28	1763.15	$5/2^+$				
		6648.6	21	1219.38	$1/2^+$				
		7867.8	100	0.0	$3/2^+$				
		7873.03							
7873.03	$13/2^+$	1212.8		6660.2	$(11/2^-)$				E_γ : from ($^{28}\text{Si}, \alpha p \gamma$) only.
		1786.2 5	9.6 4	6087.31	$13/2^-$	D(+Q)	-0.6 6		I_γ : weighted average of 8.2 18 from ($^{28}\text{Si}, \alpha p \gamma$) and 9.7 4 from ($^{16}\text{O}, \alpha p \gamma$).
		1946.35 30	48.0 26	5926.65	$(11/2)^-$	E1+M2	+0.2 1	0.000594 22	Mult., δ : from $\gamma\gamma(\text{DCO})$ in ($^{28}\text{Si}, \alpha p \gamma$). E_γ : weighted average of 1946.2 5 from ($^{16}\text{O}, \alpha p \gamma$) and 1946.40 30 from ($^{14}\text{N}, \alpha p n \gamma$). I_γ : weighted average of 46 5 from ($^{28}\text{Si}, \alpha p \gamma$)

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[‡]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[#]</u>	<u>δ[#]</u>	<u>α^a</u>	<u>Comments</u>
7873.03	13/2 ⁺	2465.94 30	100.0 20	5407.07	11/2 ⁻	E1+M2	-0.25 10	0.00091 4	and 48.5 26 from (¹⁶ O, $\alpha\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, $\alpha\gamma\gamma$), $\Delta J=1$. E _{γ} : weighted average of 2466.2 5 from (¹⁶ O, $\alpha\gamma$) and 2465.85 30 from (¹⁴ N, $\alpha p\gamma$). I _{γ} : weighted average of 100 9 from (²⁸ Si, $\alpha\gamma$) and 100.0 20 from (¹⁶ O, $\alpha\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, $\alpha\gamma\gamma$), $\Delta J=1$.
7880.8	3/2 ⁺ ,5/2 ⁺	2226	40	5655	(3/2) ⁺				
		3041.6	5.7	4839.10	(1/2 ⁺ ,3/2)				
		3256.3	6.7	4624.3	3/2,5/2				
		3702.7	10	4177.9	3/2 ⁻				
		3768.2	4.3	4112.4	7/2 ⁺				
		3821.5	6.7	4059.1	(3/2) ⁻				
		3962.1	6.7	3918.47	3/2 ⁺				
		4877.8	37	3002.64	5/2 ⁺				
		5186.4	83	2694.00	3/2 ⁺				
		6117.1	100	1763.15	5/2 ⁺				
		7879.9	33	0.0	3/2 ⁺				
7899.2	(5/2)	2253	2.2	5646	(5/2,7/2,9/2 ⁺)				
		2735.8	1.3	5163.34	7/2 ⁻				
		3018.1	1.2	4880.96	7/2 ⁽⁺⁾				
		3274.7	0.74	4624.3	3/2,5/2				
		3721.1	12	4177.9	3/2 ⁻				
		3725.6	12	4173.43	(5/2 ⁺)				
		3839.9	4.4	4059.1	(3/2) ⁻				
		4736.0	5.9	3162.87	7/2 ⁻				
		5204.8	1.9	2694.00	3/2 ⁺				
		6135.5	100	1763.15	5/2 ⁺				
		7898.2	5.9	0.0	3/2 ⁺				
7923.3	(3/2 ⁺ ,5/2 ⁺)	3084.1	3.4	4839.10	(1/2 ⁺ ,3/2)				
		3153.3	4.9	4769.86	(5/2 ⁻)				
		3298.8	4.6	4624.3	3/2,5/2				
		3745.2	7.3	4177.9	3/2 ⁻				
		3749.7	12	4173.43	(5/2 ⁺)				
		3864.0	4.2	4059.1	(3/2) ⁻				
		3955.5	3.7	3967.6	1/2 ⁺				
		4004.6	2.9	3918.47	3/2 ⁺				
		4920.3	2.2	3002.64	5/2 ⁺				
		5228.9	27	2694.00	3/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	
7923.3	$(3/2^+, 5/2^+)$	5277.2	2.2	2645.67	$7/2^+$		
		6159.6	3.7	1763.15	$5/2^+$		
		6703.2	66	1219.38	$1/2^+$		
		7922.3	100	0.0	$3/2^+$		
7970.3	$(5/2^-)$	2324	15	5646	$(5/2, 7/2, 9/2^+)$		
		2384.2	15	5586.0	$5/2^+$		
		2755.7	15	5214.5	$(5/2^+)$		
		3089.2	50	4880.96	$7/2^{(+)}$		
		3200.3	25	4769.86	$(5/2^-)$		
		3622.3	40	4347.76	$9/2^-$		
		3792.2	30	4177.9	$3/2^-$		
		3857.7	30	4112.4	$7/2^+$		
		4807.1	50	3162.87	$7/2^-$		
		4967.3	25	3002.64	$5/2^+$		
		5324.2	100	2645.67	$7/2^+$		
		6206.6	100	1763.15	$5/2^+$		
		6750.2	5	1219.38	$1/2^+$		
		7987.8	$3/2^+$	1881.6	0.83	6106.2	$(3/2, 5/2^+)$
		2333	2.8	5655	$(3/2)^+$		
		3809.7	<1.3	4177.9	$3/2^-$		
		3814.2	<1.3	4173.43	$(5/2^+)$		
		3928.5	1.7	4059.1	$(3/2)^-$		
		6767.7	100	1219.38	$1/2^+$	M1+E2	$\delta: +0.21$ 4 or -3.1 4 from (p, γ).
		7986.8	60	0.0	$3/2^+$	M1+E2	$\delta: -0.36$ 2 or -11 3 from (p, γ).
7995.6	$(5/2)$	2188.9	0.77	5806.6	$(1/2, 3/2, 5/2)$		
		2397.1	1.4	5598.4	$3/2^+, 5/2^+$		
		2592.2	3.9	5403.3	$3/2^-$		
		2832.1	1.9	5163.34	$7/2^-$		
		3817.5	5.1	4177.9	$3/2^-$		
		3936.3	1.4	4059.1	$(3/2)^-$		
		4832.4	9	3162.87	$7/2^-$		
		4992.6	2.6	3002.64	$5/2^+$		
		5301.2	0.26	2694.00	$3/2^+$		
		5349.5	0.77	2645.67	$7/2^+$		
		6231.9	1.2	1763.15	$5/2^+$		
		7994.6	100	0.0	$3/2^+$		
8001.0	$(7/2^+)$	2355	2.6	5646	$(5/2, 7/2, 9/2^+)$		
		2786.4	1.1	5214.5	$(5/2^+)$		
		3119.9	1	4880.96	$7/2^{(+)}$		
		3231.0	11	4769.86	$(5/2^-)$		
		3653.0	2.7	4347.76	$9/2^-$		
		3827.4	14	4173.43	$(5/2^+)$		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)											
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments		
8001.0	(7/2 ⁺)	4057.9	4.3	3942.9	9/2 ⁺						
		4837.8	1.1	3162.87	7/2 ⁻						
		5354.9	2.3	2645.67	7/2 ⁺						
		6237.3	100	1763.15	5/2 ⁺						
		8000.0	2	0.0	3/2 ⁺						
8005.0	5/2 ⁺	2350	3.3	5655	(3/2) ⁺						
		3123.9	1.1	4880.96	7/2 ⁽⁺⁾						
		3165.8	5.5	4839.10	(1/2 ⁺ , 3/2)						
		3826.9	<2.4	4177.9	3/2 ⁻				I_γ : for decays to 4173 and/or 4178 levels.		
		3831.3	<2.4	4173.43	(5/2 ⁺)				I_γ : for decays to 4173 and/or 4178 levels.		
		4841.8	27	3162.87	7/2 ⁻						
		5002.0	3.6	3002.64	5/2 ⁺						
		5310.6	3.1	2694.00	3/2 ⁺						
		5358.9	7.3	2645.67	7/2 ⁺						
		6241.3	27	1763.15	5/2 ⁺						
		6784.9	1.1	1219.38	1/2 ⁺						
		8004.0	100	0.0	3/2 ⁺						
		8035.7	1/2 ⁻ , 3/2 ⁻	2229.0	10.7 11	5806.6	(1/2, 3/2, 5/2)				
				3857.6	26.8 27	4177.9	3/2 ⁻				
3976.4	19.6 20			4059.1	(3/2) ⁻						
5341.3	9 5			2694.00	3/2 ⁺						
6272.0	9 5			1763.15	5/2 ⁺						
6815.6	100 11			1219.38	1/2 ⁺						
8034.7	3.6 18			0.0	3/2 ⁺						
8038.4	1/2 ⁺ , 3/2 ⁺	3183.8	1.1	4854.4	(1/2, 3/2)						
		3199.1	2.8	4839.10	(1/2 ⁺ , 3/2)						
		3413.9	5.3	4624.3	3/2, 5/2						
		3860.3	2.8	4177.9	3/2 ⁻						
		3979.1	5.3	4059.1	(3/2) ⁻						
		4070.6	1.4	3967.6	1/2 ⁺						
		4119.7	2.5	3918.47	3/2 ⁺						
		5035.4	7	3002.64	5/2 ⁺						
		5344.0	100	2694.00	3/2 ⁺	M1+E2	-0.20 2	1.44×10 ⁻³ 2	Mult., δ : D+Q with δ =-0.20 2 or +19 8 from (p, γ); E1+M2 and the larger δ (E2/M1) is ruled out by RUL. B(M1)(W.u.)=52.1 +43-37 exceeds RUL=10. B(E2)(W.u.)=2.8×10 ² +6-5 exceeds RUL=100.		
		6274.7	26	1763.15	5/2 ⁺						
		6818.3	7	1219.38	1/2 ⁺						
		8037.4	14	0.0	3/2 ⁺	M1+E2	-0.13 4		B(M1)(W.u.)=2.20 44; B(E2)(W.u.)=2.2 +16-12		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a
Comments							
Mult., δ : D+Q with $\delta=-0.13$ 4 or $+8.1$ 26 from (p, γ); E1+M2 and the larger $\delta(\text{E2/M1})$ is ruled out by RUL.							
8075.9	(5/2)	2430	20	5646	(5/2,7/2,9/2 ⁺)		
		2489.8	20	5586.0	5/2 ⁺		
		2861.3	14	5214.5	(5/2 ⁺)		
		2912.4		5163.34	7/2 ⁻		
		3194.8		4880.96	7/2 ⁽⁺⁾		
		3305.9	7.1	4769.86	(5/2 ⁻)		
		3897.8	14	4177.9	3/2 ⁻		
		3963.3	3.6	4112.4	7/2 ⁺		
		5072.9	3.6	3002.64	5/2 ⁺		
		6312.1	100	1763.15	5/2 ⁺		
		8074.9		0.0	3/2 ⁺		
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2)		
		2450	13	5646	(5/2,7/2,9/2 ⁺)		
		2497.6	13	5598.4	3/2 ⁺ ,5/2 ⁺		
		2881.5	3.3	5214.5	(5/2 ⁺)		
		2932.6	5.1	5163.34	7/2 ⁻		
		3256.8	7.7	4839.10	(1/2 ⁺ ,3/2)		
		3326.1	5.1	4769.86	(5/2 ⁻)		
		3918.0	4.9	4177.9	3/2 ⁻		
		4036.8	5.1	4059.1	(3/2) ⁻		
		4177.4	4.6	3918.47	3/2 ⁺		
		4932.9	13	3162.87	7/2 ⁻		
		5401.7	13	2694.00	3/2 ⁺		
		5450.0	44	2645.67	7/2 ⁺		
		6332.3	21	1763.15	5/2 ⁺		
8106.4	3/2 ⁺	8095.1	100	0.0	3/2 ⁺		
		2507.9	1.4	5598.4	3/2 ⁺ ,5/2 ⁺		
		3251.8	1.6	4854.4	(1/2,3/2)		
		3267.1	1.4	4839.10	(1/2 ⁺ ,3/2)		
		3928.3	4.3	4177.9	3/2 ⁻		
		4047.1	3.5	4059.1	(3/2) ⁻		
		4138.5	8.1	3967.6	1/2 ⁺		
		5103.4	24	3002.64	5/2 ⁺		
		5412.0	24	2694.00	3/2 ⁺	M1+E2	0.00153 8 δ : +4.9 10 or -0.05 4.
		6342.6	8.1	1763.15	5/2 ⁺		
		6886.3	95	1219.38	1/2 ⁺		
		8105.4	100	0.0	3/2 ⁺	M1+E2	δ : +1.2 1 or -0.46 4.
8113.3	(1/2 ⁺ ,3/2,5/2 ⁺)	1621	19	6492	(3/2 ⁺)		
		2007.0	19	6106.2	(3/2,5/2 ⁺)		
		2306.6	3.4	5806.6	(1/2,3/2,5/2)		

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8113.3	$(1/2^+, 3/2, 5/2^+)$	2458	2.8	5655	$(3/2)^+$
		2514.8		5598.4	$3/2^+, 5/2^+$
		2898.7	5.6	5214.5	$(5/2^+)$
		3939.6	22	4173.43	$(5/2^+)$
		4054.0	4.1	4059.1	$(3/2)^-$
		4145.4	16	3967.6	$1/2^+$
		5110.3	5.9	3002.64	$5/2^+$
		5418.9	41	2694.00	$3/2^+$
		6349.5	16	1763.15	$5/2^+$
		6893.2	100	1219.38	$1/2^+$
		8112.3	78	0.0	$3/2^+$
8147.2	$3/2^-$	2464.9	1.6	5682.2	$1/2^-, 3/2^-$
		3138.0	4.8	5009.1	$(1/2, 3/2)$
		3969.1	13	4177.9	$3/2^-$
		4087.8	4.8	4059.1	$(3/2)^-$
		4179.3	3.2	3967.6	$1/2^+$
		4228.5	4.8	3918.47	$3/2^+$
		5452.7	4.8	2694.00	$3/2^+$
		6927.1	24	1219.38	$1/2^+$
		8146.2	100	0.0	$3/2^+$
8156.8	$(5/2^+)$	2511	7.8	5646	$(5/2, 7/2, 9/2^+)$
		2570.7	16	5586.0	$5/2^+$
		2942.2	12	5214.5	$(5/2^+)$
		3386.8	2.0	4769.86	$(5/2^-)$
		3978.7	9.8	4177.9	$3/2^-$
		4044.2	14	4112.4	$7/2^+$
		4097.4	2.0	4059.1	$(3/2)^-$
		4213.6	3.9	3942.9	$9/2^+$
		4993.6	9.8	3162.87	$7/2^-$
		5153.8	12	3002.64	$5/2^+$
		5510.7	100	2645.67	$7/2^+$
		6393.0	2.0	1763.15	$5/2^+$
		8155.8	5.9	0.0	$3/2^+$
8179.4	$(3/2^-, 5/2, 7/2^+)$	3554.9	13.2 <i>I3</i>	4624.3	$3/2, 5/2$
		5016.1	11.8 <i>I2</i>	3162.87	$7/2^-$
		5176.4	5.9 <i>30</i>	3002.64	$5/2^+$
		5484.9	4.4 <i>22</i>	2694.00	$3/2^+$
		6415.6	100 <i>I0</i>	1763.15	$5/2^+$
		8178.4	11.8 <i>I2</i>	0.0	$3/2^+$
8208.2	$5/2^+$	2553	0.9	5655	$(3/2)^+$
		3327.1	1.0	4880.96	$7/2^{(+)}$
		3368.9	0.51	4839.10	$(1/2^+, 3/2)$

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
8208.2	$5/2^+$	4148.8	1.9	4059.1	$(3/2)^-$				
		4289.5		3918.47	$3/2^+$				
		5205.1	1.4	3002.64	$5/2^+$				
		5513.7	0.64	2694.00	$3/2^+$				
		6444.4	18	1763.15	$5/2^+$	M1+E2	+0.33 4		B(M1)(W.u.)=0.75 +35-22; B(E2)(W.u.)=7.5 +41-25
		6988.1	3.9	1219.38	$1/2^+$				
		8207.2	100	0.0	$3/2^+$	M1+E2	-0.36 4		B(M1)(W.u.)=2.0 +9-5; B(E2)(W.u.)=15 +7-4
		8216.3	$5/2^+$	2561	$(3/2)^+$				
		3446.3		4769.86	$(5/2)^-$				
		4038.2		4177.9	$3/2^-$				
		4042.6		4173.43	$(5/2^+)$				
		5053.0		3162.87	$7/2^-$	(E1)		2.04×10^{-3} 3	B(E1)(W.u.)=0.062 +21-12 Mult.: D+Q with $\delta=+0.11$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi=\text{yes}$ from level scheme; but M2 component is unlikely for this strong primary transition.
		5213.2		3002.64	$5/2^+$				
		5521.8		2694.00	$3/2^+$				
		6452.5		1763.15	$5/2^+$				
		8215.3		0.0	$3/2^+$	M1+E2	+0.06 3		B(M1)(W.u.)=0.55 +18-9; B(E2)(W.u.)=0.11 +16-8
		8242.4		4173.43	$(5/2^+)$				
		5239.3		3002.64	$5/2^+$				
		7022.3		1219.38	$1/2^+$				
		8251		4138	$7/2^+$				
		5248		3002.64	$5/2^+$				
		7031		1219.38	$1/2^+$				
		8250		0.0	$3/2^+$				
		8269.0		4769.86	$(5/2)^-$				
		4090.8		4177.9	$3/2^-$				
		4350.2		3918.47	$3/2^+$				
8251	$(3/2^+, 5/2^+)$	5105.7	16	3162.87	$7/2^-$				
		5265.9	29	3002.64	$5/2^+$				
		5574.5	13	2694.00	$3/2^+$				
		5622.9	8.9	2645.67	$7/2^+$				
		6505.2	18	1763.15	$5/2^+$				
		8268.0	100	0.0	$3/2^+$				
		8277.2	$(5/2)^+$	2622	$(3/2)^+$				
		3396.1		4880.96	$7/2^{(+)}$				
		3437.9		4839.10	$(1/2^+, 3/2)$				
		3507.2		4769.86	$(5/2)^-$				
		3652.7		4624.3	$3/2, 5/2$				
		3929.2		4347.76	$9/2^-$				
		4103.5		4173.43	$(5/2^+)$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]
8277.2	$(5/2)^+$	4217.8	5.2	4059.1	$(3/2)^-$	
		4358.4	17	3918.47	$3/2^+$	
		5274.1	35	3002.64	$5/2^+$	
		5582.7	10	2694.00	$3/2^+$	
		5631.0	10	2645.67	$7/2^+$	
		6513.4	79	1763.15	$5/2^+$	
		8276.2	100	0.0	$3/2^+$	
		2696.3	29	5586.0	$5/2^+$	
		3067.8	29	5214.5	$(5/2^+)$	
		3118.9	11	5163.34	$7/2^-$	
8282.4	$(3/2^-, 5/2)$	4104.2	50	4177.9	$3/2^-$	
		4223.0	21	4059.1	$(3/2)^-$	
		5119.1	93	3162.87	$7/2^-$	
		6518.6	25	1763.15	$5/2^+$	
		8281.4	100	0.0	$3/2^+$	
		2633	4.6	5655	$(3/2)^+$	
		3278.6	6.8	5009.1	$(1/2, 3/2)$	
		4109.7	21	4177.9	$3/2^-$	
		4228.5	4.6	4059.1	$(3/2)^-$	
		4319.9	6.8	3967.6	$1/2^+$	
8287.82	$1/2^-$	4369.06	2.3	3918.47	$3/2^+$	
		5593.3	52	2694.00	$3/2^+$	
		7067.67	100	1219.38	$1/2^+$	(E1)
						B(E1)(W.u.)=0.069 +24-14
						Mult.: D+Q with $\delta(Q/D)=+0.21$ 2 or -3.2 2 in (p, γ); but M2 component is unlikely for this primary transition based on resonance $\Gamma=0.04$ keV.
		8286.77	30	0.0	$3/2^+$	
		3288.9	13	5009.1	$(1/2, 3/2)$	
		3417.1	17	4880.96	$7/2^{(+)}$	
		3443.6	21	4854.4	$(1/2, 3/2)$	
		4124.5	25	4173.43	$(5/2^+)$	
8298.2	$3/2^+$	4238.8	42	4059.1	$(3/2)^-$	
		5134.9	4.2	3162.87	$7/2^-$	
						This γ is questionable if $J^\pi(8298)=3/2^+$, since it would require Mult=M2 which is probably unlikely; however $J^\pi(8298)=3/2^-$ would make the strong 7078.0 γ and 8297.1 γ less likely.
		5295.1	4.2	3002.64	$5/2^+$	
		5603.7	42	2694.00	$3/2^+$	
		5652.0	25	2645.67	$7/2^+$	
		6534.4	33	1763.15	$5/2^+$	
		7078.1	100	1219.38	$1/2^+$	M1+E2
		8297.1	54	0.0	$3/2^+$	M1+E2
						δ : -1.7 1 or -0.02 2 from $\gamma(\theta)$ in (p, γ).
8318.1	$(5/2^+)$					δ : -0.15 4 or $+9.5$ 36 from $\gamma(\theta)$ in (p, γ).
		2511.4	0.9	5806.6	$(1/2, 3/2, 5/2)$	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8318.1	$(5/2^+)$	2672	0.9	5646	$(5/2, 7/2, 9/2^+)$				
		2719.6	1.9	5598.4	$3/2^+, 5/2^+$				
		3548.1	0.39	4769.86	$(5/2^-)$				
		3693.6	2.3	4624.3	$3/2, 5/2$				
		4139.9	<3.9	4177.9	$3/2^-$				
		4144.4	<3.9	4173.43	$(5/2^+)$				
		4258.7	1.5	4059.1	$(3/2)^-$				
		4350.2	3.9	3967.6	$1/2^+$				
		5154.8	0.64	3162.87	$7/2^-$				
		5315.0	3.9	3002.64	$5/2^+$				
		5623.6	2.1	2694.00	$3/2^+$				
		5671.9	1.2	2645.67	$7/2^+$				
		6554.3	2.3	1763.15	$5/2^+$				
		7098.0	2.6	1219.38	$1/2^+$				
		8317.0	100	0.0	$3/2^+$				
8319.6	$15/2^-$	2232.7 6	57.2 28	6087.31	$13/2^-$	[M1,E2]		0.00040 4	E_γ, I_γ : from ($^{16}\text{O}, \alpha p \gamma$). B(M1)(W.u.)=0.0577 4I if M1, B(E2)(W.u.)=43.8 3I if E2.
		2911.9 8	100 5	5407.07	$11/2^-$	E2		0.000753 1I	B(E2)(W.u.)=20.3 12 $E_\gamma, I_\gamma, \text{Mult.}$: from ($^{16}\text{O}, \alpha p \gamma$). Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{DCO})$; M2 ruled out by RUL.
8323.1	$(3/2^+, 5/2^+)$	4210.4	1.2	4112.4	$7/2^+$				
		5320.0	3.7	3002.64	$5/2^+$				
		5676.9	4.9	2645.67	$7/2^+$				
		6559.3	<2.4	1763.15	$5/2^+$				
		7103.0	9.8	1219.38	$1/2^+$				
		8322.0	100	0.0	$3/2^+$				
8381.8	$5/2^+$	3611.7		4769.86	$(5/2^-)$				
		3757.3		4624.3	$3/2, 5/2$				
		4269.1	23.5 24	4112.4	$7/2^+$				
		4463.0	71 7	3918.47	$3/2^+$	M1+E2	+0.15 3	1.19×10^{-3} 2	B(M1)(W.u.)=2.9 +13-7; B(E2)(W.u.)=13 +8-5
		5378.7	74 8	3002.64	$5/2^+$	M1+E2	+0.02 3	1.44×10^{-3} 2	B(M1)(W.u.)=1.8 +14-7
		5687.3	2.9 15	2694.00	$3/2^+$	M1+E2	-0.72 3I	0.00157 4	B(M1)(W.u.)=0.039 +32-2I; B(E2)(W.u.)=2.4 +23-18
		5735.6	15 8	2645.67	$7/2^+$				
		6618.0	100 10	1763.15	$5/2^+$	M1+E2	+0.06 3		B(M1)(W.u.)=1.3 +6-3; B(E2)(W.u.)=0.4 +6-3
		8380.7	9 5	0.0	$3/2^+$	D+Q	1.65		
		3632.8	16 8	4769.86	$(5/2^-)$				
8402.9	$5/2^-$	4224.7	29 16	4177.9	$3/2^-$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8402.9	$5/2^-$	4343.5	74 20	4059.1	$(3/2)^-$
		5708.4	32 16	2694.00	$3/2^+$
		5756.7	23 13	2645.67	$7/2^+$
		6639.1	48 13	1763.15	$5/2^+$
		8401.8	100 26	0.0	$3/2^+$
8405.8	$(5/2^-)$	3524.7	35 9	4880.96	$7/2^{(+)}$
		4232.1	3.7 19	4173.43	$(5/2^+)$
		4293.1	3.3 16	4112.4	$7/2^+$
		5402.7	35 9	3002.64	$5/2^+$
		6642.0	100 26	1763.15	$5/2^+$
		8404.7	56 14	0.0	$3/2^+$
		3407.4	14	5009.1	$(1/2,3/2)$
8416.7	$1/2^+$	4243.0	31	4173.43	$(5/2^+)$
		4473.5	8.6	3942.9	$9/2^+$
		5413.6	20	3002.64	$5/2^+$
		5770.5	23	2645.67	$7/2^+$
		6652.9	63	1763.15	$5/2^+$
		7196.5	26	1219.38	$1/2^+$
		8415.6	100	0.0	$3/2^+$
		2809	47 24	5655	$(3/2)^+$
		3249.6	34 17	5214.5	$(5/2^+)$
		3455.0	47 24	5009.1	$(1/2,3/2)$
8464.3	$(1/2^+,3/2,5/2^+)$	3609.7	19 10	4854.4	$(1/2,3/2)$
		3839.8	25 13	4624.3	$3/2,5/2$
		4286.1	31 15	4177.9	$3/2^-$
		4290.6	32 17	4173.43	$(5/2^+)$
		4496.4	31 15	3967.6	$1/2^+$
		4545.5	15 8	3918.47	$3/2^+$
		5461.2	100 24	3002.64	$5/2^+$
		5769.8	94 24	2694.00	$3/2^+$
		7244.1	53 24	1219.38	$1/2^+$
		8463.2	59 18	0.0	$3/2^+$
		2829	12 6	5655	$(3/2)^+$
		3603.2	15 8	4880.96	$7/2^{(+)}$
		3859.9	2.2 11	4624.3	$3/2,5/2$
8484.4	$(3/2)^+$	4516.5	2.6 13	3967.6	$1/2^+$
		4565.6	11 6	3918.47	$3/2^+$
		5481.3	15 9	3002.64	$5/2^+$
		5789.9	44 11	2694.00	$3/2^+$
		5838.2	6.1 31	2645.67	$7/2^+$
		6720.6	100 26	1763.15	$5/2^+$
		8483.3	9 5	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8486.2	$3/2^-$	2728	2.4 <i>13</i>	5758	(1/2,3/2 ⁻)				
		3861.7	3.9 <i>20</i>	4624.3	3/2,5/2				
		4308.0	10 5	4177.9	3/2 ⁻				
		4312.5	10 5	4173.43	(5/2 ⁺)				I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		4426.8	4.1 <i>22</i>	4059.1	(3/2) ⁻				I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		4518.3	15 9	3967.6	1/2 ⁺				
		5483.1	17 9	3002.64	5/2 ⁺				
		5791.7	41 <i>11</i>	2694.00	3/2 ⁺				
		6722.4	8 4	1763.15	5/2 ⁺				
		7266.0	14 7	1219.38	1/2 ⁺				
		8485.1	100 26	0.0	3/2 ⁺				
		2399.8 8	100.0 <i>34</i>	6087.31	13/2 ⁻	(M1(+E2))	+0.2 +3-2	0.000427 17	Mult., δ : D+Q and δ from $\gamma\gamma(\text{DCO})$ in ($^{28}\text{Si},\alpha p\gamma$); $\Delta\pi=\text{no}$ from level scheme.
		3080.0 7	31.8 20	5407.07	11/2 ⁻	E2		0.000825 12	$E_\gamma, I_\gamma, \text{Mult.}$: from ($^{16}\text{O},\alpha p\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
8514.3	$1/2^-$	3659.7	16 8	4854.4	(1/2,3/2)				
		4336.1	7 4	4177.9	3/2 ⁻				
		4454.9	100 26	4059.1	(3/2) ⁻				
		4546.4	85 22	3967.6	1/2 ⁺				
		4595.5	24 12	3918.47	3/2 ⁺				
		7294.1	37 <i>11</i>	1219.38	1/2 ⁺				
		8513.2	100 26	0.0	3/2 ⁺				
8558.2?	(13/2 ⁻)	2631.4 ^b	100	5926.65	(11/2) ⁻	D(+Q)	-0.2 +2-3		$E_\gamma, \text{Mult.}, \delta$: from ($^{28}\text{Si},\alpha p\gamma$); $\Delta\pi=\text{no}$ from level scheme; Mult=Q, $\Delta J=1$.
8571.9	(5/2) ⁺	2848.3	3.1 <i>16</i>	5723.52	5/2 ⁺				
		2917	1.9 <i>10</i>	5655	(3/2) ⁺				
		2973.4	1.0 6	5598.4	3/2 ⁺ , 5/2 ⁺				
		3408.4	2.6 <i>13</i>	5163.34	7/2 ⁻				
		3690.7	1.3 7	4880.96	7/2 ⁽⁺⁾				
		4398.2	1.0 6	4173.43	(5/2 ⁺)				
		4459.2	1.6 9	4112.4	7/2 ⁺				
		4653.1	1.6 9	3918.47	3/2 ⁺				
		5408.6	2.3 <i>12</i>	3162.87	7/2 ⁻				
		5568.8	6.4 <i>33</i>	3002.64	5/2 ⁺				
		5877.4	0.9 4	2694.00	3/2 ⁺				
		5925.7	11 6	2645.67	7/2 ⁺				
		7351.7	8 4	1219.38	1/2 ⁺				
		8570.8	100 <i>10</i>	0.0	3/2 ⁺				
8580.5	1/2 ⁺	2088	8 4	6492	(3/2 ⁺)				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8580.5	$1/2^+$	3741.2	2.4 13	4839.10	$(1/2^+, 3/2)$
		4402.3	8 4	4177.9	$3/2^-$
		4521.1	3.7 19	4059.1	$(3/2)^-$
		4661.7	4.3 22	3918.47	$3/2^+$
		5886.0	10 5	2694.00	$3/2^+$
		7360.3	100 11	1219.38	$1/2^+$
		8579.4	48 13	0.0	$3/2^+$
		2832	22 11	5758	$(1/2, 3/2^-)$
8590.1	$5/2^+$	2935	17 9	5655	$(3/2)^+$
		3820.0	73 18	4769.86	$(5/2^-)$
		4416.4	59 14	4173.43	$(5/2^+)$
		4477.4	16 8	4112.4	$7/2^+$
		5426.8	36 18	3162.87	$7/2^-$
		5587.0	100 27	3002.64	$5/2^+$
		5943.9	25 13	2645.67	$7/2^+$
		6826.2	64 18	1763.15	$5/2^+$
		8589.0	46 14	0.0	$3/2^+$
		6848.1	100 10	1763.15	$5/2^+$
8612.0	$3/2^+, 5/2^+$	7391.8	34 8	1219.38	$1/2^+$
		8610.9	66 16	0.0	$3/2^+$
		3732.5	2.9 14	4880.96	$7/2^{(+)}$
8613.7	$5/2^+$	4554.3	10 5	4059.1	$(3/2)^-$
		4670.5	7.1	3942.9	$9/2^+$
		4694.9	13 7	3918.47	$3/2^+$
		5610.6	4.5 23	3002.64	$5/2^+$
		5919.2	8 4	2694.00	$3/2^+$
		5967.5	41 11	2645.67	$7/2^+$
		6849.8	16	1763.15	$5/2^+$
		7393.5	3.6	1219.38	$1/2^+$
		8612.6	100 11	0.0	$3/2^+$
		2894.6	6.4 33	5723.52	$5/2^+$
8618.2	$3/2^+, 5/2^+$	2963	5.2 27	5655	$(3/2)^+$
		3763.6	4.2 21	4854.4	$(1/2, 3/2)$
		3993.7	4.6 24	4624.3	$3/2, 5/2$
		4650.3	9 5	3967.6	$1/2^+$
		4699.4	7 4	3918.47	$3/2^+$
		5615.1	49 12	3002.64	$5/2^+$
		5923.7	42 12	2694.00	$3/2^+$
		5972.0	5.2 27	2645.67	$7/2^+$
		6854.3	70 18	1763.15	$5/2^+$
		8617.1	100 24	0.0	$3/2^+$
		2906.5	0.6 4	5723.52	$5/2^+$
8630.1	$7/2^-$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8630.1	$7/2^-$	3466.6	17 9	5163.34	$7/2^-$	M1+E2	+0.60 10	0.000884 15	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		3860.0	3.0 15	4769.86	$(5/2^-)$				
		4282.1	26 7	4347.76	$9/2^-$	M1+E2	-0.184 14	1.13×10^{-3} 2	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		4456.4	8 4	4173.43	$(5/2^+)$				
		4517.4	21 11	4112.4	$7/2^+$	(E1+M2)	-0.06 2	1.86×10^{-3} 3	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4686.9	21 11	3942.9	$9/2^+$				
		5466.8	7 4	3162.87	$7/2^-$				
		5627.0	3.0 15	3002.64	$5/2^+$				
		5983.9	5.5 28	2645.67	$7/2^+$				
		6866.2	100 26	1763.15	$5/2^+$	(E1+M2)	-0.011 3		Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
8640	$(5/2^+)$	8629.0	1.1 7	0.0	$3/2^+$				
		4697	61	3942.9	$9/2^+$				
		5477	53	3162.87	$7/2^-$				
		5637	16	3002.64	$5/2^+$				
		5946	34	2694.00	$3/2^+$				
		6876	100	1763.15	$5/2^+$				
8641.4	$3/2^+, 5/2^+$	2986	10 6	5655	$(3/2^+)$				
		3055.3	9 5	5586.0	$5/2^+$				
		3237.9	17 9	5403.3	$3/2^-$				
		3802.1	9 5	4839.10	$(1/2^+, 3/2)$				
		3871.3	86 22	4769.86	$(5/2^-)$				
		4016.9	7 4	4624.3	$3/2, 5/2$				
		4463.2	50 14	4177.9	$3/2^-$				I $_\gamma$: final level is uncertain, either 4173 or 4178 (1976Sp08).
		4467.7	50 14	4173.43	$(5/2^+)$				
		4582.0	32 18	4059.1	$(3/2)^-$				
		4722.6	21 11	3918.47	$3/2^+$				
8686.2	$5/2^-$	5946.9	19 9	2694.00	$3/2^+$				
		8640.3	100 25	0.0	$3/2^+$				
		3100.1	2.8 15	5586.0	$5/2^+$				
		3282.7	4.1 21	5403.3	$3/2^-$				
		4508.0	2.6 13	4177.9	$3/2^-$				
		4626.8	4.6 24	4059.1	$(3/2)^-$				
		5522.9	4.9 25	3162.87	$7/2^-$				
		5991.7	3.7 19	2694.00	$3/2^+$				
		6922.3	6.3 32	1763.15	$5/2^+$				
		7466.0	2.5 13	1219.38	$1/2^+$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	α^a	Comments
8686.2	5/2 ⁻	8685.0	100 11	0.0	3/2 ⁺			
8696.9	3/2 ⁻	2890.2	3.6 19	5806.6	(1/2,3/2,5/2)			
		3098.4	6.0 30	5598.4	3/2 ⁺ ,5/2 ⁺			
		7476.7	15 4	1219.38	1/2 ⁺			
		8695.7	100 10	0.0	3/2 ⁺			
8750.9	3/2 ⁻	2944.2	4.2 21	5806.6	(1/2,3/2,5/2)			
		3347.4	9 5	5403.3	3/2 ⁻			
		3536.2	4.5 23	5214.5	(5/2 ⁺)			
		4572.7	4.0 21	4177.9	3/2 ⁻			
		4577.2	4.3 23	4173.43	(5/2 ⁺)			
		6056.3	5.7 28	2694.00	3/2 ⁺			
		6987.0	57 15	1763.15	5/2 ⁺			
		7530.7	3.8	1219.38	1/2 ⁺			
		8749.7	100 10	0.0	3/2 ⁺			
8779.8	3/2 ⁻	3181.2	16 8	5598.4	3/2 ⁺ ,5/2 ⁺			
		3565.1	24 12	5214.5	(5/2 ⁺)			
		4601.6	56 15	4177.9	3/2 ⁻			I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		4606.0	56 15	4173.43	(5/2 ⁺)			I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		4861.0	16 8	3918.47	3/2 ⁺			
		5776.7	100 26	3002.64	5/2 ⁺			
		6085.2	44 11	2694.00	3/2 ⁺			
		6133.6	48 11	2645.67	7/2 ⁺			
		7015.9	25 13	1763.15	5/2 ⁺			
		8778.6	41 11	0.0	3/2 ⁺			
8787.2	(5/2 ⁺)	3063.5	27 14	5723.52	5/2 ⁺			
		3623.7	76 20	5163.34	7/2 ⁻			
		4017.1	27 14	4769.86	(5/2 ⁻)			
		4609.0	36 20	4177.9	3/2 ⁻			
		5623.8	23 12	3162.87	7/2 ⁻			
		5784.1	80 20	3002.64	5/2 ⁺			
		7023.3	100 24	1763.15	5/2 ⁺			
		8786.0	32 16	0.0	3/2 ⁺			
8788.3	(15/2 ⁺)	915.4 4	100	7873.03	13/2 ⁺	(M1)	4.82×10 ⁻⁵ 7	E_γ ,Mult.: from (¹⁶ O, α γ), with Mult=D, $\Delta J=1$ from $\gamma\gamma$ (ADO); $\Delta\pi$ =(no) from level scheme.
8798.4	(3/2 ⁺ ,5/2 ⁺)	4739.0	13 7	4059.1	(3/2 ⁻)			
		5795.2	26 7	3002.64	5/2 ⁺			
		7034.5	11 6	1763.15	5/2 ⁺			
		7578.1	11 6	1219.38	1/2 ⁺			
		8797.2	100 10	0.0	3/2 ⁺			
8820.9	(1/2 ⁺ to 7/2 ⁺)	5817.7	17 10	3002.64	5/2 ⁺			
		6126.3	35 10	2694.00	3/2 ⁺			
		7057.0	40 10	1763.15	5/2 ⁺			

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	$\gamma(^{35}\text{Cl})$ (continued)		Comments
						Mult. #	α^a	
8820.9	(1/2 ⁺ to 7/2 ⁺)	8819.7	100 10	0.0	3/2 ⁺			
8824.2		5821.0	100 24	3002.64	5/2 ⁺			
		6129.6	16.2 8	2694.00	3/2 ⁺			
		7060.3	43 11	1763.15	5/2 ⁺			
		7603.9	60 16	1219.38	1/2 ⁺			
	1/2 ⁻	8823.0	51 14	0.0	3/2 ⁺			
8829.3		3174	30 15	5655	(3/2 ⁺)			
		3990.0	3.9 21	4839.10	(1/2 ⁺ , 3/2)			
		4059.2	4.6 24	4769.86	(5/2 ⁻)			
		4910.5	14 7	3918.47	3/2 ⁺			
		5826.1	67 18	3002.64	5/2 ⁺			
		6134.7	39 9	2694.00	3/2 ⁺			
		6183.0	100 24	2645.67	7/2 ⁺			
		7065.4	27 15	1763.15	5/2 ⁺			
		7609.0	8 4	1219.38	1/2 ⁺			
		8828.1	12 6	0.0	3/2 ⁺			
8833.1		3618.4	30 15	5214.5	(5/2 ⁺)			
		4063.0	11 6	4769.86	(5/2 ⁻)			
		4659.3	61 18	4173.43	(5/2 ⁺)			
		4720.4	20 11	4112.4	7/2 ⁺			
		4773.7	48 13	4059.1	(3/2 ⁻)			
	5/2 ⁻ , 7/2 ⁻	4914.3	48 13	3918.47	3/2 ⁺			
		5669.7	16 8	3162.87	7/2 ⁻			
		5829.9	29 15	3002.64	5/2 ⁺			
		6138.5	20 10	2694.00	3/2 ⁺			
		6186.8	100 26	2645.67	7/2 ⁺			
		7069.2	39 22	1763.15	5/2 ⁺			
		8831.9	13 7	0.0	3/2 ⁺			
8838.1		3674.6		5163.34	7/2 ⁻			
		3956.9		4880.96	7/2 ⁽⁺⁾			
		4068.0	13 7	4769.86	(5/2 ⁻)			
	17/2 ⁺	4490.0		4347.76	9/2 ⁻			
		4778.7		4059.1	(3/2 ⁻)			
		5674.7	46 11	3162.87	7/2 ⁻			
		7074.2	100 10	1763.15	5/2 ⁺			
8844.4		524.8		8319.6	15/2 ⁻			
		971.38 20	100.0 30	7873.03	13/2 ⁺	E2	6.16×10 ⁻⁵ 9	B(E2)(W.u.)=16.3 +37-26 E _γ : weighted average of 971.4 5 from (¹⁶ O,αpγ) and 971.38 20 from (¹⁴ N,αpny).
	5/2 ⁺	2757.0		6087.31	13/2 ⁻			
8856.2		3975.0	6.7 33	4880.96	7/2 ⁽⁺⁾			

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8856.2	$5/2^+$	4231.6	10 5	4624.3	$3/2, 5/2$
		4678.0	7 4	4177.9	$3/2^-$
		4937.4	22 6	3918.47	$3/2^+$
		5692.8	100 10	3162.87	$7/2^-$
		6161.6	5.5 28	2694.00	$3/2^+$
		6209.9		2645.67	$7/2^+$
		7092.3	14 8	1763.15	$5/2^+$
		8855.0	31 8	0.0	$3/2^+$
		4690.4	5.3 27	4177.9	$3/2^-$
		5865.4	33 8	3002.64	$5/2^+$
8868.6	$3/2^+, 5/2^+$	6174.0	25 6	2694.00	$3/2^+$
		7104.7	9 5	1763.15	$5/2^+$
		7648.3	33 8	1219.38	$1/2^+$
		8867.4	100 25	0.0	$3/2^+$
		3231	23 12	5655	$(3/2)^+$
		3287.5	25 13	5598.4	$3/2^+, 5/2^+$
		3671.4	9 4	5214.5	$(5/2^+)$
		3722.6	35 16	5163.34	$7/2^-$
		4116.0	17 9	4769.86	$(5/2^-)$
		4707.9	31 16	4177.9	$3/2^-$
8886.1	$(5/2)$	4967.3	16 8	3918.47	$3/2^+$
		5882.9	31 16	3002.64	$5/2^+$
		6239.8	100 27	2645.67	$7/2^+$
		7122.2	46 12	1763.15	$5/2^+$
		8884.9	54 16	0.0	$3/2^+$
		3307.1	4.3 22	5586.0	$5/2^+$
		4780.6	24 11	4112.4	$7/2^+$
		4950.0	11 6	3942.9	$9/2^+$
		5729.9	78 19	3162.87	$7/2^-$
		6198.7	100 24	2694.00	$3/2^+$
8893.3	$(5/2^+, 7/2^+)$	7129.4	51 14	1763.15	$5/2^+$
		8892.1	2.7 14	0.0	$3/2^+$
		2724	15 8	6181	$(1/2^+ \text{ to } 7/2^+)$
		3250	12 6	5655	$(3/2)^+$
		3895.5	13 7	5009.1	$(1/2, 3/2)$
		4050.2	21 11	4854.4	$(1/2, 3/2)$
		4065.5	16 8	4839.10	$(1/2^+, 3/2)$
		4726.6	62 17	4177.9	$3/2^-$
		4845.3	20 10	4059.1	$(3/2)^-$
		6210.2	52 14	2694.00	$3/2^+$
8904.8	$(1/2, 3/2, 5/2^+)$	7684.5	35 10	1219.38	$1/2^+$
		8903.6	100 24	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$\gamma(^{35}\text{Cl})$ (continued)		E_f	J_f^π	$\gamma(^{35}\text{Cl})$ (continued)		Comments
		E_γ †	I_γ ‡			Mult. #	δ #	
8906.8	$5/2^+$	3252		5655	$(3/2)^+$			
		3261	5.4 28	5646	$(5/2, 7/2, 9/2^+)$			
		3308.2	3.2 16	5598.4	$3/2^+, 5/2^+$			
		3320.6	5.8	5586.0	$5/2^+$			
		3376	3.6 19	5531				
		3692.1	6.4 32	5214.5	$(5/2^+)$			
		4136.7	8 4	4769.86	$(5/2^-)$			
		4963.5	19 5	3942.9	$9/2^+$			
		7142.9	100 10	1763.15	$5/2^+$			
8919.8	$(5/2^-, 7/2, 9/2^+)$	3274	10 5	5646	$(5/2, 7/2, 9/2^+)$			
		4571.7	14 7	4347.76	$9/2^-$			
		5756.4	100 11	3162.87	$7/2^-$			
		6273.5	46 12	2645.67	$7/2^+$			
		7155.9	7 4	1763.15	$5/2^+$			
8953.1	$3/2^+$	7732.8		1219.38	$1/2^+$	M1+E2	-0.34 2	Mult., δ : or $\delta=-5.0$ 4 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
		8951.9		0.0	$3/2^+$	M1+E2	-0.39 5	Mult., δ : or $\delta=-8.2$ 7 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
8984.2	$5/2^+$	3260.5	1.3 7	5723.52	$5/2^+$			
		4144.8	2.0 10	4839.10	$(1/2^+, 3/2)$			
		4214.1	0.8 5	4769.86	$(5/2^-)$			
		4359.6	0.8 5	4624.3	$3/2, 5/2$			
		4806.0	2.5 13	4177.9	$3/2^-$			
		4810.4	2.5 13	4173.43	$(5/2^+)$			
		5065.3	15 8	3918.47	$3/2^+$			
		5820.8	10 5	3162.87	$7/2^-$			
		6289.6	3.0 15	2694.00	$3/2^+$			
		6337.9	2.0 10	2645.67	$7/2^+$			
		7220.3	27 7	1763.15	$5/2^+$			
		8983.0	100 10	0.0	$3/2^+$			
		3357.8	1.7	5723.52	$5/2^+$			
		4200.3	1.7	4880.96	$7/2^{(+)}$			
9081.5	$5/2^+$	4242.1		4839.10	$(1/2^+, 3/2)$			
		4311.4		4769.86	$(5/2^-)$			
		4456.9		4624.3	$3/2, 5/2$			
		4903.2	3.3	4177.9	$3/2^-$			
		4907.7	3.3	4173.43	$(5/2^+)$			
		5162.6	15	3918.47	$3/2^+$	D(+Q)	-0.002 9	
		5918.1	10	3162.87	$7/2^-$			
		6386.9	3.3	2694.00	$3/2^+$			
		6435.2	1.7	2645.67	$7/2^+$			
		7317.5	27	1763.15	$5/2^+$	M1+E2	+0.11 2	B(M1)(W.u.)=1.16 +3/-25; B(E2)(W.u.)=0.99 +50/-38

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
9081.5	5/2 ⁺	9080.2	100	0.0	3/2 ⁺	M1+E2	+0.089 9		B(M1)(W.u.)=2.26 +49-33; B(E2)(W.u.)=0.82 +27-19
9157.1	5/2 ⁻	3942.4	17	5214.5	(5/2 ⁺)				
		4386.9	16	4769.86	(5/2 ⁻)				
		4978.8	30	4177.9	3/2 ⁻				I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		4983.3	30	4173.43	(5/2 ⁺)				I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
		5993.7	22	3162.87	7/2 ⁻	M1+E2	+0.10 2	1.59×10 ⁻³ 2	
		6153.9	20	3002.64	5/2 ⁺				
		6462.5	3.0	2694.00	3/2 ⁺				
		6510.8	6.7	2645.67	7/2 ⁺				
		7393.1	17	1763.15	5/2 ⁺				
		9155.8	100	0.0	3/2 ⁺	(E1)			Mult.: D+Q with $\delta=+0.09$ I from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely.
10181.1	19/2 ⁻	1336.3 5	100.0 34	8844.4	17/2 ⁺	(E1)		0.0001637 23	B(E1)(W.u.)=0.00235 +21-18 I_γ : weighted average of 100 17 from (²⁸ Si, $\alpha\gamma$) and 100.0 34 from (¹⁶ O, $\alpha\gamma$). E_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with $\Delta J=1$; $\Delta\pi$ =(yes) from level scheme.
		1693.5 5	84 9	8487.5	15/2 ⁻	E2		0.0001880 26	B(E2)(W.u.)=44.4 46 E_γ , I_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
		1861.3 5	83 6	8319.6	15/2 ⁻	E2		0.000262 4	B(E2)(W.u.)=27.4 +26-24 E_γ , I_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
10221.9	(17/2 ⁻)	1377.8 10	100	8844.4	17/2 ⁺				
10858.5	19/2 ⁺	2014.7 9	100 5	8844.4	17/2 ⁺	(M1)		0.000265 4	E_γ , I_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with $\Delta J=1$; $\Delta\pi$ =no from level scheme.
		2069.8 10	32.5 22	8788.3	(15/2 ⁺)	E2		0.000361 5	E_γ , I_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$) with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO); M2 ruled out by RUL.
11459.1	21/2 ⁺	2614.5 5	100	8844.4	17/2 ⁺	E2		0.000619 9	B(E2)(W.u.)=15.9 8 E_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO); M2 ruled out by RUL.
12572.2	23/2 ⁻	1113.3 5	24.4 18	11459.1	21/2 ⁺	(E1)		3.60×10 ⁻⁵ 5	B(E1)(W.u.)=0.00272 +41-35 E_γ , I_γ ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with $\Delta J=1$; $\Delta\pi$ =yes from level scheme.
		2390.8 5	100 4	10181.1	19/2 ⁻	E2		0.000515 7	B(E2)(W.u.)=26.0 +36-29

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}[†]</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
16306.4	(27/2 ⁻)	3734.0	12572.2	23/2 ⁻	E _{γ} , I _{γ} , Mult.: from (¹⁶ O, $\alpha\gamma\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO); M2 ruled out by RUL.

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p, γ), unless otherwise noted.

From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in (p, γ) for γ rays originating from low-spin levels ($J < 9/2$) and from $\gamma\gamma(\theta)$ (DCO) and γ (lin pol) in (²⁸Si, $\alpha\gamma\gamma$) for γ rays originating from high-spin ($J \geq 9/2$) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ (lin pol) or RUL with measured T_{1/2}, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p, γ), a D+Q with definite $\delta(Q/D) > 0$ from $\gamma(\theta)$ is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ . The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the ³⁴S(p, γ) dataset.

@ Primary transitions probably from the unresolved 7501+7503 doublet in [1976Me12](#) in (p, γ).

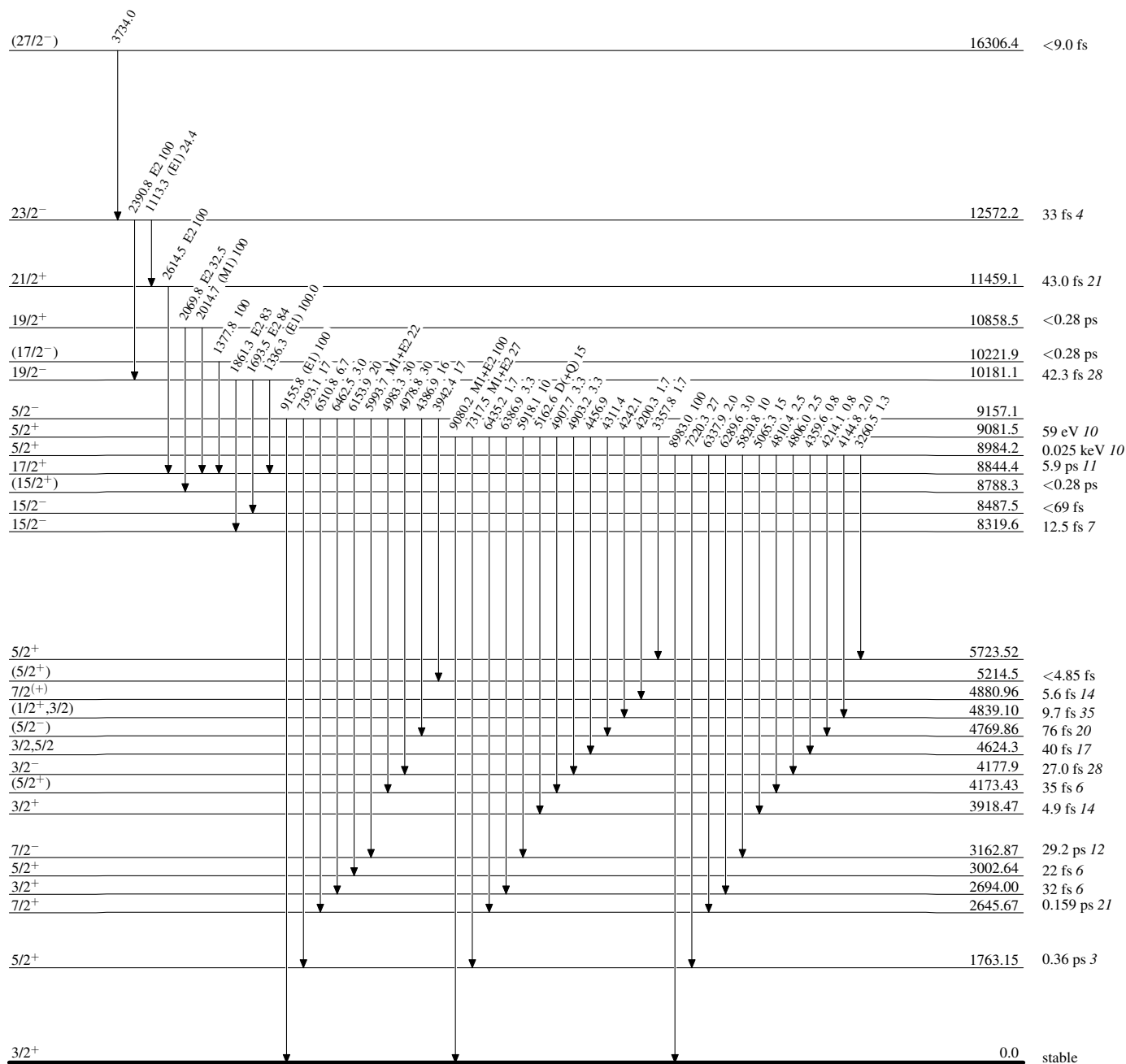
& Primary transitions probably from the unresolved 7837+7839 doublet in [1976Me12](#) in (p, γ).

^a Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^b Placement of transition in the level scheme is uncertain.

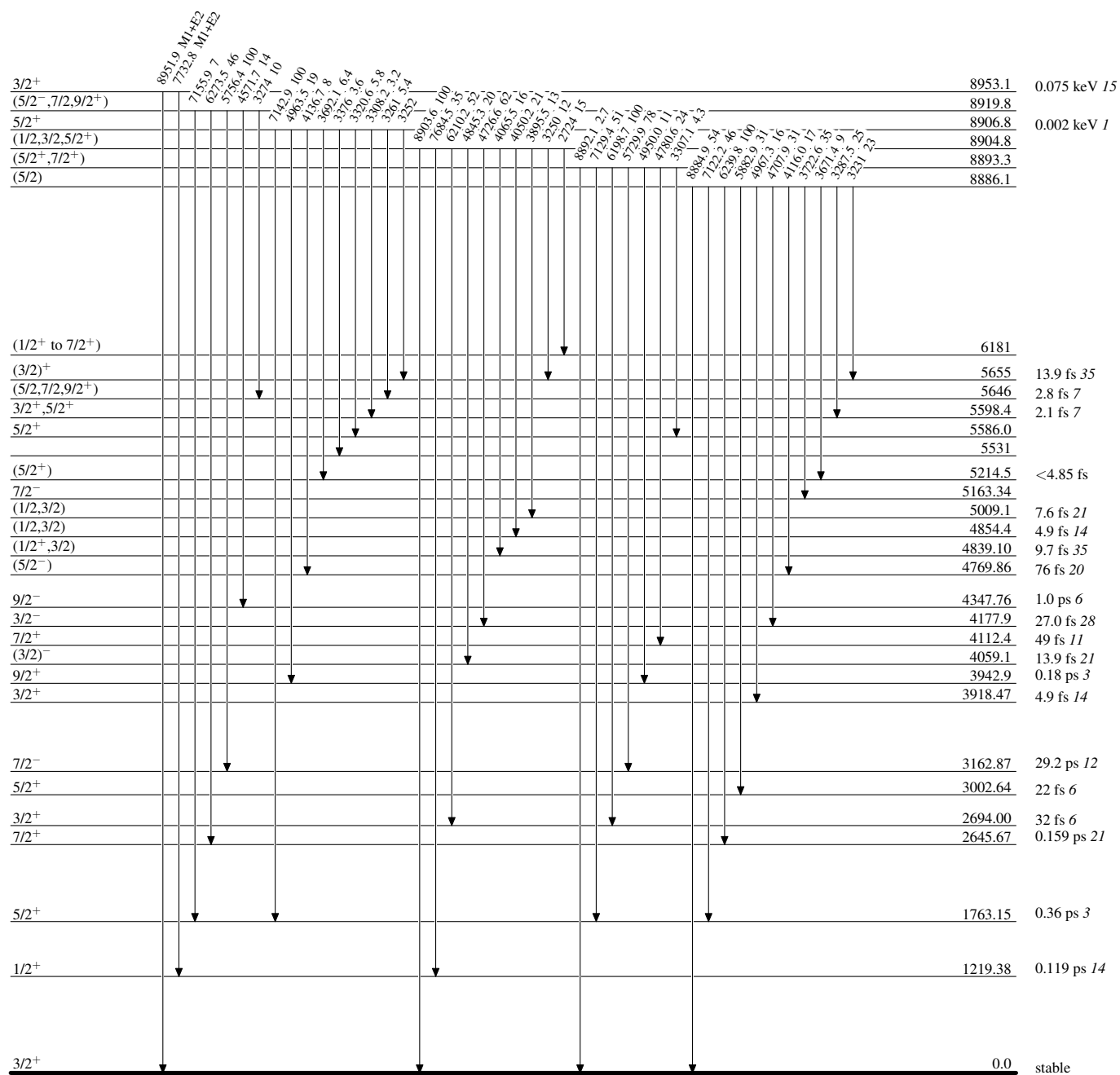
Adopted Levels, Gammas**Level Scheme**

Intensities: Relative photon branching from each level



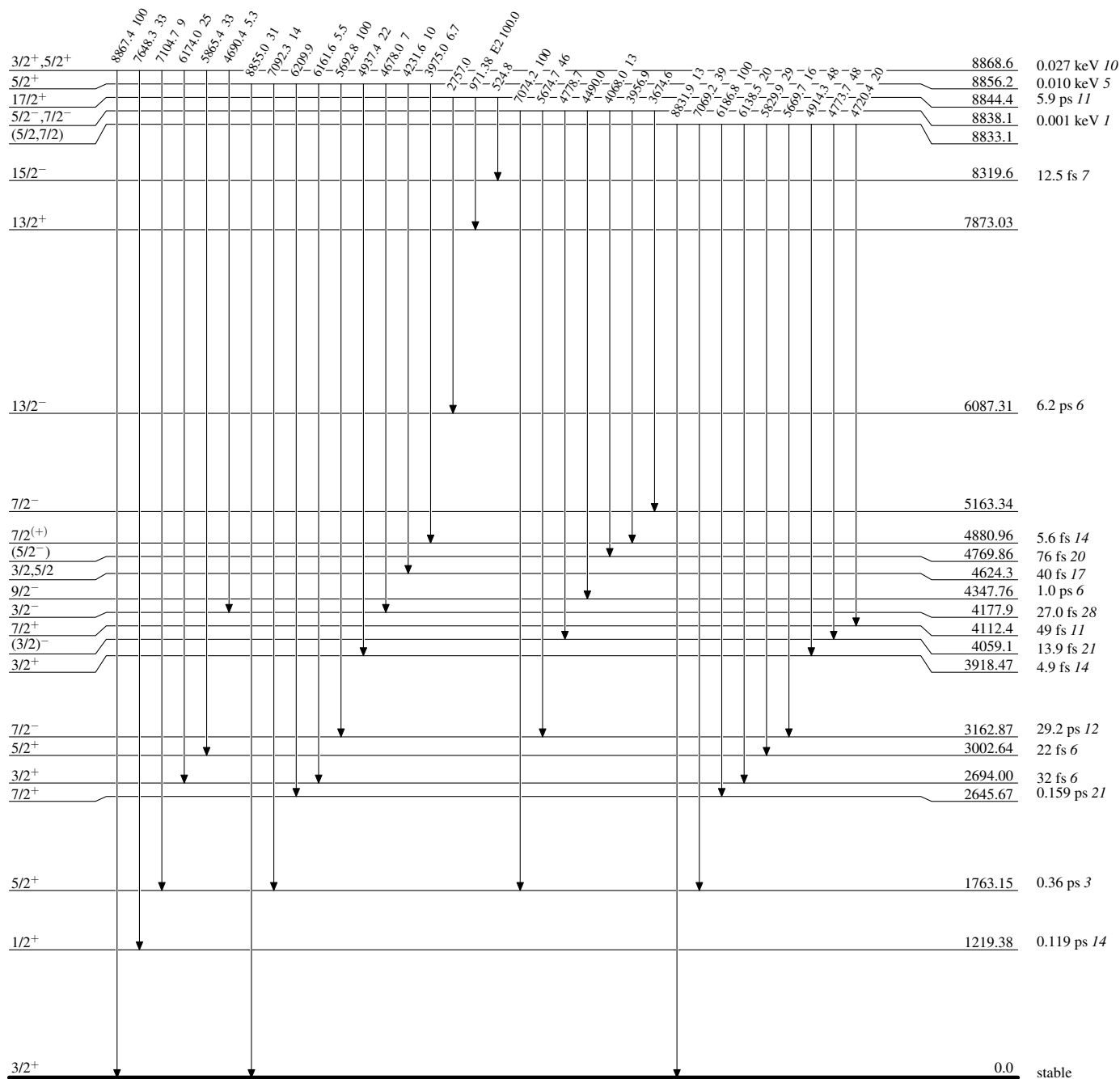
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

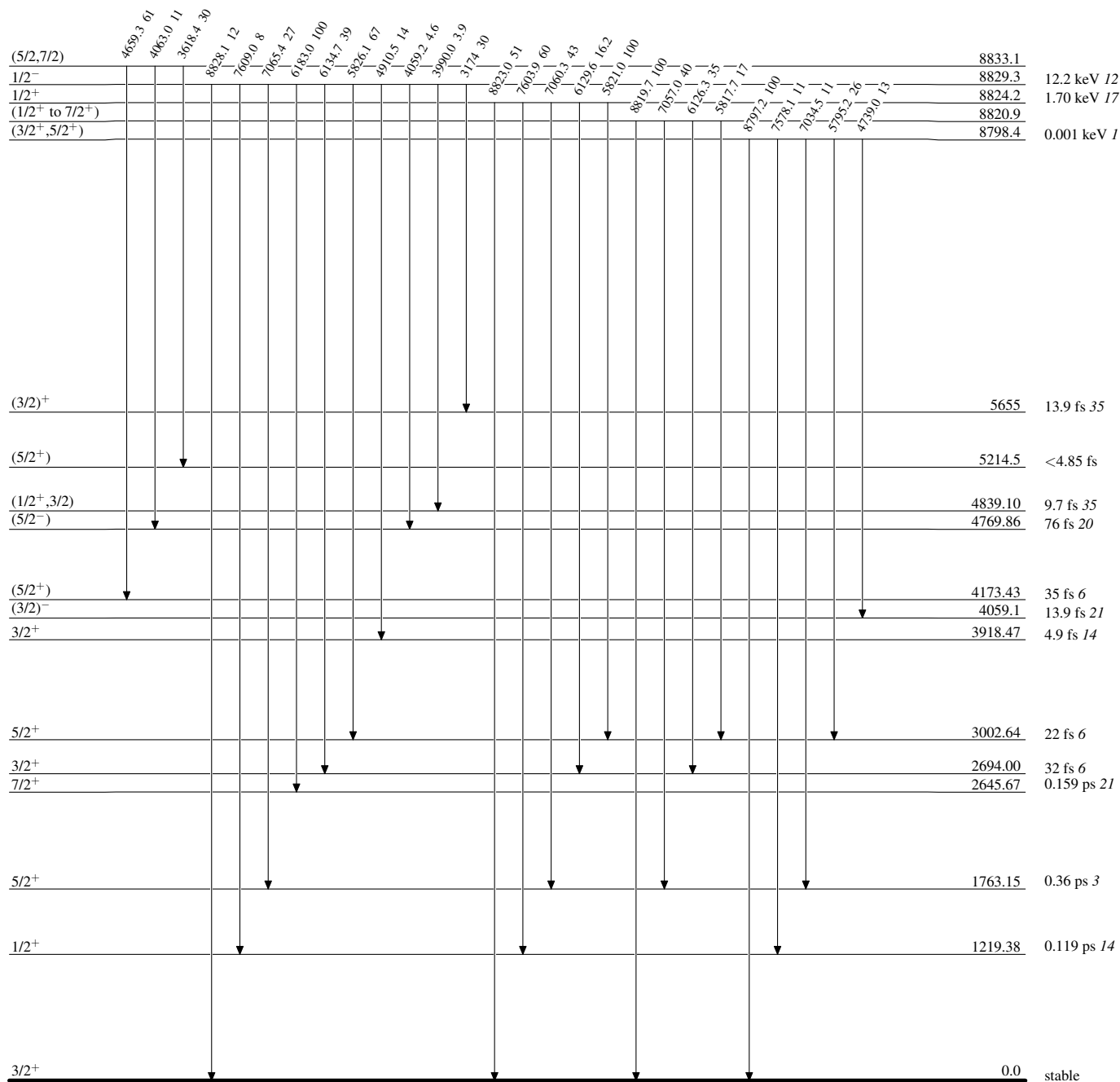
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



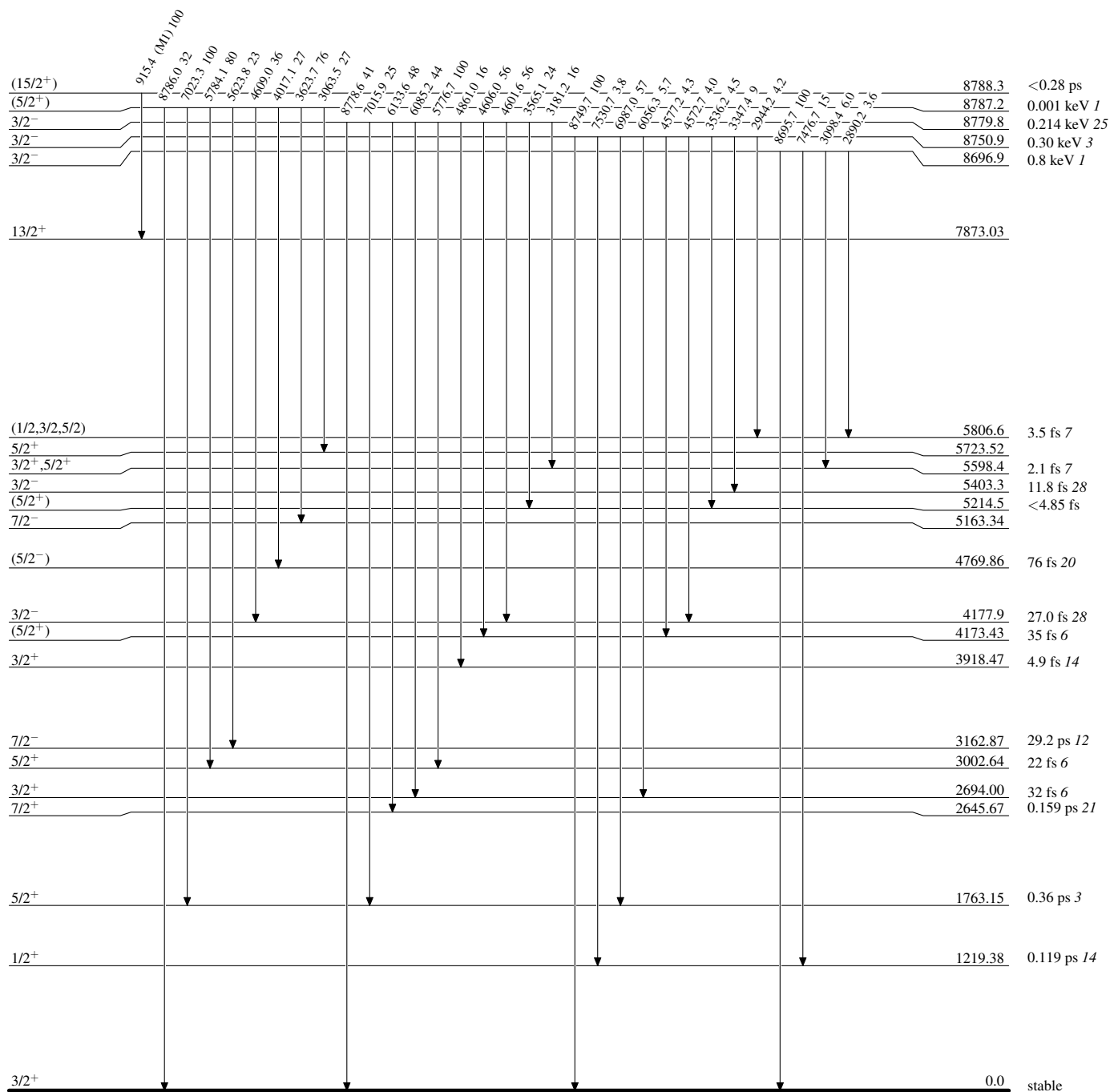
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



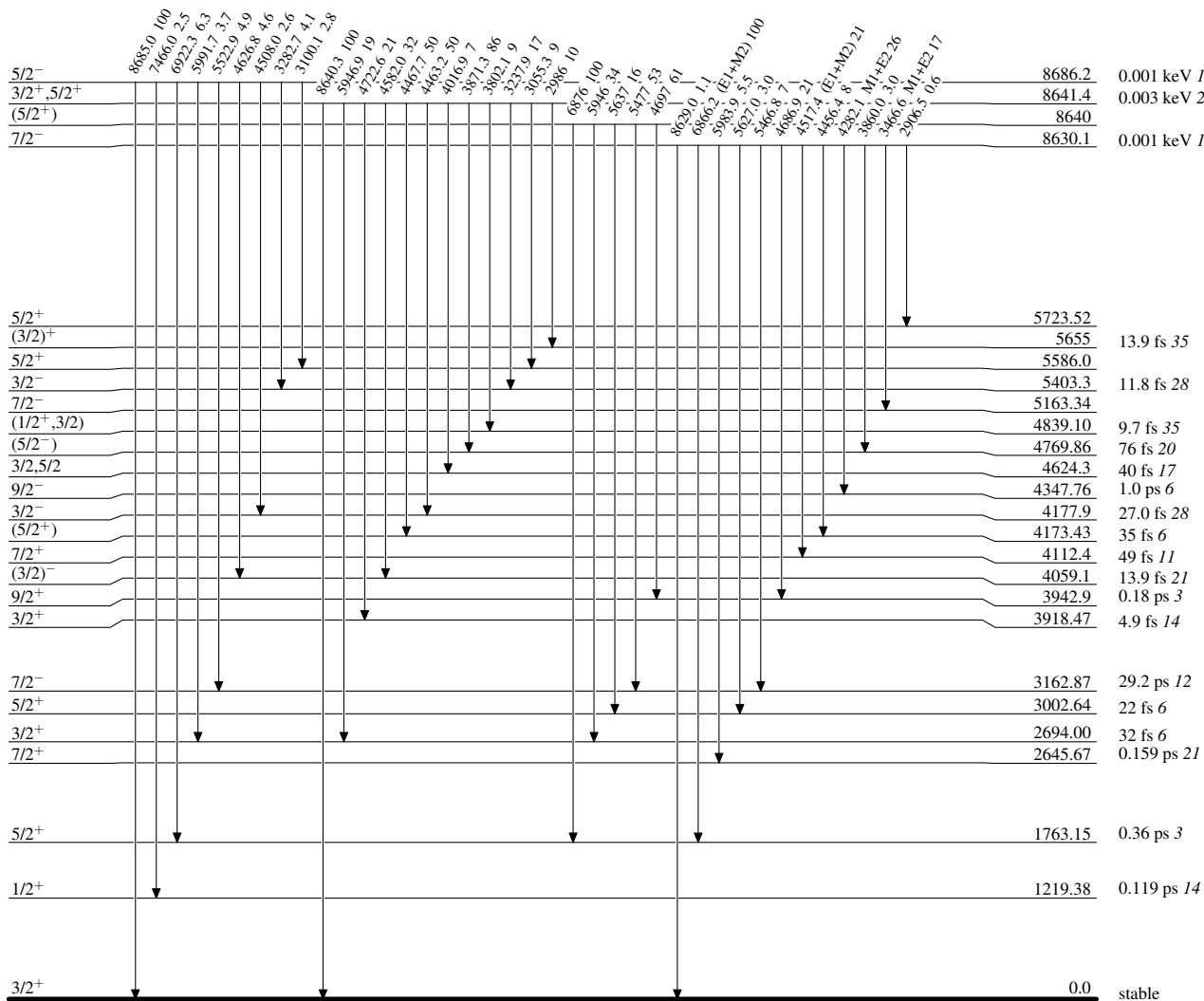
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



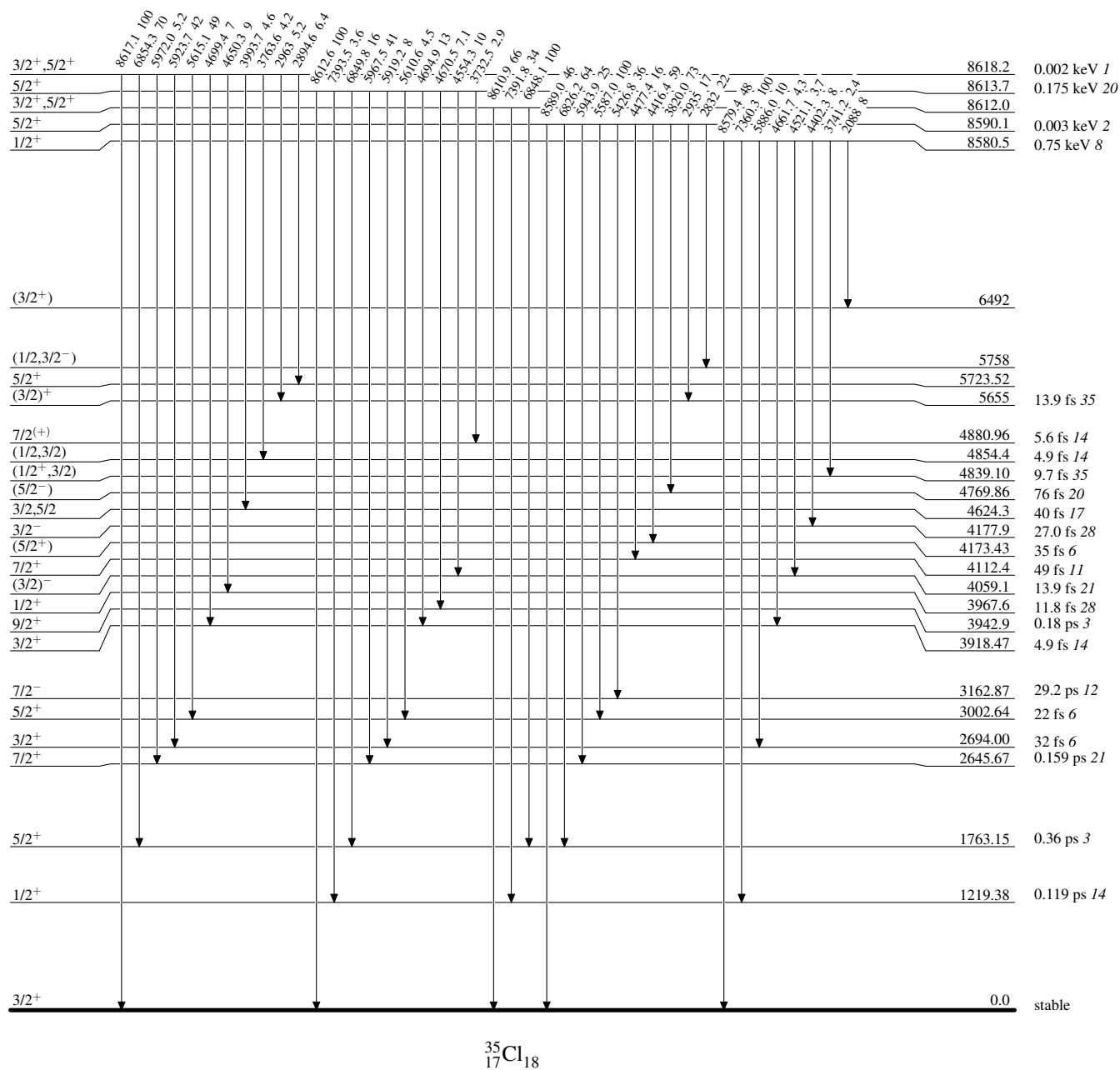
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

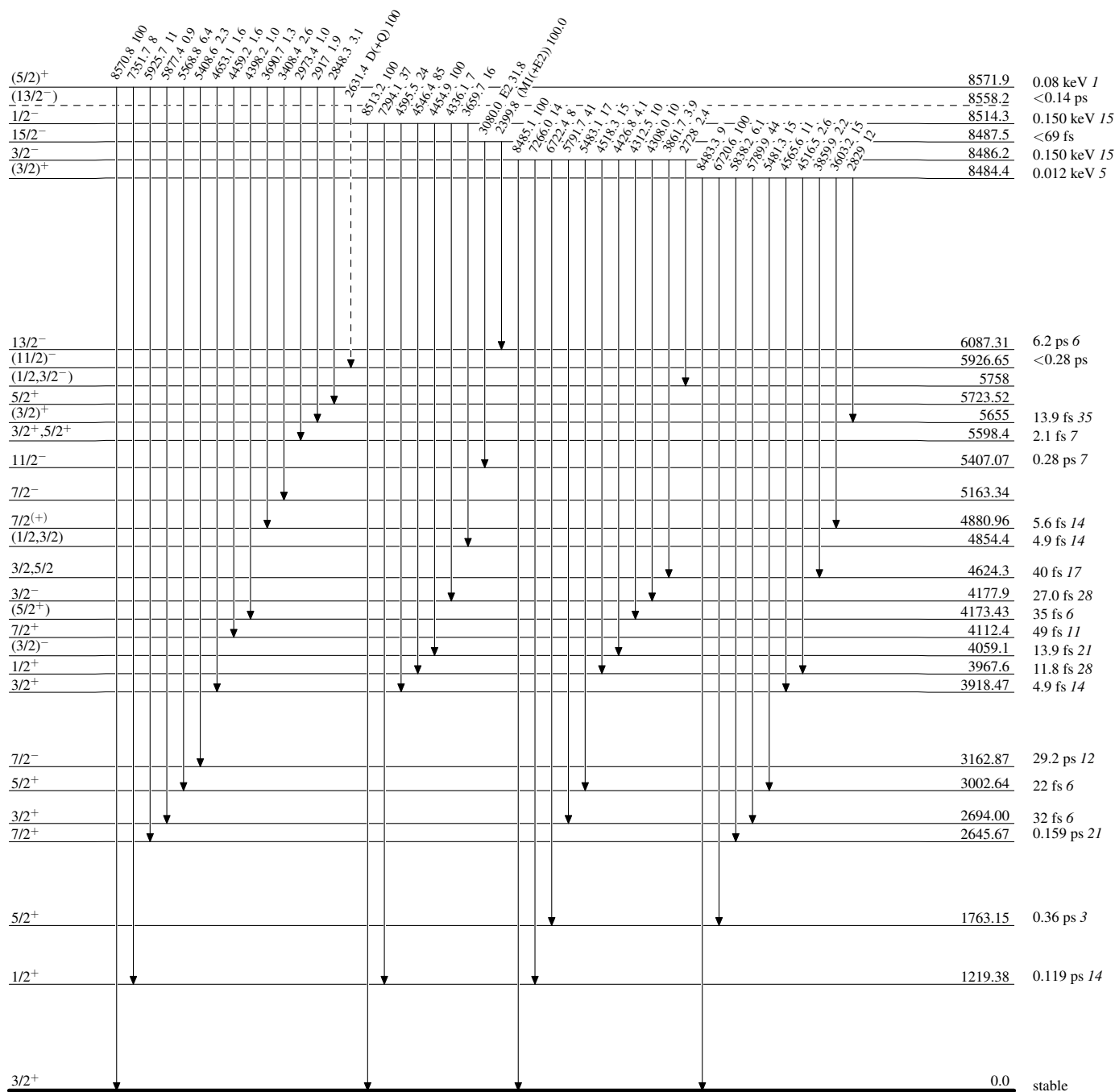
 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Legend

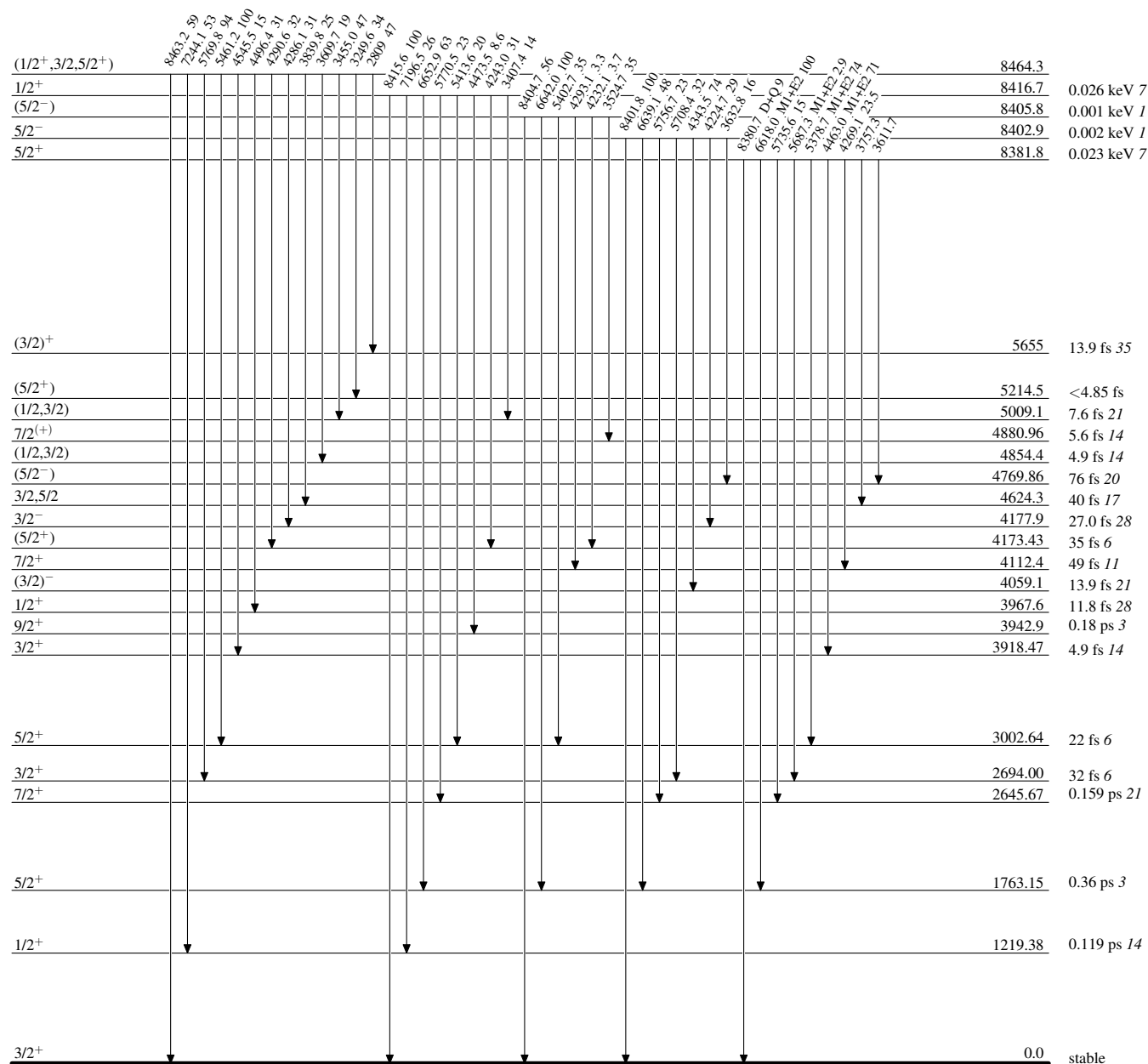
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

Adopted Levels, GammasLevel Scheme (continued)

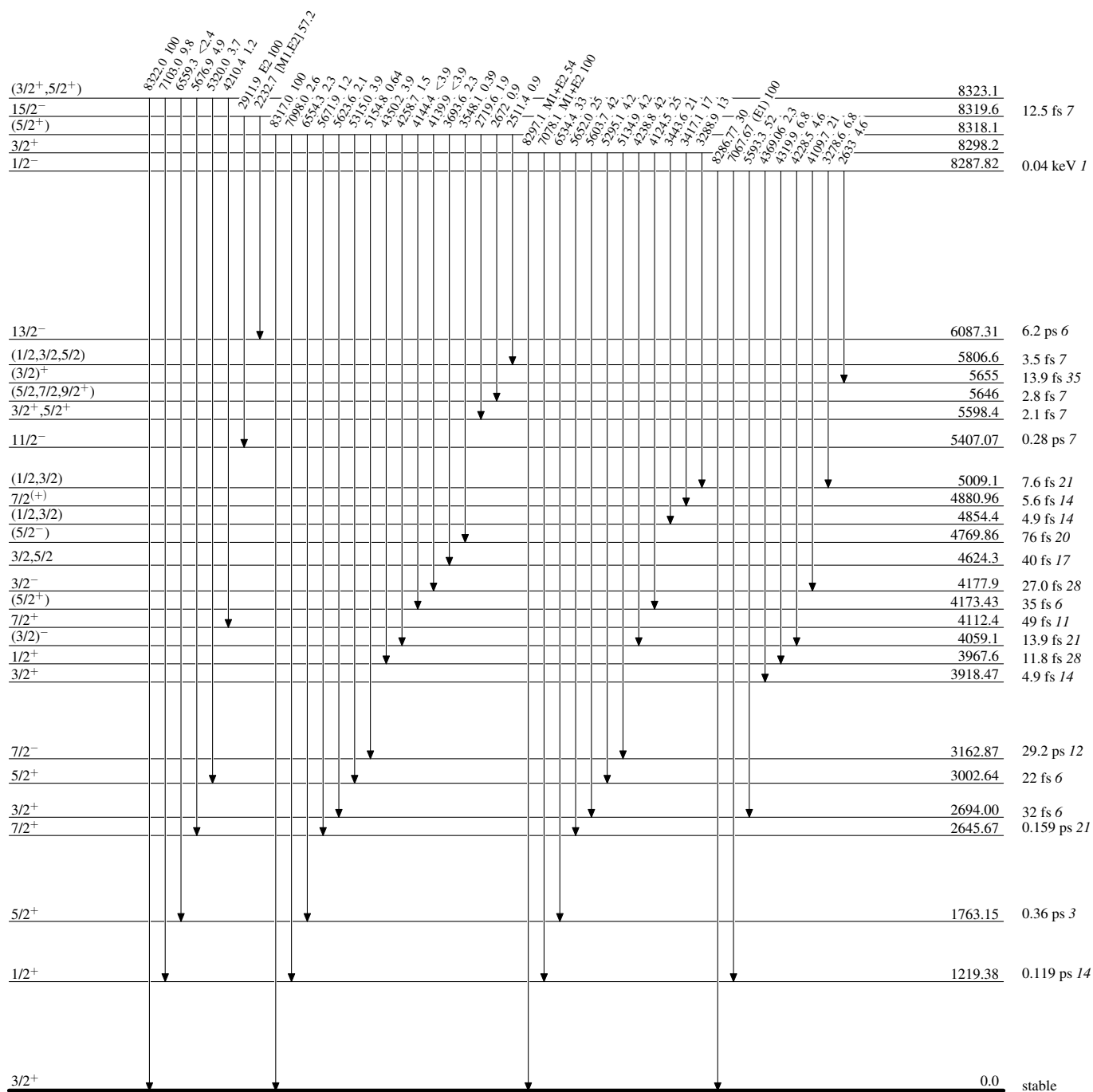
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

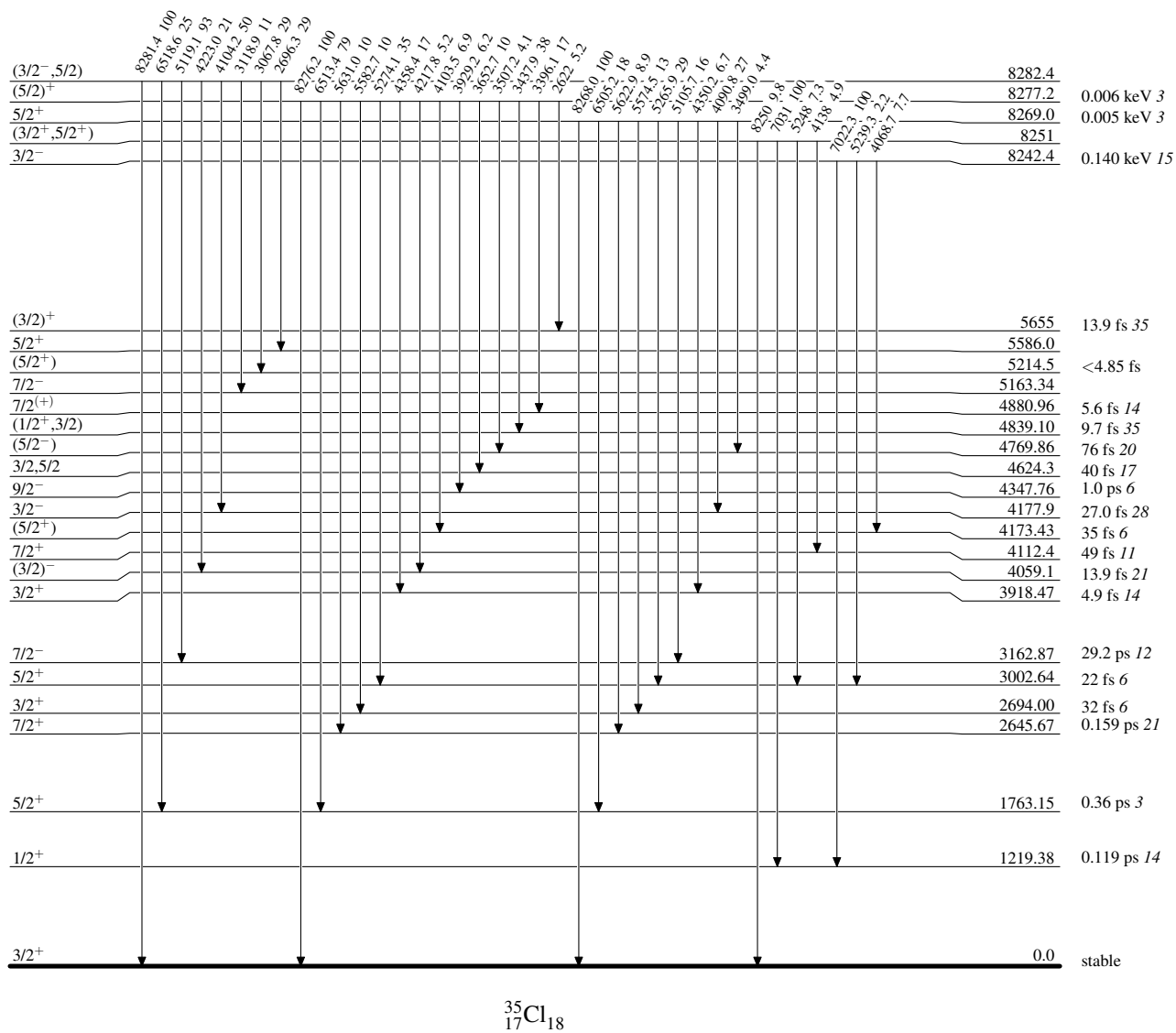
Level Scheme (continued)

Intensities: Relative photon branching from each level



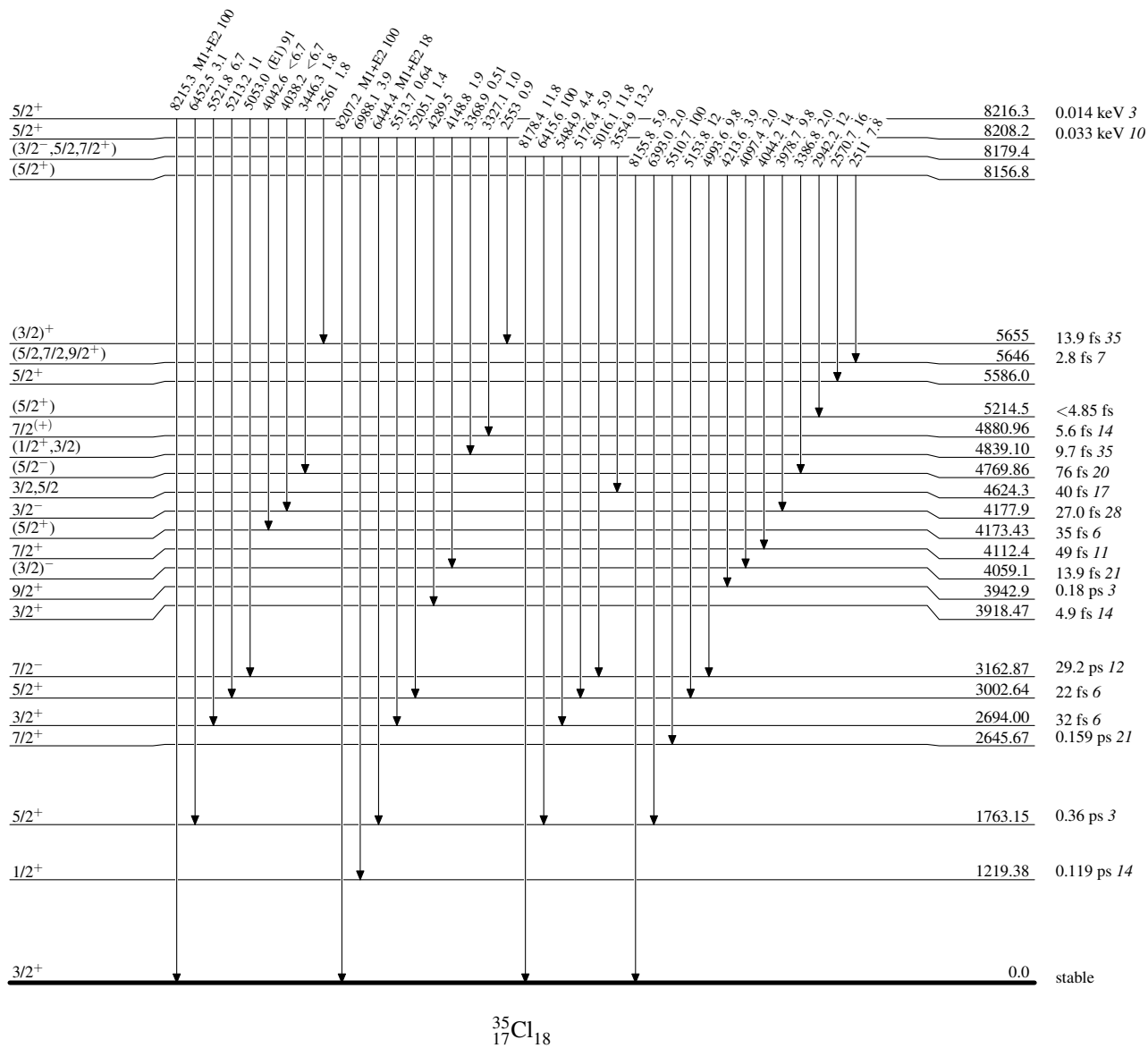
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

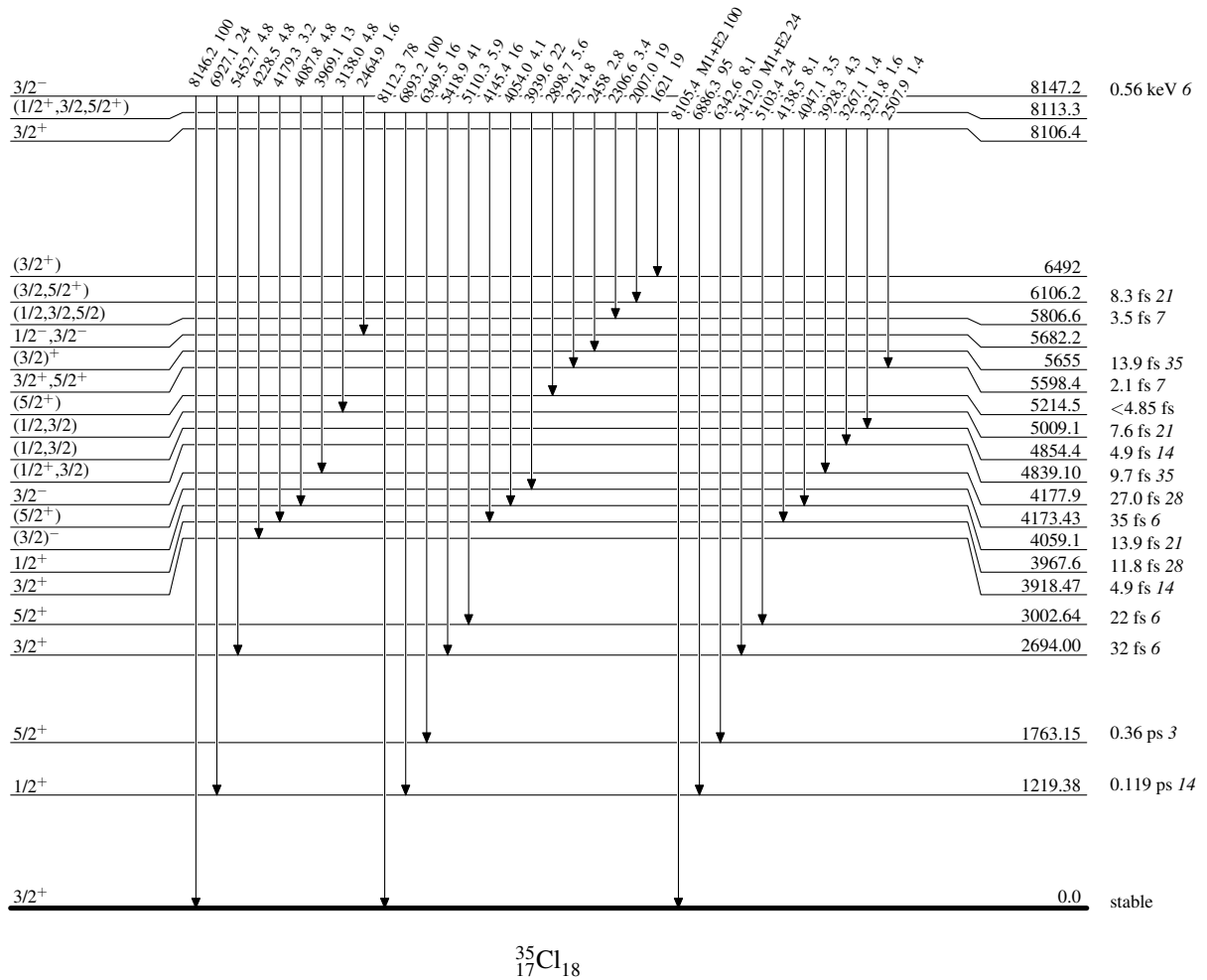
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

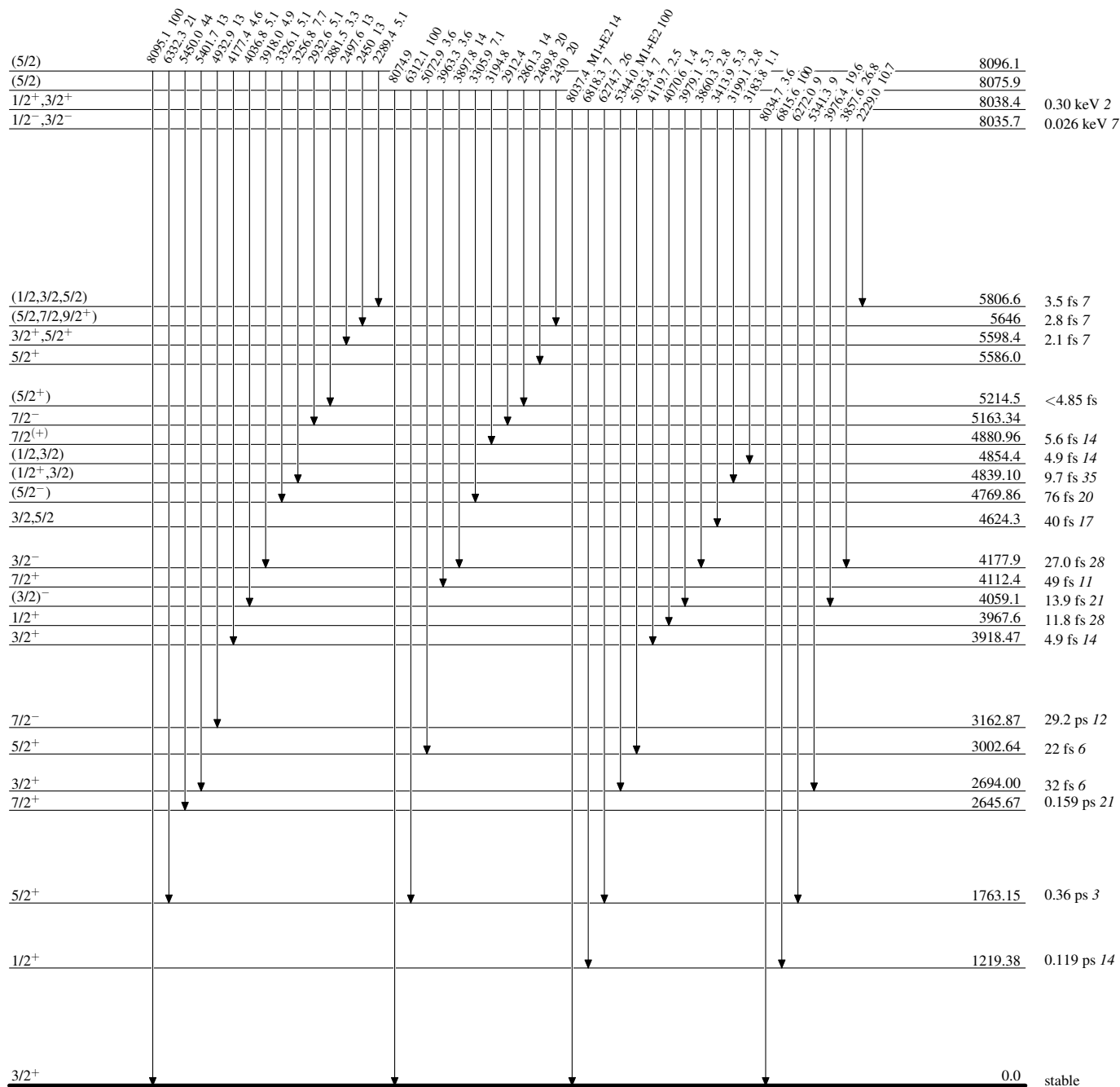
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

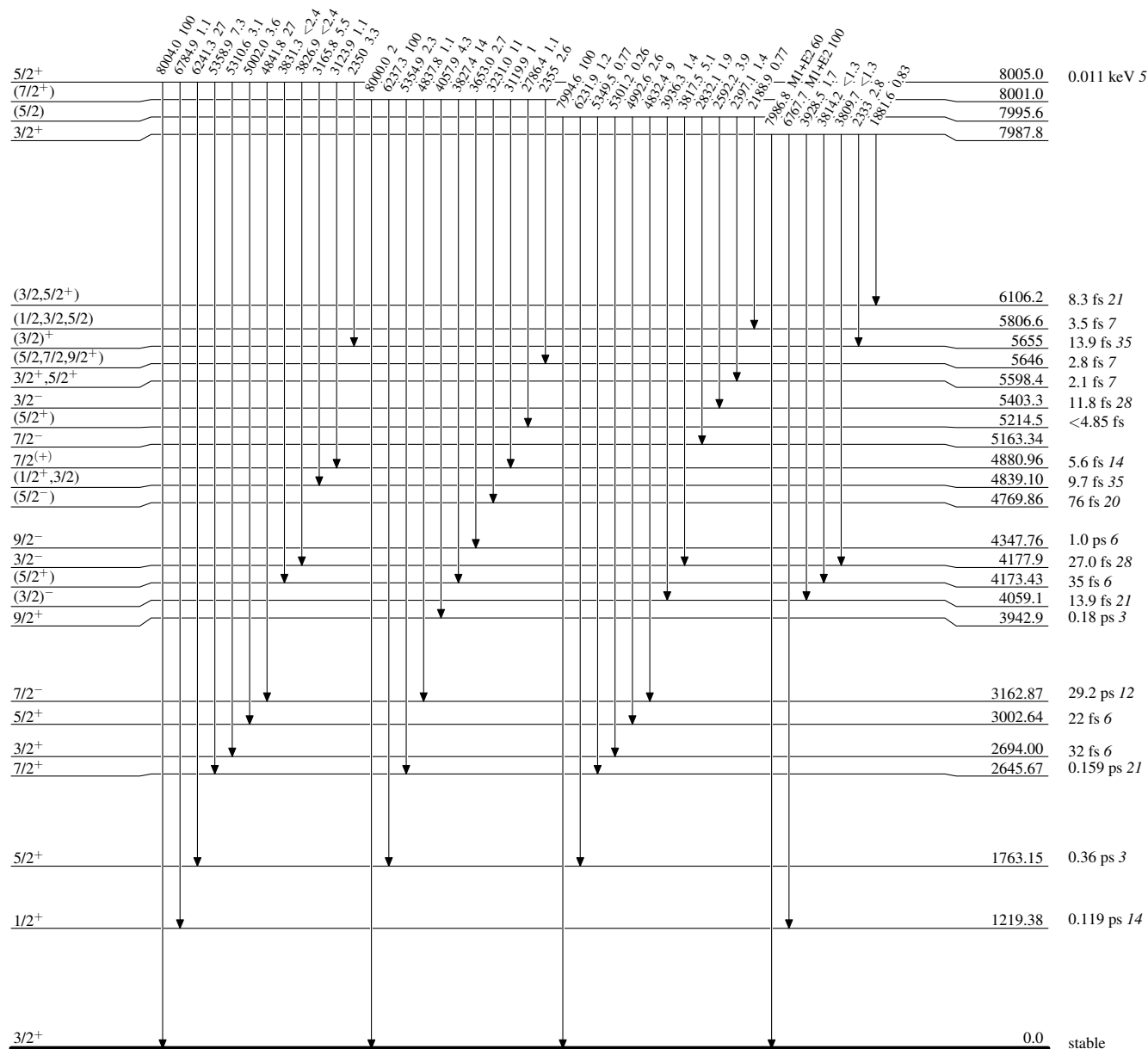
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

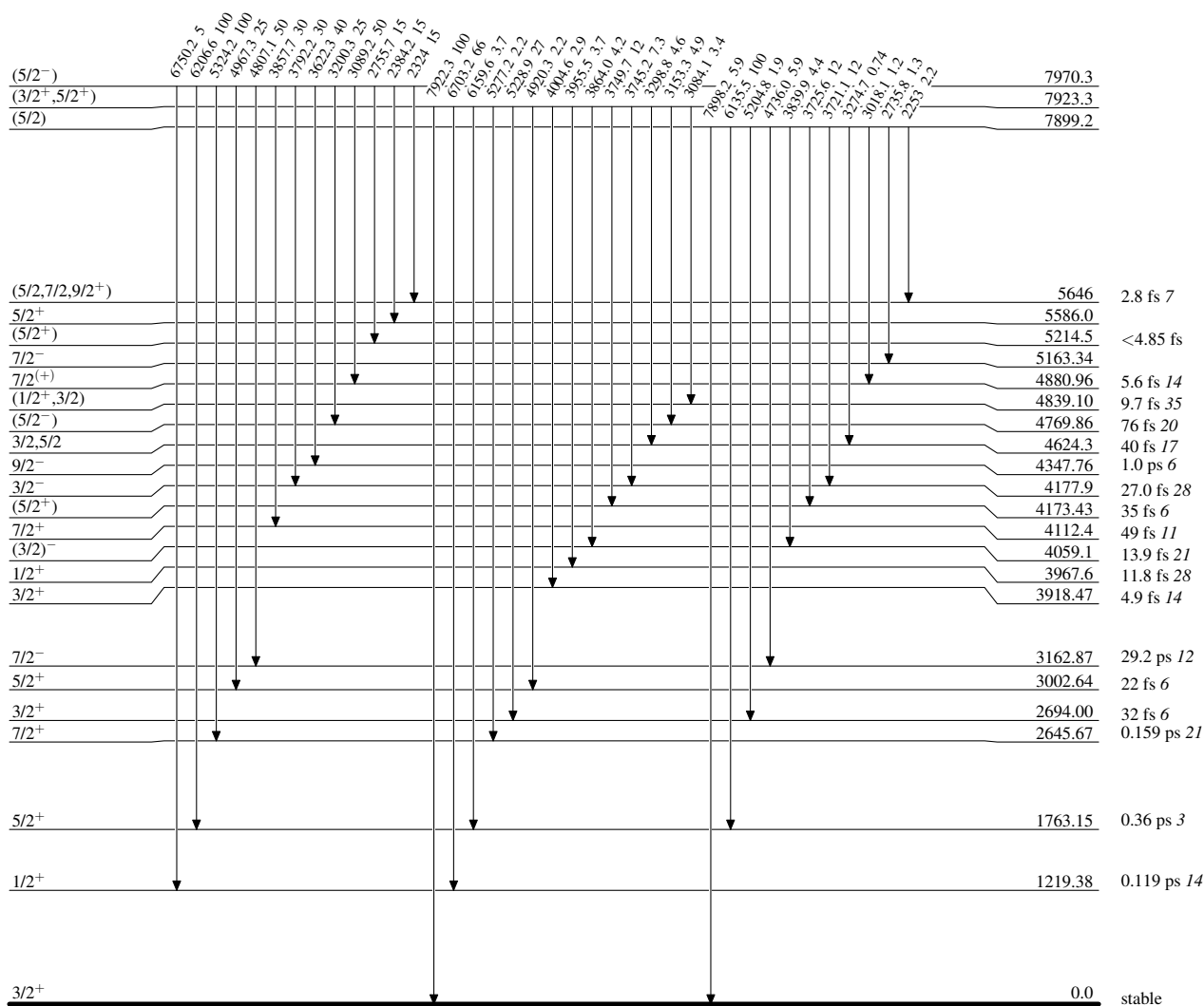
Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

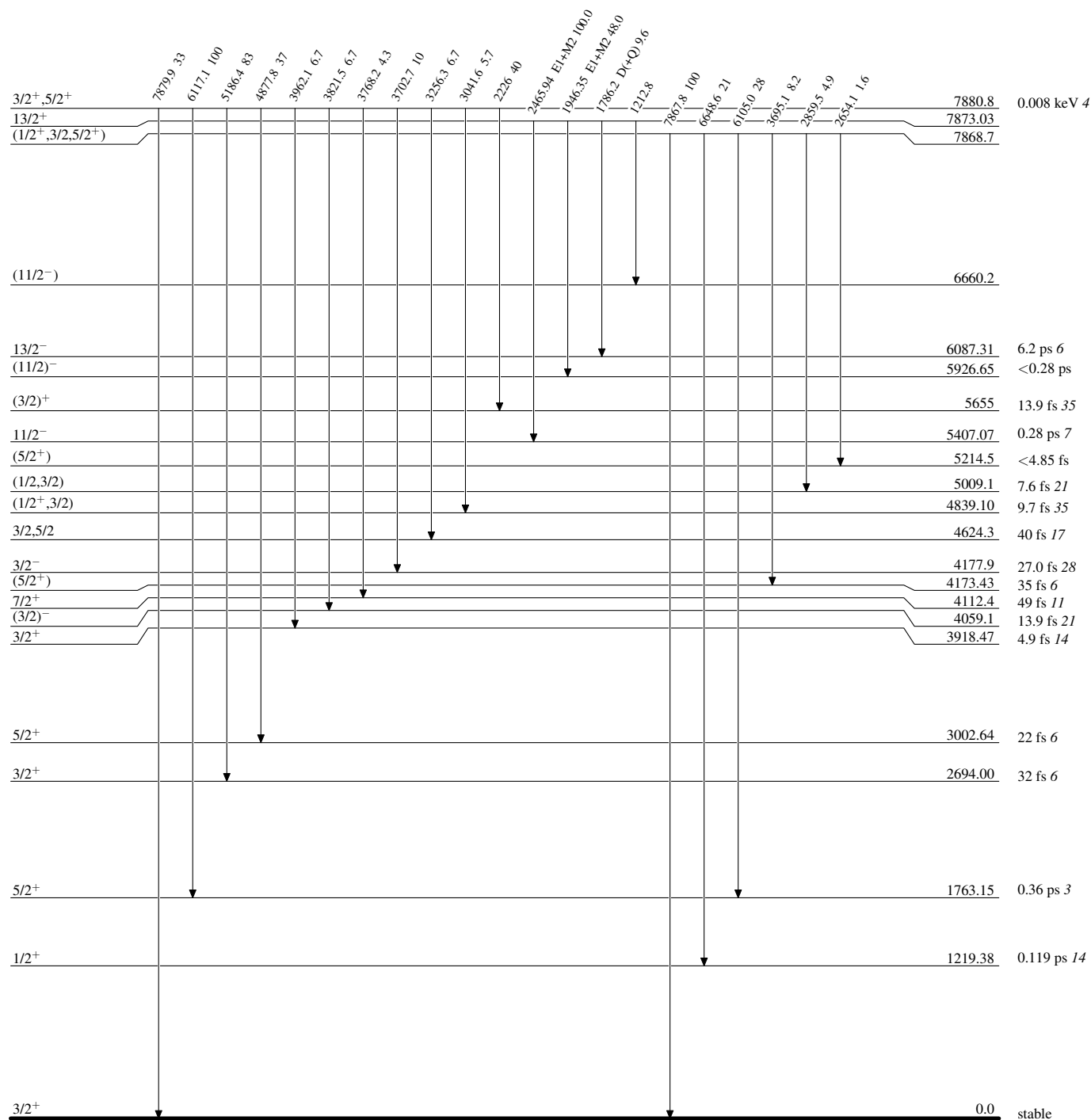
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

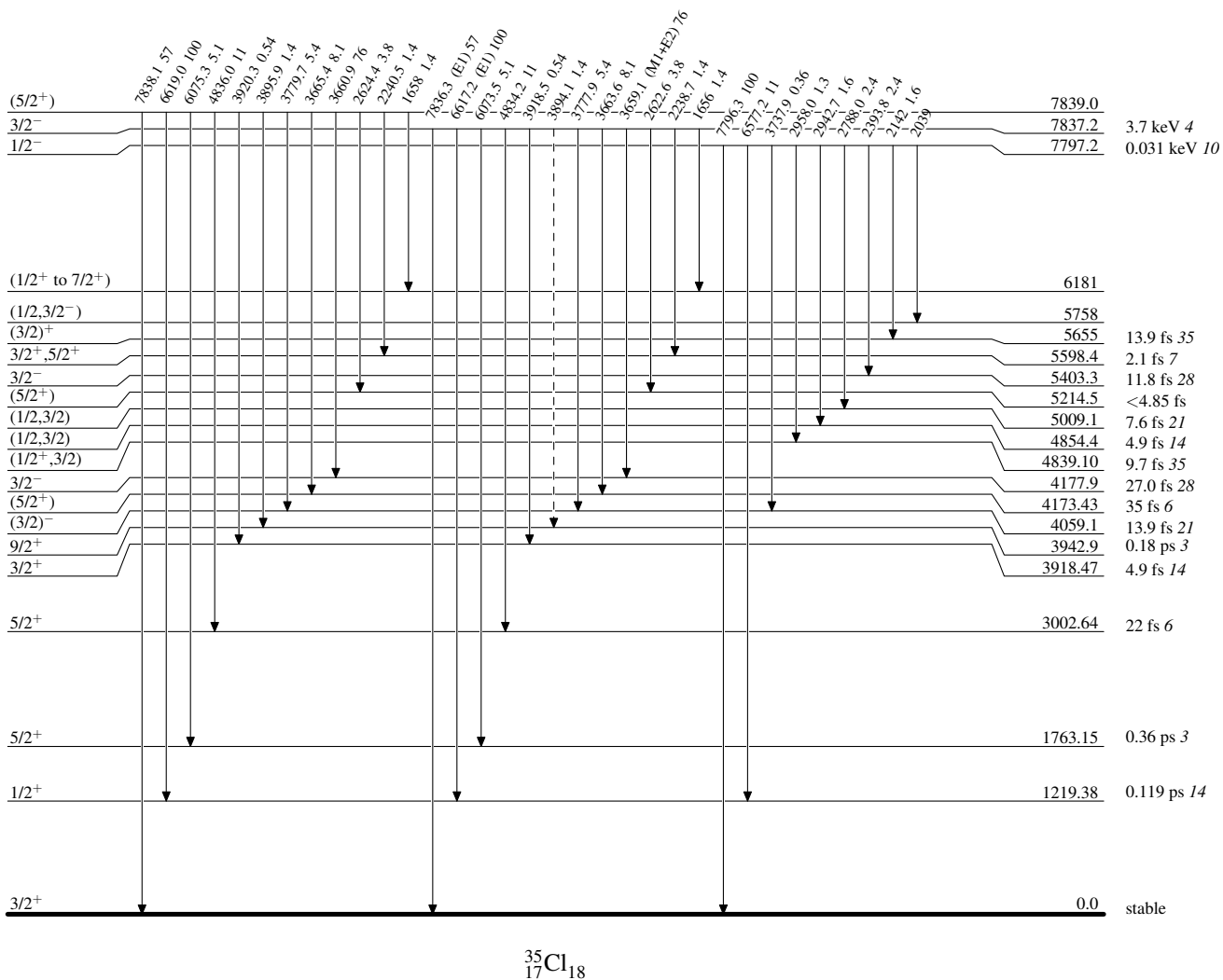


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

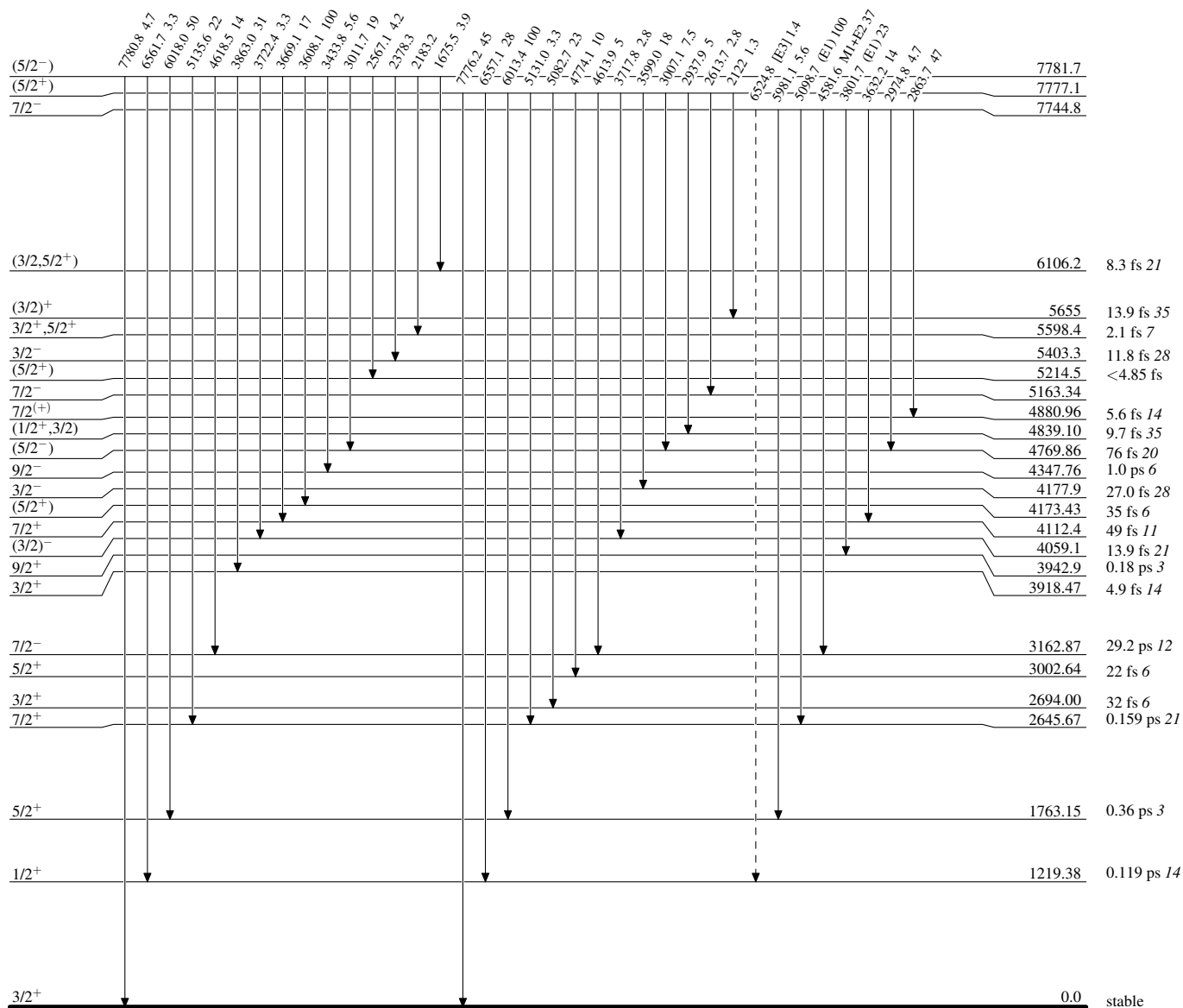
-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Legend

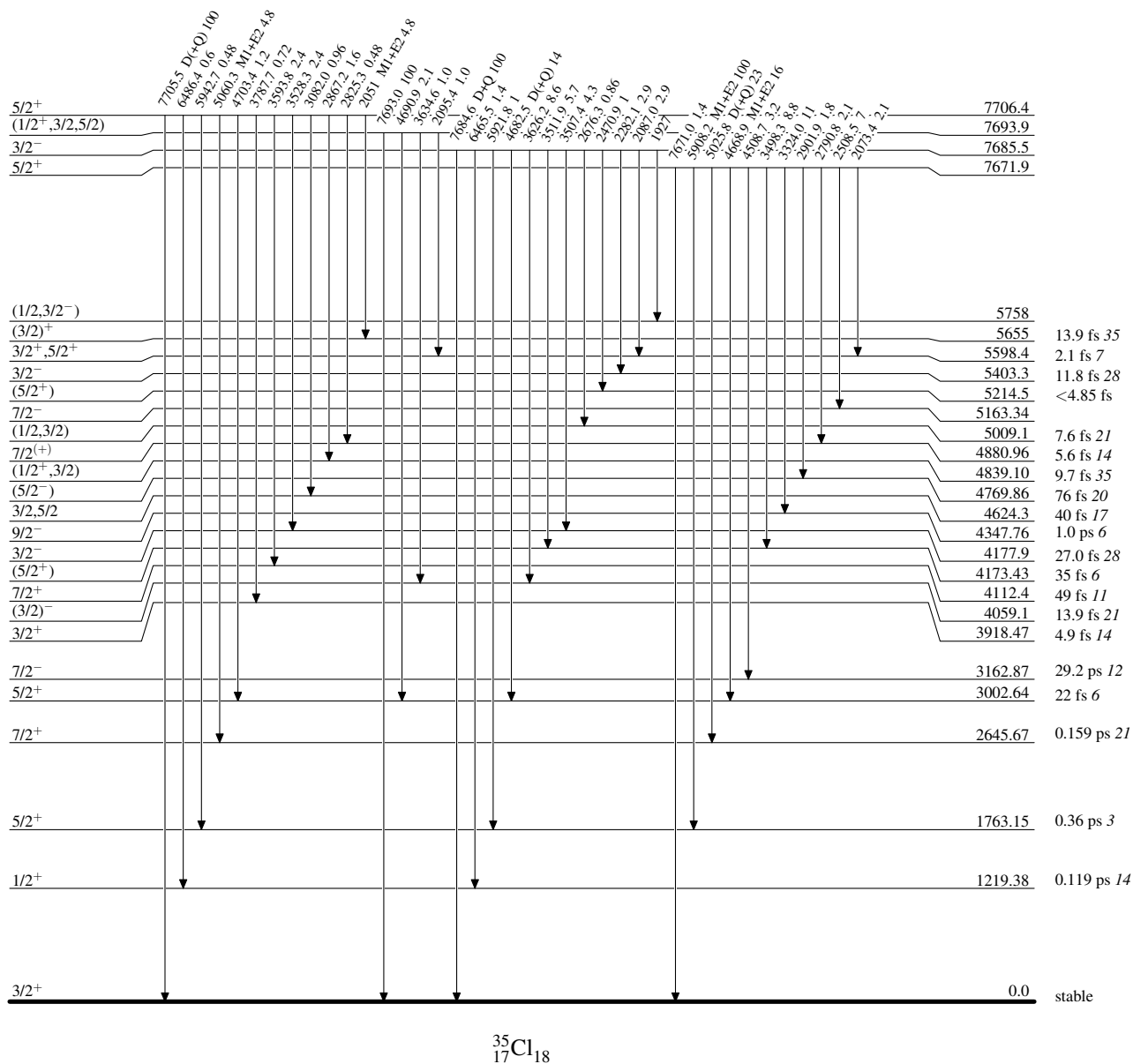
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

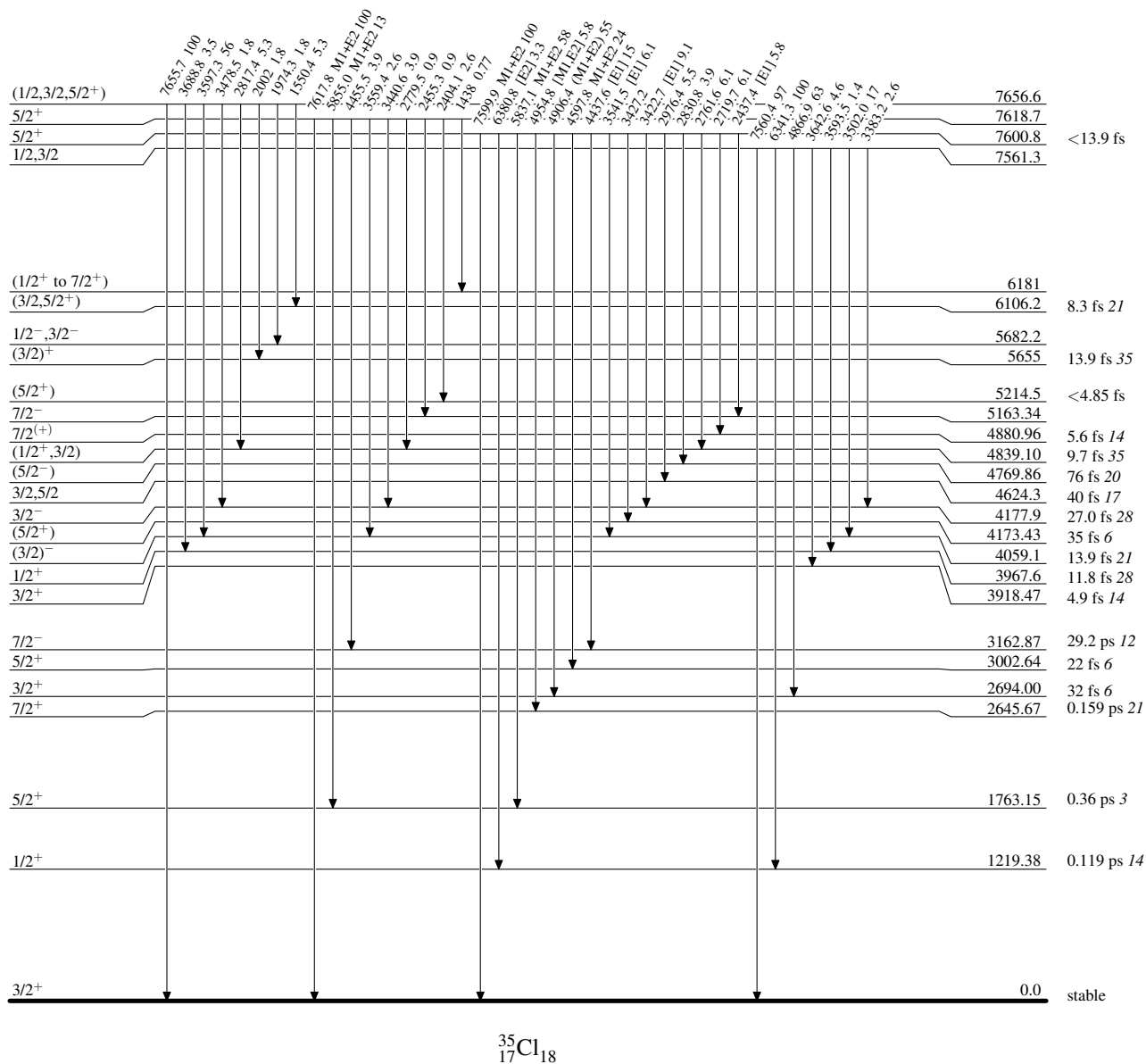
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



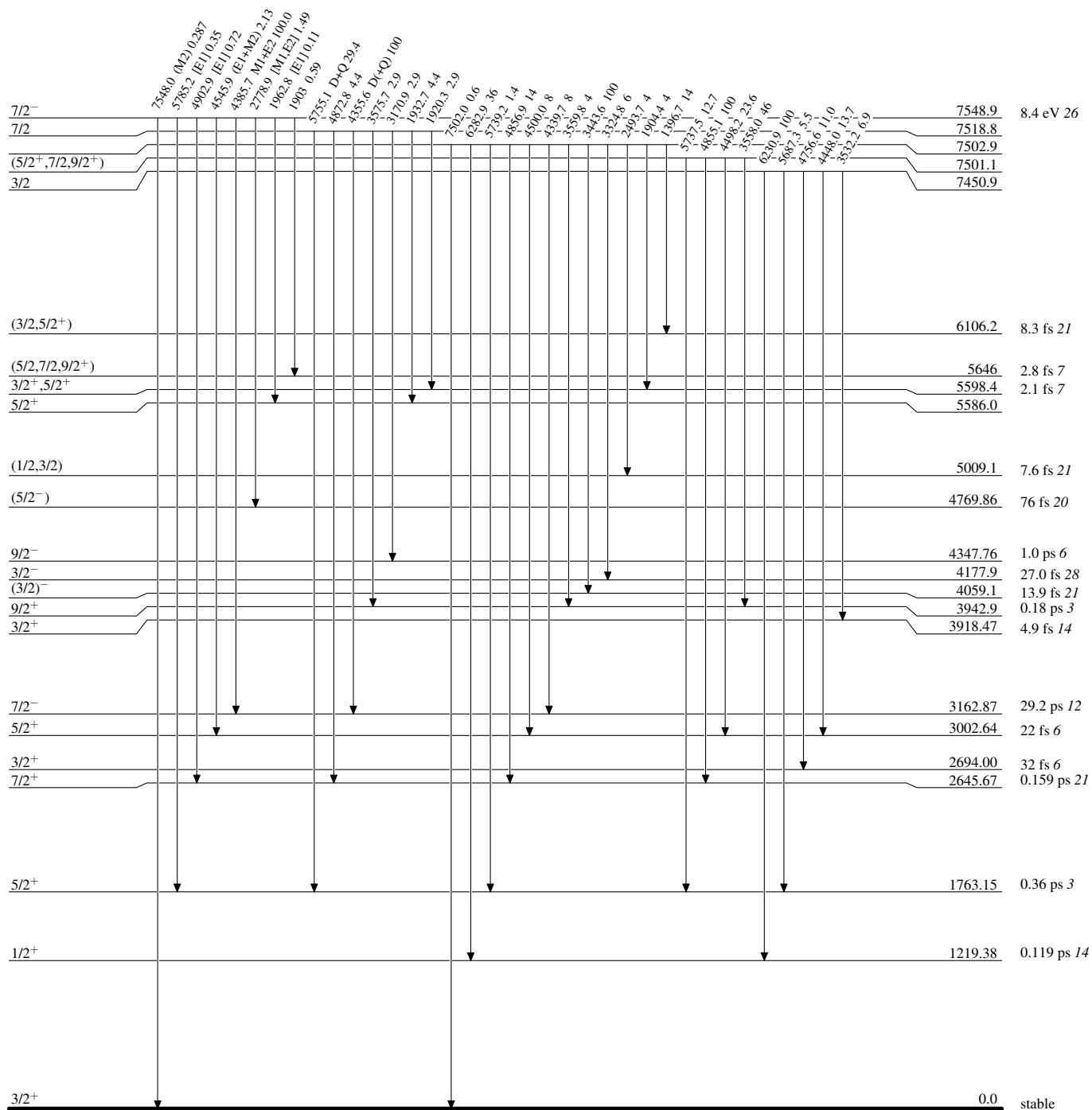
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

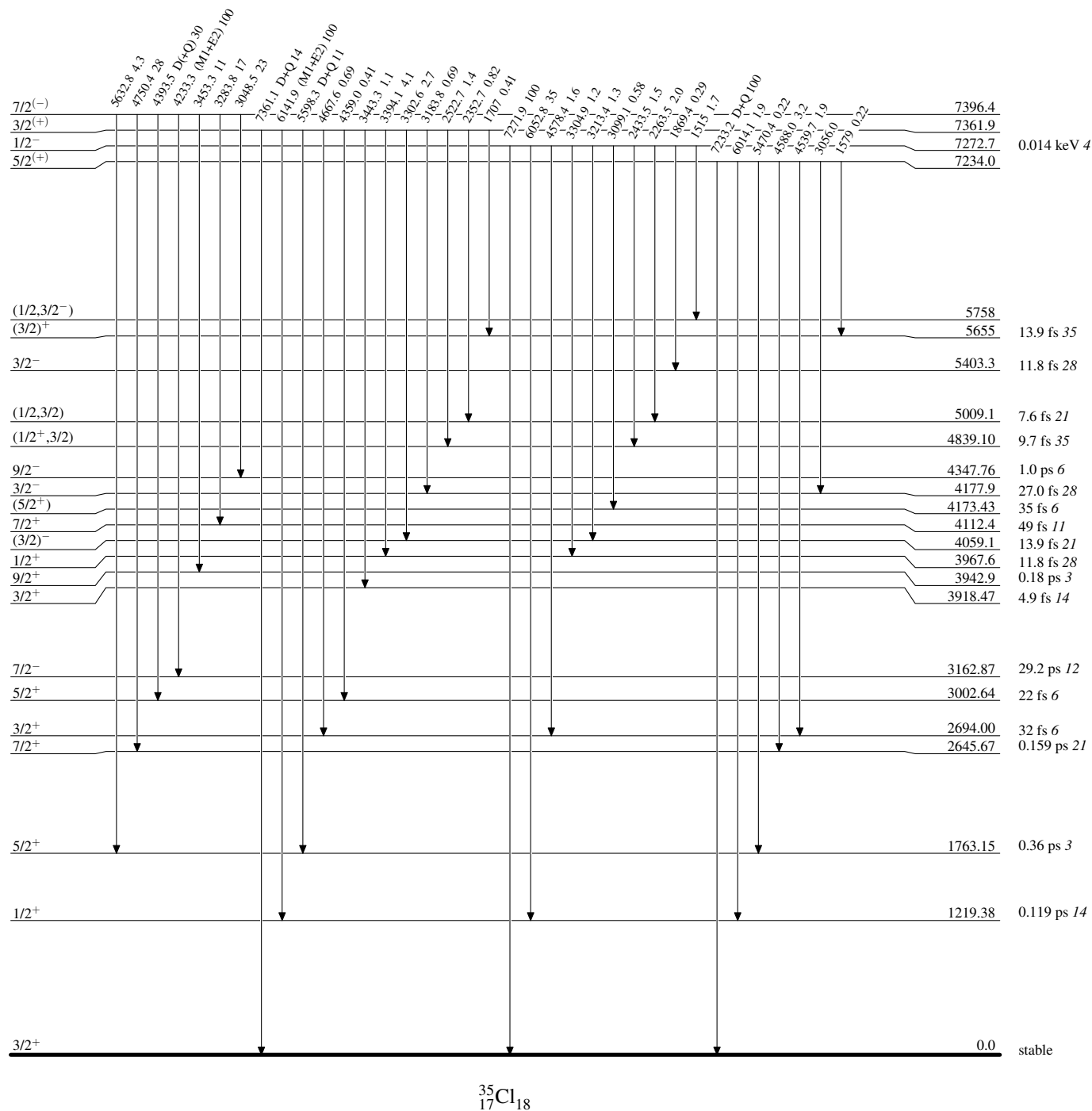
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

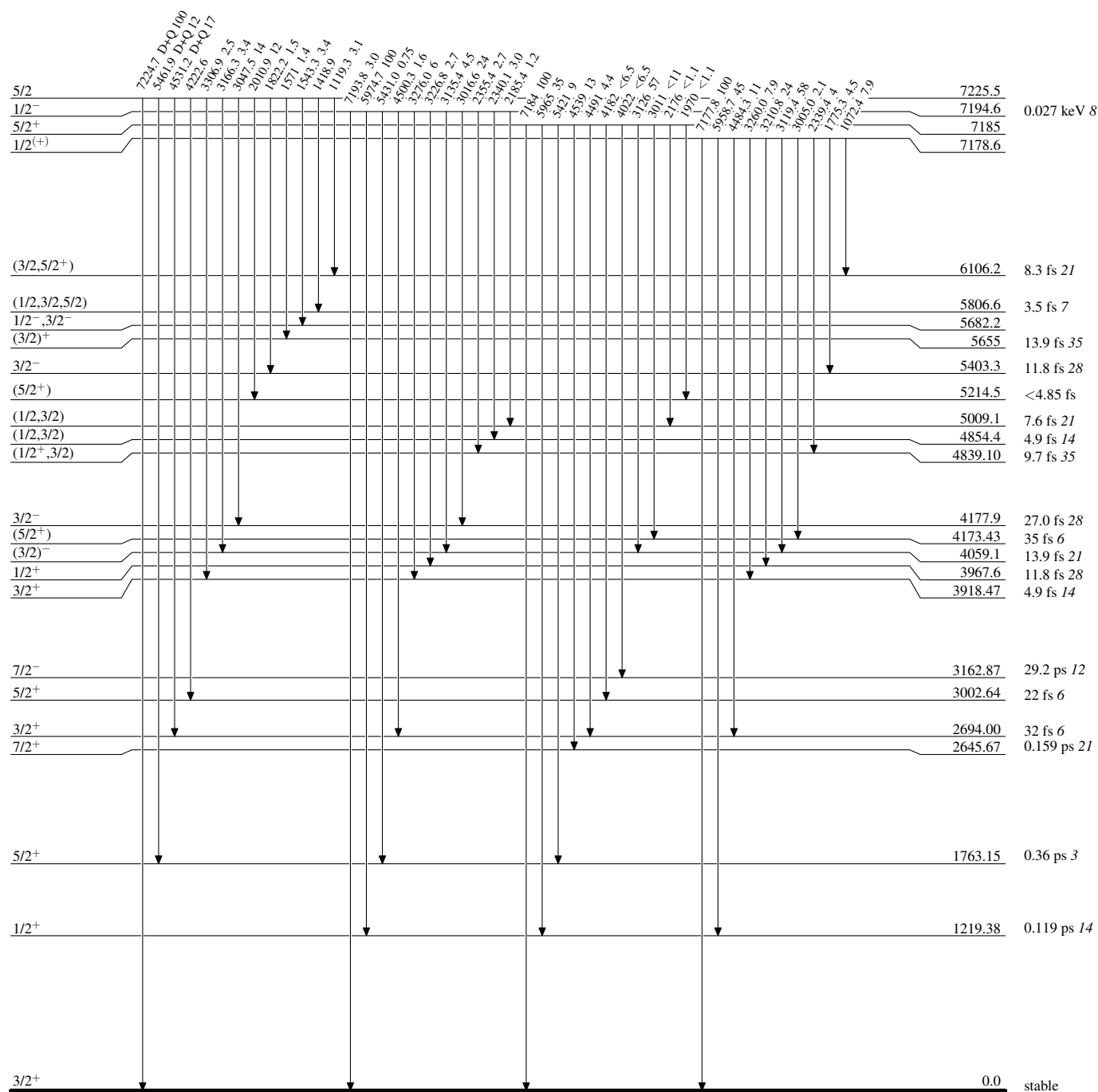
Level Scheme (continued)

Intensities: Relative photon branching from each level



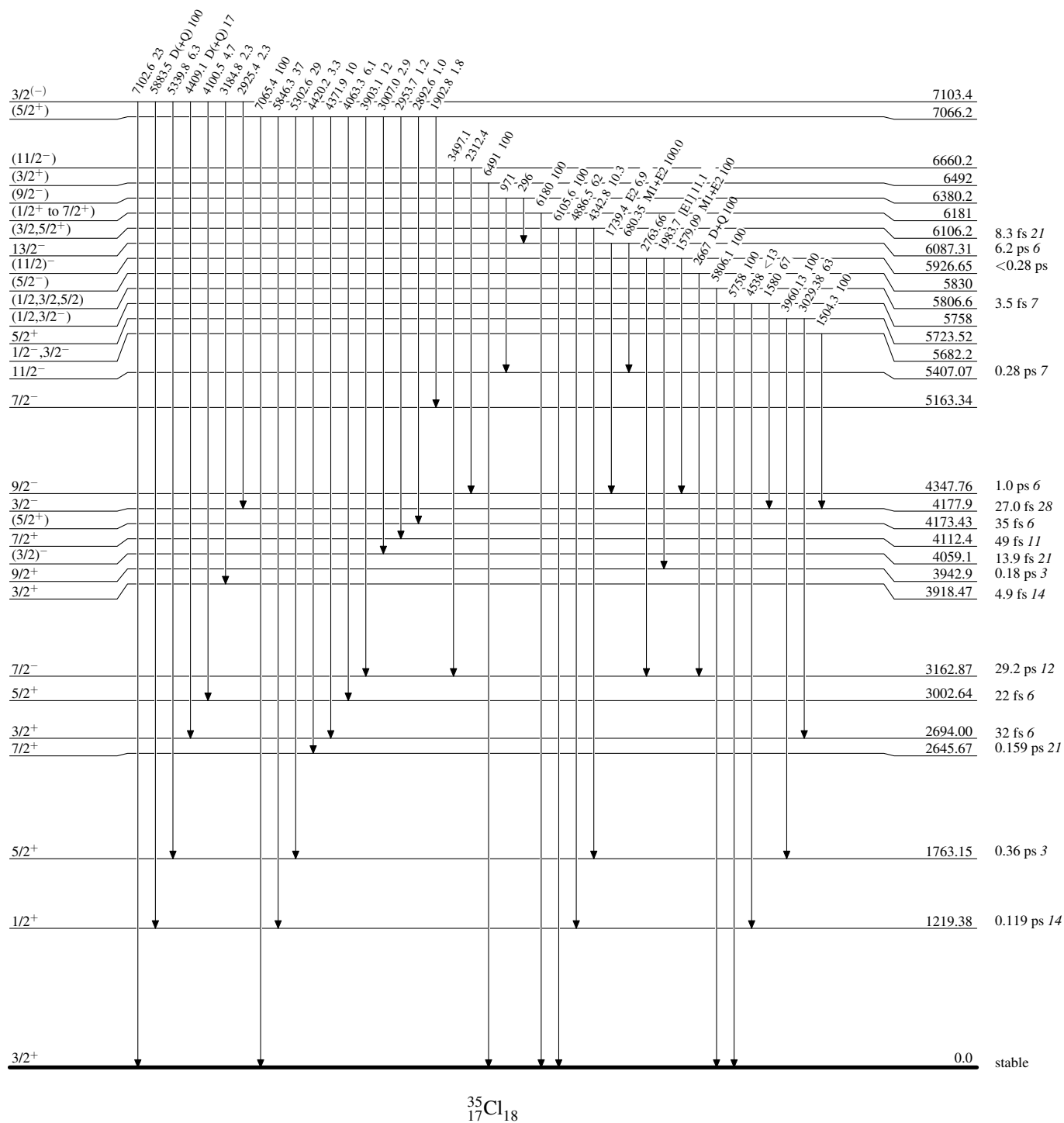
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

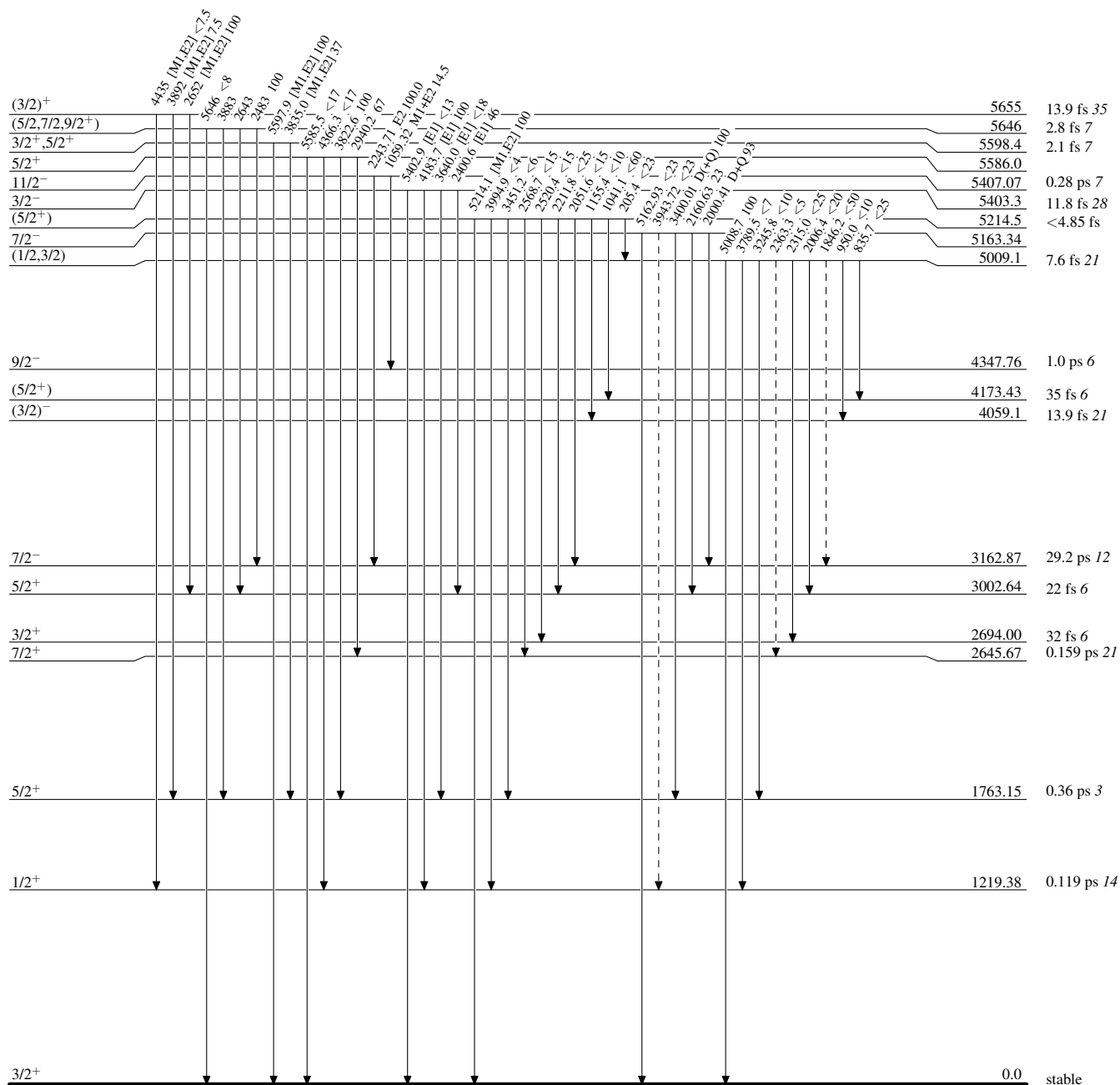


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

 -----► γ Decay (Uncertain)


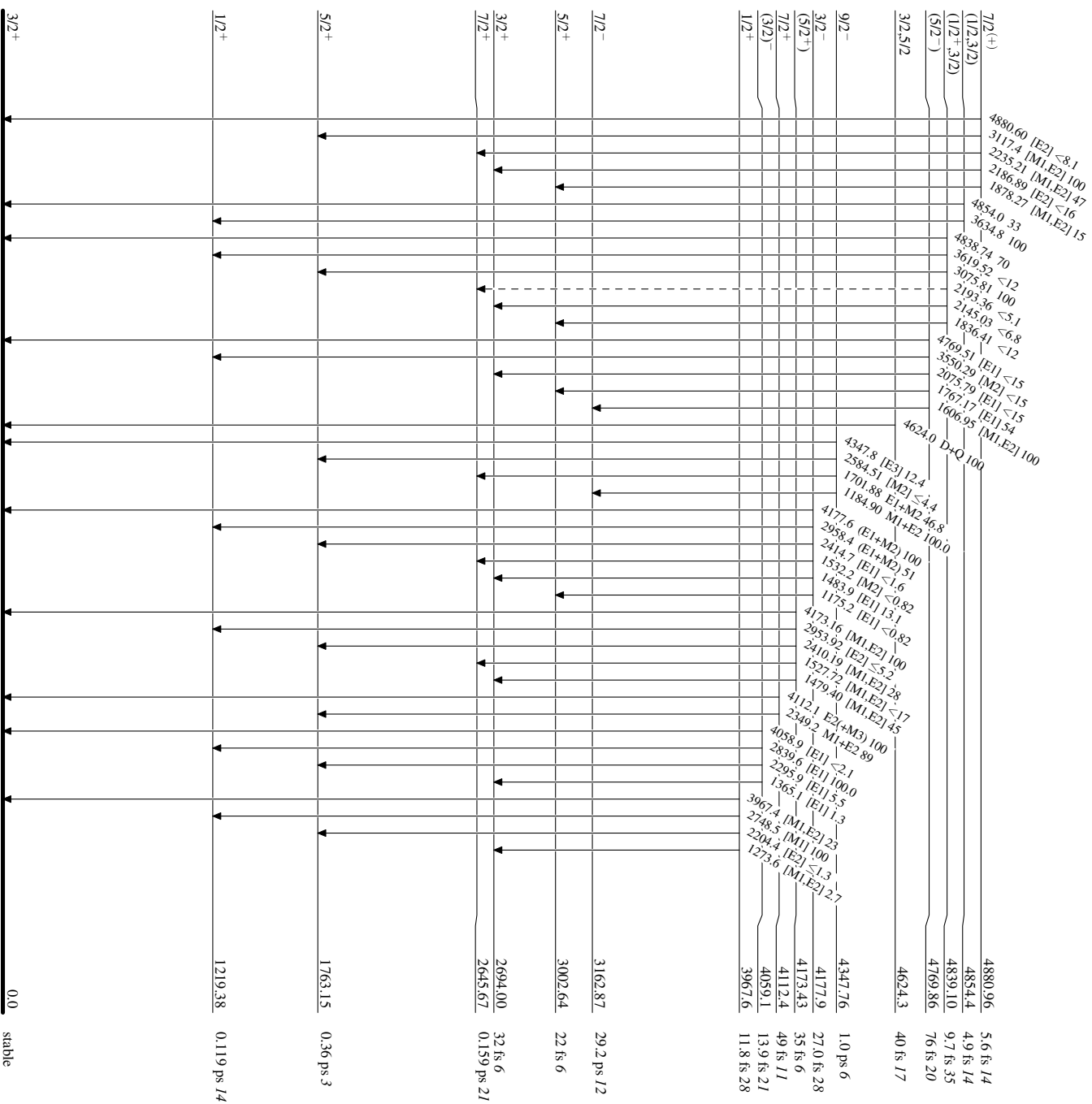
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

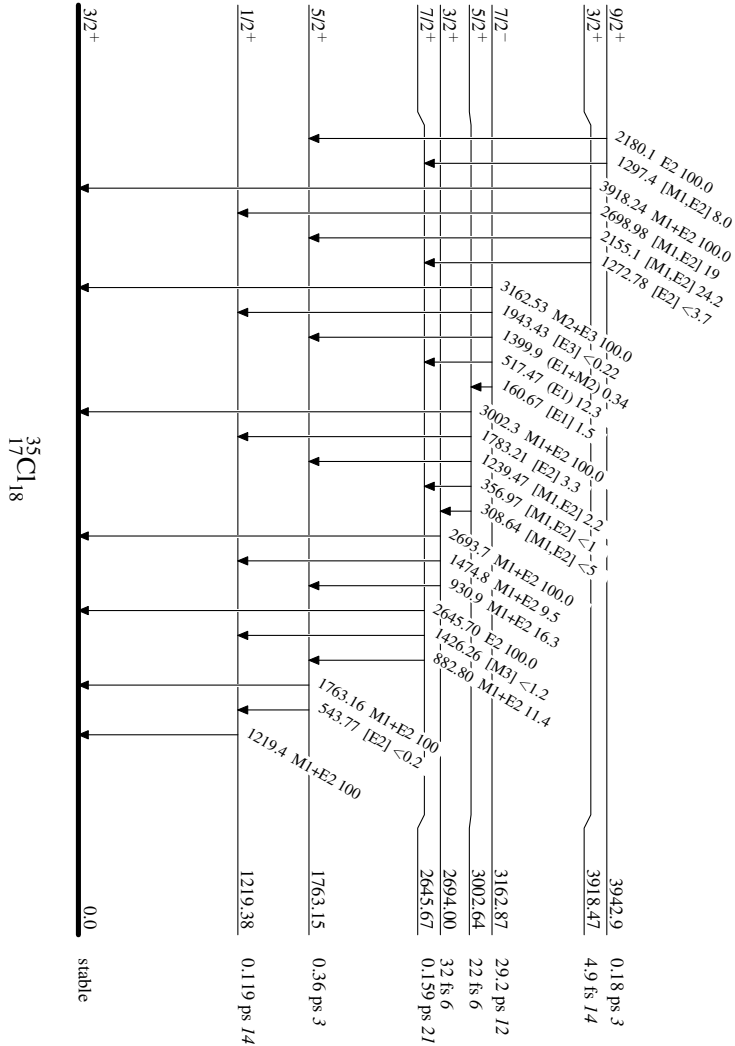
-----▶ γ Decay (Uncertain)



Adopted Levels, Gammas

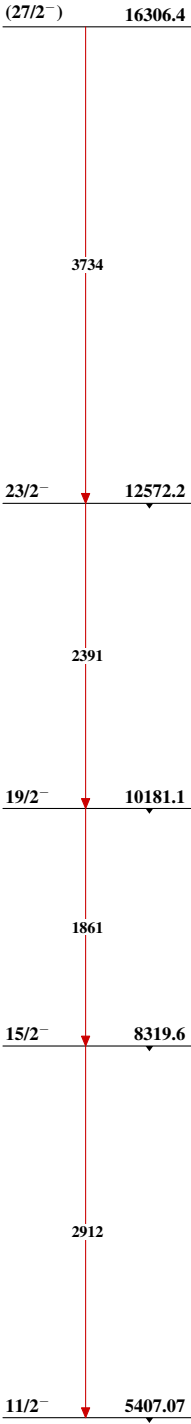
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Band(A): Band based on $f_{7/2}$
orbital



$^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas (continued)Band(B): Band based on $13/2^+$ $21/2^+$ 11459.1 $19/2^+$ 10858.5

2615

2070

2015

 $17/2^+$ 8844.4
($15/2^+$) 8788.3

971

915

 $13/2^+$ 7873.03 $^{35}_{17}\text{Cl}_{18}$